When Is Label-Free Adaptation Knowable? Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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***Abstract*—Test-time adaptation (TTA) updates a model on unlabeled data to counter distribution shift, but the same update that recovers accuracy under one shift can quietly destroy it under another—and with no target labels, the system cannot tell which case it faces. We study the decision that comes *before* adaptation—should the model adapt at all?—and cast it as a three-way choice: *adapt* when adaptation is knowably helpful, *freeze* when it is knowably harmful, and *abstain* when the unlabeled evidence cannot decide. Our central result is that this question is sometimes *provably unanswerable*: two target worlds can induce identical label-free evidence yet opposite adaptation benefit, so no label-free rule is correct in both and abstention becomes necessary, not merely cautious. When the evidence *is* informative, we give an exact success condition—the sign of the benefit is recoverable iff an observable confidence margin exceeds how far calibration can drift (the "calibration-drift budget")—together with a finite-sample certificate that acts on it while holding the false-adapt and false-freeze rates at a chosen level α. We package this as KGA, a drop-in wrapper around existing adapters (Tent, EATA, SAR) that decides *whether* to trust them rather than how to adapt. On a pre-registered, un-cherry-picked panel of natural and corruption shifts—each scored once on held-out test with bootstrap confidence intervals—KGA behaves as the theory predicts: on pre-registered mixed harmful+helpful streams it beats POEM and AETTA (the no-harm SOTA) with Holm-corrected regret gaps at** $\mathrm{FA_u}$**=0; it strictly beats both always-adapt and always-freeze on synthetic stress grids where harmful adaptation is frequent and detectable (CIFAR-10-C, ImageNet-C SAR), prevents damage where adaptation mostly hurts, and is uniformly no-harm on every natural shift, where adapting is already near-optimal. KGA is a safety layer, not a universal accuracy booster: its wins are regime-specific, and we report—rather than hide—the regimes and honest nulls where it only ties. Full per-dataset tables and the complete benchmark battery appear in the body and appendix.**

***Index Terms*—test-time adaptation, distribution shift, selective prediction, identifiability, label-free risk estimation**

Scope. We report positive and negative results; see §IX and §VIII for honest limitations and excluded alternate wins.

I. Introduction

Deployed models rarely encounter the same distribution they were trained on. Test-time adaptation (TTA) addresses this mismatch by updating a model on unlabeled target data, often through entropy minimization, sample filtering, or stability-aware updates. In favorable shifts, such updates can recover lost accuracy. In unfavorable shifts, however, the same unlabeled objective can silently degrade performance: the model may sharpen wrong predictions, adapt to label imbalance, or collapse under non-stationary streams. Because target labels are unavailable at deployment, the system often cannot tell whether adaptation is helping or hurting.

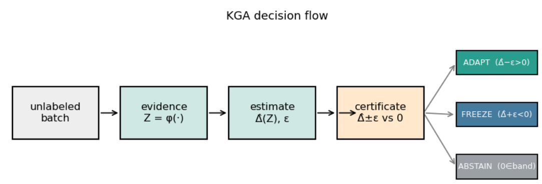

Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (sign unidentifiable), with false-adapt/false-freeze $\leq \alpha$.

Most work on TTA asks how to design a better adaptation mechanism. This paper asks a prior question: should the system adapt at all? We study label-free adaptation as a decision problem. Given a frozen baseline, an adapted candidate, and an unlabeled target batch, the system must choose one of three actions: adapt when adaptation is knowably helpful, freeze when adaptation is knowably harmful, and abstain when the available evidence cannot identify the sign of the benefit. Figure 1 previews this decision certificate.

This decision is constrained by identifiability. We show that there exist target worlds with identical label-free evidence but opposite adaptation benefit. In such cases, no label-free rule can certify the correct decision in both worlds; abstention is not merely conservative, but information-theoretically necessary. Conversely, when the label-free evidence is risk-aligned and the benefit estimate admits a valid uncertainty radius, a finite-sample certificate can safely route between adapt, freeze, and abstain while controlling false-adapt and false-freeze errors.

Our main theoretical result is an exact benefit-sign frontier. The sign of the adaptation benefit is identifiable from label-free observables precisely when an observable margin on the disagreement region exceeds a calibration-drift budget. Intuitively, the margin is how confidently the label-free evidence points toward "adapt" or "freeze," and the budget is how far that signal can be thrown off by calibration drift; the decision is trustworthy exactly when confidence outweighs drift. This characterizes why common label-free heuristics can work in benign regimes and fail under drift: they implicitly assume that this drift is zero or negligible. We further show that no label-free-checkable assumption can remove this obstruction in full generality; the remaining ambiguity is an irreducible one-bit orientation choice, which can only be supplied through structural assumptions that are falsifiable but not verifiable.

We instantiate the framework as KGA, a mechanism-agnostic wrapper around existing adaptation candidates such as Tent, EATA, and SAR. KGA does not introduce a new adaptation objective. Instead, it estimates the label-free benefit of a candidate, attaches a calibrated uncertainty radius, and returns adapt, freeze, or abstain. This makes KGA a safety layer for TTA rather than a replacement for TTA methods.

Empirically, KGA behaves according to the theory. On a pre-registered CIFAR-10-C stress grid, where harmful adaptation is frequent and detectable, KGA beats both always-adapt and always-freeze for Tent and EATA while making zero false-adapt decisions, and it ties the collapse-resistant SAR; on the same mixed stream it **beats POEM and AETTA** (Table VIII) with Holm-corrected paired-bootstrap CIs. In online non-stationary CIFAR-10 streams, always-adapt collapses in the harm-dominated regime and always-freeze forgoes gains in the helpful regime, while KGA remains near-best across both. At ImageNet scale, KGA also improves over both trivial policies on a harmful SAR stream. On natural-shift audits, under a valid out-of-fold conformal radius KGA is *uniformly no-harm*: it matches the better fixed policy and beats the worse one at false-adapt $\leq \alpha$, with *no* natural-shift beats-both. Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt (CIs exclude zero) but *tie* always-freeze (Tables XIV, XV); their earlier in-sample-radius "beats-both" was a calibration bug, corrected here to no-harm. On the one-sided WILDS shifts Camelyon17 (Protocol G) and RxRx1, KGA likewise matches the better fixed policy at false-adapt $\leq \alpha$ without beating both; the earlier Camelyon17 Protocol-G beats-both was a separate pooling artifact (in-distribution `id_val` cells mixed into the held-out set, §VII-B). ImageNet-R remains a split-specific lead, not a stable win.

a) Contributions.:

- We formalize label-free TTA as an adapt/freeze/abstain decision problem governed by the sign of adaptation benefit.
- We prove an impossibility result showing that some adaptation decisions are fundamentally unknowable from label-free evidence.
- We derive a finite-sample certificate that controls false adaptation and false freezing under risk-aligned evidence.
- We characterize an exact benefit-sign frontier based on observable disagreement margin and calibration drift.
- We validate the framework on a seven-dataset core panel (Office-Home, Camelyon17, CIFAR-10-C, ImageNet-C, iWildCam, RxRx1, ImageNet-R) with bootstrap confidence intervals: beats-both on the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR), a pre-registered mixed head-to-head win over POEM and AETTA (Table VIII), and uniform no-harm on every natural shift; controlled synthetic validations and an additional breadth dataset (PACS) are reported in Appendix B, with CIFAR-10.1 retained in-body as a supporting low-margin check (§VII-A). A constructed cross-protocol aggregate (n=143) beats both fixed policies under out-of-fold calibration, demonstrating per-condition routing under heterogeneity without claiming a universal mixed-deployment win (§VII-C).

Two bars, stated once. Throughout, the **pre-registered headline bar** is a point-estimate beats-both at false-adapt $\leq \alpha$; the **CI-robust** criterion (both bootstrap regret-gap CIs exclude zero) is a strictly stronger layer we report as a separate column. Under a valid out-of-fold radius, no natural shift clears either bar: Office-Home and iWildCam beat always-adapt with CIs excluding zero but *tie* always-freeze, so both are no-harm, not wins. The only beats-both rows are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR).

K-Bound does not claim universal improvement. When adaptation is already safe, a certificate can only tie or slightly trail always-adapt. When freezing is already optimal, it should match freeze rather than force an unnecessary update. Its distinctive value appears when harmful adaptation is frequent, detectable, and costly—the regime where blind adaptation is most dangerous.

II. Related work and positioning

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting.

A. Test-time adaptation methods

Entropy-based TTA updates a frozen model on unlabeled target data. **Tent** [1] minimizes prediction entropy over the batch-norm affine parameters; **SHOT** [2] combines information maximization with pseudo-labeling for source-free adaptation; **TTT** [3] trains a self-supervised auxiliary task and updates it at test time; **EATA** [4] adds entropy

TABLE I

Pre-registered evaluation panel (un-cherry-picked): nine datasets, one bar. Regret-to-oracle (lower is better) for always-freeze, always-adapt, and KGA; oracle regret is 0 by construction. **Pre-reg WIN** applies the single headline bar uniformly: held-out test scored once, false-adapt $\leq \alpha$=0.10, *and* $\text{regret}_{\text{KGA}} < \text{regret}_{\text{ADAPT}}$ *and* $< \text{regret}_{\text{FREEZE}}$ (point estimate). **CI-robust** is the strictly stronger layer (`statistical_policy_v1.md`): both regret-gap 95% bootstrap CIs exclude zero. Under a valid out-of-fold radius the only beats-both rows are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR); every natural shift is no-harm (matches the better fixed policy, beats the worse, false-adapt $\leq \alpha$). KGA regret values are reported under the valid out-of-fold radius; earlier in-sample-radius Office-Home/iWildCam "wins" were a calibration bug and are corrected to no-harm (§VII-B). The seven core datasets plus a supporting null (CIFAR-10.1) and a domain-generalization suite (PACS) are all shown; heuristic / AETTA / ATC gate comparisons appear in App. B (Table XIX).

Dataset (protocol)	regime	always-freeze	always-adapt	**KGA**	false-adapt	pre-reg WIN	CI-robust
CIFAR-10-C stress (Tent/EATA)	synthetic, harmful-detectable	0.1232/0.1311	0.0086/0.0037	**0.0016/0.0015**	$0 \leq \alpha$	**yes**	yes
ImageNet-C (SAR)	synthetic, harmful-detectable	0.0267	0.1124	**0.0229**	$0 \leq \alpha$	**yes**	re-run[a]
Office-Home (Prot. M v2)	covariate, freeze-favorable	0.01582	0.04681	0.01570	$0 \leq \alpha$	no[b]	no
iWildCam (Prot. H v2)	natural, freeze-favorable	0.00410	0.10283	0.00410	$0 \leq \alpha$	no[b]	no
Camelyon17 (Prot. G)	natural, adapt-favorable	0.1381	0.0000	0.0000	$0 \leq \alpha$	no[c]	no
RxRx1 (Prot. J)	natural, harmful-detectable	0.0000	0.2531	0.0000	$0 \leq \alpha$	no[d]	—
ImageNet-R (Prot. D)	natural, *unknowable*	0.0180	0.0052	0.0042[e]	$0.071 \leq \alpha$	no	no
CIFAR-10.1 (Prot. K)	natural, low-margin	0.00171	0.01902	0.00206	$0.444 > \alpha$	no[f]	no
PACS (Prot. P)	domain-gen, freeze-favorable	0.0259	0.0274	0.0191	≤ 0.056	no[g]	no

[a]Synthetic corruption; robustness is a mechanism-faithful SAR re-run (Table XX), not a bootstrap CI. [b]No-harm under a valid out-of-fold radius: beats always-adapt with CI excluding zero (Office-Home +0.031 [0.004, 0.062]; iWildCam +0.099 [0.080, 0.118]) but *ties* always-freeze (Office-Home +0.0001 [0, 0.0003]; iWildCam exactly +0.0000 [0, 0]). The earlier in-sample-radius beats-both was a calibration bug; corrected here (§VII-B). [c]On held-out OOD test (hospital domain, n=18) online EATA is helpful-dominated (0% harmful), so KGA adapts and *ties* always-adapt. The previously reported pooled Protocol-G beats-both (regret 3.6×10^{-5}, FA 2.6%, n=54) mixed in-distribution `id_val` cells into the held-out set; on genuine OOD it vanishes (§VII-B). [d]KGA freezes all 360 conditions, *tying* the freeze oracle; adapting is catastrophic (0.253). No-harm, not beats-both. [e]Beats both on 1/4 splits only (0/10 backbones, App. B); not a stable win. [f]$\text{regret}_{\text{KGA}} >$ always-freeze and FA$>\alpha$ cross-seed. [g]PACS row is the mean over 4 leave-one-domain-out splits: no-harm on 3/4 (art_painting, cartoon, sketch; FA 0), CI-robustly beating always-adapt on cartoon; the photo split is a null where adapting is oracle-optimal and KGA safely partial-adapts (FA $0.056 \leq \alpha$). Per-domain in Table XXI.

filtering of unreliable samples plus a Fisher penalty against forgetting; **SAR** [5] adds sharpness-aware updates with a reset that aborts incipient collapse; **CoTTA** [6] stabilizes continual adaptation with teacher-student ensembling; and **MEMO** [7] adapts via test-time augmentation and entropy minimization. These methods make adaptation *more robust*, and several detect or suppress instability, but all commit to adapting: none emits a *pre-adaptation* certificate separating helpful, harmful, and unknowable evidence, and none can *abstain* on the adapt/freeze decision itself. K-Bound is mechanism-agnostic—it *wraps* any such candidate (we evaluate Tent/EATA/SAR). Tempora [12] further shows that unconstrained offline rankings can invert under latency and compute pressure, with no fixed policy robust across temporal scenarios. K-Bound is complementary: rather than selecting one always-on method, it certifies per-condition ADAPT/FREEZE/ABSTAIN decisions.

B. Protected and guarded adaptation

A second line guards adaptation *during* the update. **POEM** [11] protects TTA by online entropy matching—a betting-style monitor that halts or dampens a single failure mode—and **Schirmer et al.** [9] run anytime-valid risk monitors over the TTA stream, raising a sequential alarm when adaptation degrades a proxy risk; both descend from sequential harmful-shift tracking for deployed models (**Podkopaev & Ramdas** [10]). These are the nearest neighbours of our certificate, and the delta is precise: they act *reactively, during or after* the update and control a false-*alarm* rate in one failure direction, whereas K-Bound decides *before* committing, poses the *three-way* decision with an explicit *unknowable* region (where abstention is mandatory, Cor. 1), controls the false-*adapt* rate on the benefit sign itself, and proves the matching impossibility floor (Prop. 1). The two are complementary: a pre-adaptation certificate can gate what a runtime monitor then tracks.

C. Label-free accuracy and error estimation

A third line predicts target *performance* without labels: **ATC** [13] thresholds softmax confidence calibrated on source; **Agreement-on-the-Line** [15], [14] exploits the linear correlation between in- and out-of-distribution model *agreement* to extrapolate accuracy; **AETTA** [17] estimates TTA accuracy from dropout-prediction disagreement. Closest in spirit, **Kim et al.** [16] use agreement-on-the-line to decide *when* TTA helps and to select TTA hyperparameters without labels—a genuine decision use of an accuracy estimate, but a plug-in one: it is exactly the β=0 face of our frontier (Thm. 3(iv)), sound only when calibration transfers, with no finite-sample false-adapt control and no abstain region. Estimating $R_T(f)$—or detecting that TTA has failed *post hoc*—is strictly different from certifying sign $\big(R_T(f_0) - R_T(f_a)\big)$ *before* adapting with a controlled false-adapt rate, the object K-Bound bounds.

D. Unsupervised risk estimation and identifiability

The hardness of our problem is inherited from unsupervised risk estimation. **Steinhardt & Liang** [20] estimate risk without labels only under a conditional-independence structure; **Ben-David et al.** [21], [22] bound target risk by source risk plus an $\mathcal{H}$-divergence; **Lipton et al.** [23] (BBSE) correct label shift under an invertible confusion matrix;

Rosenfeld & Garg [24] bound target error under an explicit, unverifiable assumption (their Remark 3.9 asserts informally the impossibility our Theorem 4 proves); the rank-one agreement algebra underlying our Theorem 4 is classical [25], [26], [28], [29], its dependent-classifier diagnostics appear in [27], and label-free algebraic evaluation with consistency alarms is developed in the NTQR program [30]. Each shows risk is identifiable only under structural assumptions and impossible in general—precisely the boundary our Theorem 1(ii) makes quantitative. We weaken the target from a *risk* to the *sign of a difference* of risks, which is identifiable on a strictly larger set (the observable disagreement region, Lemma 1).

E. Selective prediction, conformal, and anytime-valid inference

Selective prediction [34] abstains on individual *predictions*; we instead abstain on the *adaptation decision*. Our finite-sample certificate reuses split-conformal prediction (**Vovk et al.** [35]; **Angelopoulos & Bates** [36]) to build the radius ε, and its streaming form (Appendix E0c) is a testing-by-betting e-process in the safe-anytime-valid-inference tradition (**Shafer** [37]; **Ramdas et al.** [38]); confidence-sequence and concentration tools (**Howard et al.** [40], **Maurer & Pontil** [44], **Hoeffding** [45]); and Vovk's conformal test martingales [39] embody, for exchangeability, the falsify-not-verify stance our budget audit (Thm. 4(iv)) takes for the drift budget.

The surviving gap. No prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee. Table II makes the gap precise.

F. Benchmarks, code, and reproducibility

TTA progress is benchmarked on standardized corruption suites—CIFAR-10/100-C and ImageNet-C [49], plus ImageNet-R [50]—and on cross-domain shift suites including WILDS/Camelyon17/RxRx1/Office-Home [51], [52], [53], [54]. The field increasingly emphasizes standardized, executable evaluation (**RobustBench** [56]; **DomainBed** [55]) and reproducibility checks on fresh test draws such as CIFAR-10.1 [57]. Most TTA methods ship official code, but evaluations vary in batch regime, stream composition, and update budget—*precisely* the axis along which adaptation flips between helpful and harmful. We therefore release K-Bound as a *pre-registered, executable* artifact: a public repository with pinned environments, a one-command reproduction harness, auto-downloaded standard corruption data, fixed seeds, machine-checked theorem validators, and the full stratified result manifests behind every table and figure (§XII). Our certificate builds on split-conformal prediction (Vovk et al.; Angelopoulos & Bates) and, for the streaming variant, testing-by-betting e-processes (Shafer; Ramdas et al.). The CAN benchmark formalizes a practical "adapt-or-skip" decision in chronological streams [8]; K-Bound complements that benchmark view with a theorem-level frontier and impossibility account for when such decisions are certifiable from label-free evidence.

III. Problem setup

Source $P_S(X,Y)$ has labels at train/validation; target $P_T(X,Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X),Y) - \ell(f_a(X),Y) \mid Z = z]$. The central question is when $\operatorname{sign}\Delta(z)$ is recoverable from Z alone.

a) Notation.: f_0 is the frozen model and f_a the adapted/gated candidate; $\Delta = R_T(f_0) - R_T(f_a)$ is the population adaptation benefit and $\Delta(z)$ its value conditioned on evidence $Z{=}z$; $\widehat{\Delta}$ is a label-free estimator of Δ with confidence radius ε at miscoverage level α; $g(Z) \in \{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$ is the decision rule; and $D = \{x : f_0(x) \neq f_a(x)\}$ is the observable disagreement region. *Regret-to-oracle* denotes $R(g) - R(g^\star)$ against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

IV. Definitions

Definition 1 (Observable risk alignment / Identifiability)**.** Z is *risk-aligned* if the sign of adaptation benefit $\operatorname{sign}\Delta$ is identifiable from its law (identifiability)—specifically, no two admissible worlds with opposite benefit signs induce the same evidence distribution. We treat conformal coverage of the benefit estimator separately as a property of the calibration sample (see Theorem 2).

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if Z certifies $\Delta(z) > 0$ (**adapt**); *knowably harmful* if Z certifies $\Delta(z) < 0$ (**freeze**); *unknowable* if the same Z is compatible with both signs (**abstain**).

> **What "label-free" means here.** *Target-label-free* (our setting): no labels from the target/test stream are ever used. *Validation labels allowed*: labeled source and a held-out labeled validation/probe set may calibrate the benefit estimator $\widehat{\Delta}$ and the radius ε. *Target-unsupervised evidence*: the deployment batch contributes only X, predictions, entropy, confidence, drift and other label-free statistics Z. The online-TTA results that use a clean labelled validation probe (§VII) are target-label-free in this sense—they use no *target* labels—not unsupervised of all labels.

V. Theory: four pillars

The body is organized as four pillars, with five core theorems serving as the load-bearing statements inside that scaffold. **Pillar I (impossibility floor):** Theorem 1 plus the Le Cam lower bound and regret identity establish

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

TABLE II

POSITIONING. PRIOR TTA ADAPTS AND (SOME) DETECT FAILURE; NONE GIVES A *PRE-ADAPTATION* ADAPT/FREEZE/ABSTAIN CERTIFICATE WITH A FALSE-ADAPT BOUND.

when abstention is information-theoretically forced. **Pillar II (certificate + audit):** Theorem 2 gives finite-sample false-adapt control, and the one-bit audit machinery of Theorem 4(iv) supplies a common falsifier for the legacy budget premises (formerly presented as separate propositions, now treated as corollaries of one audit framework). **Pillar III (identifiable regimes):** Theorem 3 and Theorem 4 characterize exactly what label-free evidence identifies and what one-bit supplement remains irreducible. **Pillar IV (rates):** Theorem 5 gives matching evidence-channel rates. The open piece—a weakest falsifiable class on which one declared bit suffices—is settled in Appendix J: conditionally for the single class $\mathcal{C}_{\text{mono}}$, and unconditionally as an explicit finite family of dominance polytopes (Theorem 9). Supporting variants (multiclass/regression reductions, label-shift and drift boundaries, ambiguity-reach unification, router capacity, anytime certificates, the γ-meter, and the agreement–accuracy law) are presented as corollaries or companions in the companion manuscript, each with a validator; every numeric claim here traces to a script in the public repository.

Lemma 1 (Reduction to the disagreement region)**.** *Let $D = \{x : f_0(x) \neq f_a(x)\}$ (observable). For binary Y and $0/1$ loss, with $\bar a = \mathbb{P}_T(f_a{=}Y \mid D)$, $\Delta = 2\,\mu(D)(\bar a - \frac{1}{2})$, so for $\mu(D) > 0$: $\operatorname{sign}\Delta = \operatorname{sign}(\bar a - \frac{1}{2})$. Writing s for any source-calibrated score of f_a, $M := \mathbb{E}_{\mu|D}[s] - \frac{1}{2}$ (observable) and $\gamma := \mathbb{E}_{\mu|D}[\eta_a - s]$ (unobservable), $\boxed{\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)}$. For K-class $0/1$ loss $\Delta = \mu(D)(p_a - p_0)$, and for squared loss the sign reduces to one moment on D; both extend the identity verbatim (manuscript, App. C).*

Proof sketch. Off D the losses coincide; on D exactly one of f_0, f_a is correct (binary), giving $\mathbb{E}[\delta \mid D] = 2\bar a - 1$; linearity splits $\bar a - \frac{1}{2} = M + \gamma$. Validated on synthetic draws (identity exact to machine precision; multiclass to 10^{-16} over 4000 trials). □

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat\Delta$, the gate $\hat g = \mathbf{1}[\hat\Delta > 0]$ satisfies $R(\hat g) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\hat\Delta \neq \operatorname{sign}\Delta\}\big]$ with $g^\star = \mathbf{1}[\Delta > 0]$ Bayes; on $\{|\Delta| < \varepsilon\}$ committal regret is $\le \varepsilon$, justifying abstention in the low-margin band. Validated to machine precision on synthetic regret identities.*

Proposition 1 (Finite-sample minimax regret floor for label-free gating)**.** Intuition. *The identity of Lemma 2 is an equality, not a hardness statement; pairing it with Le Cam's two-point method turns it into a finite-n lower bound on the regret of* any *label-free gate, which does not vanish as $n \to \infty$ when the two worlds are rescaled to stay equally hard. Let $\{P_\theta\}_{\theta \in \{-1,+1\}}$ be two target worlds sharing the same disagreement structure with constant benefit magnitude $|\Delta(P_\theta)| \equiv \Lambda > 0$ and opposite signs* $\operatorname{sign}\Delta(P_\theta) = \theta$*, and let $Q_\theta^{(n)}$ denote the law of* any *label-free evidence Z computed from n unlabeled observations under P_θ. Then for every label-free committal gate $\hat g : Z \mapsto \{\text{ADAPT}, \text{FREEZE}\}$ and every finite n,*

$$\inf_{\hat g} \max_{\theta \in \{\pm 1\}} \big[R_{P_\theta}(\hat g) - R_{P_\theta}(g^\star)\big] \ge \tfrac{\Lambda}{2}\Big(1 - \mathrm{TV}\big(Q_{-1}^{(n)}, Q_{+1}^{(n)}\big)\Big) \ge \tfrac{\Lambda}{4}\, e^{-\mathrm{KL}(Q_{-1}^{(n)} \| Q_{+1}^{(n)})}.$$

In particular there is an explicit Gaussian two-point family ($X \sim \mathcal{N}(\theta d_n, 1)$ on D, $f_0 = \mathbf{1}[x > 0]$, $f_a = \mathbf{1}[x < 0]$, with the target label rule giving $\Delta = \theta$, so $\Lambda = 1$) for which, taking $d_n = c/\sqrt{n}$, both bounds are constant *in n: $\mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)}) = 2\Phi(c) - 1$ and the floor equals $\Lambda\,\Phi(-c)$, with the Bretagnolle–Huber certificate $\frac{\Lambda}{4}e^{-2c^2} > 0$. Theorem 1 is the $c \to 0$ ($TV \to 0$) face, where the floor saturates at $\Lambda/2$.*

Proof. By Lemma 2, since $|\Delta| \equiv \Lambda$ and $g^\star$ is the constant correct action $\operatorname{sign}\theta$ in world θ, $R_{P_\theta}(\hat g) - R_{P_\theta}(g^\star) = \Lambda\, Q_\theta^{(n)}$($\hat g$ commits to the wrong sign). Averaging over the uniform prior on θ gives $\frac{1}{2}\sum_\theta \big[R_{P_\theta}(\hat g) - R_{P_\theta}(g^\star)\big] = \frac{\Lambda}{2} M(\hat g)$, where $M(\hat g) = Q_{+1}^{(n)}(\hat g{=}\text{FREEZE}) + Q_{-1}^{(n)}(\hat g{=}\text{ADAPT})$ is the mixed committal error of Theorem 1(ii). Le Cam's testing-affinity identity yields $\inf_{\hat g} M(\hat g) = 1 - \mathrm{TV}(Q_{-1}^{(n)}, Q_{+1}^{(n)})$, and max $\geq$ avg gives the minimax form; Bretagnolle–Huber ($1 - \mathrm{TV} \geq \frac{1}{2}e^{-\mathrm{KL}}$) gives the closed-form certificate. For the Gaussian family the sufficient statistic $\bar{X} \sim \mathcal{N}(\theta d_n, 1/n)$ has $\mathrm{TV} = 2\Phi(\sqrt{n}\, d_n) - 1$ and per-observation $\mathrm{KL} = 2d_n^2$, so $d_n = c/\sqrt{n}$ makes both n-independent; $|\Delta| = 1$ exactly because each label rule makes one of f_0, f_a correct on a probability-one event. Validated (`val_thm2_lecam_finite_n.py`, seeds fixed): across $c \in \{0.25, 0.5, 1, 1.5\}$ and $n \in \{10, 50, 200, 1000\}$ the empirical floor is constant in n, the Bayes (sample-mean threshold) gate *meets* it (so it is tight), and a brute-force minimax over a panel of label-free rules (all thresholds/polarities on five statistics) never beats it within Monte-Carlo error. □

Theorem 1 (The unknowable regime is real and tight)**.** Intuition. *Identical label-free evidence can support opposite adaptation benefits; any valid rule must then abstain rather than guess.* (i) *There exist target worlds P_T^1, P_T^2 with identical evidence laws,* $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$*, and* $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \neq 0$ *(explicit witness: $X \sim \mathcal{N}(0,1)$, $f_a = 1 - f_0$, flipped labels). No label-free rule distinguishes them; the worst-case-optimal committal-regret action is* ABSTAIN. (ii) *Quantitatively, for any committal rule g the mixed error obeys the Le Cam identity $\inf_g M(g) = 1 - \mathrm{TV}\big(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)}\big)$, equal to 1 in the witness for every n and decaying only as the evidence laws separate.* (iii) *Any rule with false-adapt and false-freeze rates $\leq \alpha$ must abstain on matched evidence with probability $\geq 1 - 2\alpha$: abstention is mandatory, not conservative.*

Proof sketch. (i) The witness makes Z a function of X alone with $P^1(X) = P^2(X)$. (ii) A committal rule is a test between the two evidence laws; Le Cam's testing affinity plus data processing. (iii) Matched evidence forces common action probabilities q_a, q_f; validity caps each at α. Validated: empirical $\inf_g M$ tracks $1 - \mathrm{TV}$ across all n, 100% abstention on the witness. □

Theorem 2 (The adapt/freeze/abstain certificate)**.** Intuition. *When a benefit estimate carries a valid uncertainty radius, committing only outside the abstain band controls false-adapt and false-freeze at level α. Suppose $\hat{\Delta}(z)$ admits a radius $\varepsilon(z)$ with $\mathbb{P}[|\hat{\Delta} - \Delta| \leq \varepsilon] \geq 1 - \alpha$. The rule* ADAPT *if $\hat{\Delta} - \varepsilon > 0$,* FREEZE *if $\hat{\Delta} + \varepsilon < 0$, else* ABSTAIN*, has marginal, unconditional false-adapt and false-freeze rates $\leq \alpha$ each (over the calibration draw under exchangeability). With $\hat{\Delta}$ an empirical-Bernstein mean of n bounded paired benefits (variance proxy σ^2, range $b - a$), the certificate commits once $n \gtrsim \big(2\sigma^2/\Delta^2\big)\log(2/\alpha) + 3(b-a)|\Delta|^{-1}\log(2/\alpha)$: sample complexity scales as Δ^{-2}, logarithmically in $1/\alpha$. A sequential, anytime-valid e-process version (valid at any stopping time; β-drift-robust variant included) is Theorem B.1 of the manuscript.*

Proof sketch. On the good event, ADAPT implies $\Delta \geq \hat{\Delta} - \varepsilon > 0$; the complement has mass $\leq \alpha$; commitment when $\varepsilon(n) < |\Delta|$ inverts the Bernstein radius. Note that this marginal conformal guarantee applies to the radius and decision rather than the accuracy of the benefit model itself, and relies on calibration-deployment exchangeability at the locked-protocol granularity, degrading under task shift. The residual burden — a calibration sample exchangeable with deployment — is exactly what Theorem 4(iv) makes auditable. Validated on synthetic e-process tests (false-adapt $\leq \alpha$ at nominal level). □

Theorem 3 (Exact benefit-sign frontier)**.** Intuition. *On the disagreement region, the benefit sign is knowable iff an observable margin $|M|$ exceeds a calibration-drift budget β; otherwise abstention is forced. Fix the observables (μ, P_S, f_0, f_a) with $\mu(D) > 0$, so M is determined, and let $\mathcal{C}_\beta = \{P_T : |\gamma| \leq \beta\}$. Under the location/alignment regularity of Appendix E0c, and noting that γ is a population quantity not observed at deploy, the statements below hold* unconditionally *for* every *drift budget $\beta \geq 0$ and every target with $\mu(D) > 0$, with β a free parameter of the frontier rather than an assumption (and (iii) shows every identifying class is of this form).* (i) *γ is a one-dimensional sufficient statistic for* $\operatorname{sign}\Delta$ *given the observables.* (ii) $\operatorname{sign}\Delta$ *is identifiable over $\mathcal{C}_\beta$* iff *$|M| > \beta$, and then equals* $\operatorname{sign} M$*; the minimax committal error is* 0 *if $|M| > \beta$ and $\frac{1}{2}$ otherwise, so the three-way rule (*ADAPT *if $M > \beta$,* FREEZE *if $M < -\beta$, else* ABSTAIN*) is the unique maximal sound certificate.* (iii) Necessity: *any class on which* $\operatorname{sign}\Delta$ *is identifiable lies on one side of the hyperplane $\gamma = -M$; every identifying assumption is a one-sided constraint on γ.* (iv) *The label-free heuristics ATC [13], DoC, GDE [18], COT [19], AETTA [17], and agreement-on-the-line [15], [16] are exactly the β=0 face: sound iff $\gamma = 0$, sharing the single failure mode $\gamma \neq 0$.*

Proof sketch. All from Lemma 1: (ii) a $\pm\beta$ perturbation crosses 0 iff $|M| \leq \beta$, plus the two-point argument of Theorem 1; (iii) constancy of $\operatorname{sign}(M + \gamma)$ is one-sidedness; (iv) each heuristic thresholds an observable surrogate, i.e. commits to $\operatorname{sign} M$. Validated: frontier law exact over 5×10^5 draws; the unconditional family characterization (identifiable over $\mathcal{C}_\beta$ *iff* $|M| > \beta$) holds for every tested β; ATC error matches the predicted event to machine precision at drift 0.04–0.28 (synthetic validators for Theorems T1–T5 and AGL). □

Corollary 1 (Knowable regimes)**.** (a) *Under covariate shift ($P_S(Y|X) = P_T(Y|X)$, bounded density ratio), importance weighting identifies Δ and its sign from labeled source plus unlabeled target (γ-budget discharged with $\beta = 0$ on the importance-weighted score); the premise itself is not label-free testable, re-entering Theorem 1 when it fails.* (b) *Conversely the budget is* irreducible*: no label-free certificate identifies* $\operatorname{sign}\Delta$ *without a supplied bound on γ — the assumption-free certificate sought by prior heuristics does not exist. (Label shift, bounded drift, smooth drift, and the unified ambiguity-reach hierarchy: manuscript, App. C.)*

Theorem 4 (One-bit dichotomy; Conjecture 1 resolved)**.** Intuition. *Label-free observables fix every adaptation statistic except one global orientation bit; no testable assumption class removes that bit in full generality. Consider a panel* $f_0, f_1, \dots$ *on* D *with correctness indicators conditionally independent given* Y *(CEI) and per-class symmetric accuracies (jointly, hypothesis* $\boldsymbol{H}$*; advantages* $b_j = 2a_j - 1$*, agreements* $c_{ij} = 2A_{ij} - 1$*; the rank-one algebra and anchor are classical [25], [26], [28], [29] — new here are the deficit, the dichotomy, and the audit).* (i) *H is minimal: under CEI alone,* $c_{ij} - b_i b_j = \pi(1-\pi)\delta_i\delta_j$ *exactly (*δ_j *the per-class accuracy gap), so the rank-one law holds for a* $K \geq 3$ *panel iff every* $\delta_j = 0$. (ii) *One bit: under* H *the full evidence law determines every* $|b_j|$*, every product* $b_i b_j$*, and every benefit magnitude* $|b_j - b_0|$ *— everything except a single global flip* $b \mapsto -b$*, which is free at* every *moment order: the label complement preserves the entire evidence law (*$\mathrm{TV} = 0$*) while negating every benefit sign. The drift* $\gamma = b_a/2 - M$ *is thereby point-identified up to that bit.* (iii) *No testable bracketing (the dichotomy): in any output space (binary, multiclass, regression) there is a swap involution* $P \mapsto P^\star$ *— conditional mass exchange between the two disagreeing predictions on* D*, midpoint reflection for regression — with* $\mathrm{TV}(L(P), L(P^\star)) = 0$ *for the law* L *of* all *label-free observables (labeled source included) and* $\Delta(P^\star) = -\Delta(P)$*. Hence no* evidence-definable *(label-free testable) assumption class identifies* $\operatorname{sign}\Delta$*; every identifying assumption is a* bit-selector *(*γ*-budgets select* $\operatorname{sign} M$*; majority-above-chance selects* $\operatorname{sign}(\operatorname{median} b)$*; the agreement-line slope selects* $\operatorname{sign}(1 - 2\bar{w})$*); and the minimal untestable supplement is exactly one bit, which suffices under* H*.* (iv) *Budgets are falsifiable, never verifiable: the bit-robust test rejecting "*$|\gamma| \leq \beta$*" when* $\min_{\mathrm{flip}} |\hat{\gamma}| > \beta + r_n$ *is level-*α *with power* $\geq 1 - \alpha$ *beyond* $2r_n$*; verification fails exactly on the blind set* $\big\{\, \big||M| - |b_a|/2\big| \leq \beta < |M| + |b_a|/2 \,\big\}$*, and the* $M \geq 4$ *rank-one residual* τ *is sound but* not *complete (strictly interior stealth laws with* $\tau = 0$ *exist that bias* b *and can flip the recovered decision).*

Proof sketch. (i) Expand c_{ij} over Y and cancel cross terms. (ii) Triple products $c_{ik}c_{il}/c_{kl} = b_i^2$; the complement $Y' = 1 - Y$ fixes predictions (maps of X) hence the whole evidence law, while $b \mapsto -b$ and H is preserved. (iii) The swap preserves the X-marginal and source, so L is fixed at every order; on D it exchanges $p_a \leftrightarrow p_0$ (or negates $f_0 + f_a - 2Y$), so Δ negates; identifiability on a swap-closed class is impossible, and selecting one member per swap orbit is by definition one bit. Algebraically: observables generate the *even* correctness subalgebra ($s_i s_j = u_i u_j$), the decision is *odd* ($s_j = u_j v$), and the swap is $v \mapsto -v$. (iv) Empirical-Bernstein for $\hat{M}$, a product-ratio perturbation bound for $\hat{b}_a$, and the flip-min for bit-robust soundness; the stealth construction is a strictly interior LP on the 2^4 correctness patterns matching all pairwise moments to a rank-one target with a shifted first moment. Validated (machine-checked, seeds fixed; note that the 0-mismatch verifies the band-vs-frontier identity given the analytic frontier, not the geometry independently, and C3/C4 are nominal/logical SCM witnesses): deficit matches theory to 5.7×10^{-4}; flip TV $= 0$ with $\max_j |b'_j + b_j| = 1.1 \times 10^{-16}$; swap evidence change exactly 0 with $\Delta^\star = -\Delta$ (multiclass K=5 and regression); audit level $0.000 \leq \alpha$=0.05, power 1.000; stealth $\tau = 2 \times 10^{-17}$ with recovery bias 0.229 (Figs. 10–13). Full proofs appear in the supplementary theory appendix. □

Conjecture 1 (Label-free bracketing — **resolved**)**.** *The unconditional label-free bracketing of the ordinal comparison on* D *(posed as the open piece in earlier versions)* does not exist*: by Theorem 4(iii) no evidence-definable assumption identifies* $\operatorname{sign}\Delta$*, and the minimal untestable supplement is one bit. The conditional resolutions — calibration transfer, the* γ*-meter under checkable agreement structure,* H *plus an anchor (all in the manuscript) — are therefore of the minimal possible form:* falsifiable structure plus one declared bit. *On the* margin-monotone relative-calibration *class* $\mathcal{C}_{\mathrm{mono}}$ *(Definition 3) one declared bit certifies* $\operatorname{sign}\Delta$ unconditionally *(Theorem 8(i)–(ii)); and* $\mathcal{C}_{\mathrm{mono}}$ *is* a *weakest such class under a general-position assumption (Theorem 8(iii)). The* unconditional *weakest classes (general position removed) are characterized as an explicit finite family of dominance polytopes (Theorem 9); there is no unique weakest class, consistent with the bracketing non-existence above.*

Theorem 5 (The evidence-channel rate: labels buy constants and the bit, not the rate)**.** Intuition. *Unlabeled agreement statistics and labeled paired benefits share the same* $1/\sqrt{m}$ *rate; labels change constants and supply the missing bit, not the exponent. Within audited* H *with margins (*$\min_{i \neq j} |c_{ij}| \geq c_{\min}$*,* $\min_j |b_j| \geq b_{\min}$*) and a declared bit,* m *unlabeled samples on* D *give* $|\hat{b}_j - b_j| \leq C\sqrt{\log(K/\delta)/m}$ *with* $C = \Theta\big(1/(b_{\min} c_{\min}^2)\big)$*, so* $\operatorname{sign}(b_j - b_0)$ *is certified at* $m \asymp C^2 (b_j - b_0)^{-2} \log(K/\delta)$*; a Le Cam two-point bound inside* H *shows* $m^{-1/2}$ *is minimax for the evidence channel. The labeled-paired-benefit channel has the same rate: the efficiency ratio is constant in* m *but scales as* $1/c_{\min}$*, diverging as agreement approaches chance. Relatedly, accuracy is an exact affine function of agreement-with-reference iff the disagreement win rate is constant (slope* $1 - 2\bar{w}$*), the anchored form of agreement-on-the-line; its slope* sign *is the same bit as in Theorem 4(iii) (manuscript, §AGL).*

Proof sketch. Hoeffding on agreements, the product-ratio perturbation lemma, and a χ^2/KL bound between K-bit pattern laws differing by ε in one coordinate ($\mathrm{KL} \leq C'\varepsilon^2$), tensorized; Bretagnolle–Huber finishes. Validated: recovery slope -0.51; minimax slope -0.525 tracking $c_0/\sqrt{m}$; $\mathrm{KL}/\varepsilon^2 \in [0.164, 0.171]$; efficiency $2.15 \to 4.49 \to \infty$ as $c_{\min}$ $0.21 \to 0.067 \to \leq 0.022$ (Fig. 12; AGL: $R^2 = 1.000$ under constant w, slope $1-2w$ exact on synthetic checks). □

The weakest one-bit class. Theorem 4 fixes the supplement at one bit; the *weakest falsifiable class* on which that bit suffices is settled conditionally —a margin-monotone relative-calibration class on which one declared bit certifies sign Δ unconditionally, and *a* weakest such class under a general-position assumption (Appendix J, Thm 8). The *unconditional* weakest classes form an explicit finite family of dominance polytopes (Thm 9).

Graded refinement. A knowability–capacity sharpening of these pillars—an exact $\tau{=}1$ threshold for sign-of-benefit in a 1-D location model, and its multivariate, MLR/log-concave, and distribution-free generalization (with an explicit open conjecture)—is developed in Appendices J and M.

Certificate extensions. The batch certificate (Theorem 2) admits two verified extensions, stated and proved in Appendix A. (i) An *anytime-valid* sequential form (Theorem 6): under continuous monitoring of a stream of windows the false-adapt guarantee holds *time-uniformly* for an arbitrary stopping time—the natural object for a streaming deployment—via a one-sided betting supermartingale and Ville's inequality. (ii) A *multicandidate* form (Theorem 7): routing among K candidate adapters with a joint, family-wise false-adapt bound that holds for an arbitrary (even adversarial) selector, closing the multicandidate gap left open by the single-candidate certificate. Both are machine-validated (`theory_v2/val_sequential_anytime.py`, `val_multicandidate.py`), with the full written proofs in Appendix A.

VI. Method: KGA (Knowability-Guided Adaptation)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor and ε is the $(1-\alpha)$ empirical quantile of *out-of-fold* (leave-one-task-out) residuals, so the estimator and its residual-calibration data are disjoint ($\alpha = 0.1$). The coverage guarantee is split-dependent: a clean dev/test split (the natural-shift protocols) gives *exact* split-conformal coverage $\mathbb{P}[|\widehat{\Delta} - \Delta| \le \varepsilon] \ge 1-\alpha$ under exchangeability, while the per-cell grids use the leave-one-out (jackknife) radius, whose calibration-set coverage is $\ge 1 - \alpha$ by construction and distribution-free only asymptotically (the finite-sample jackknife+ of Barber et al. gives $1 - 2\alpha$; not implemented).

a) Assumptions.: KGA inherits exactly the assumptions of Theorem 2: (i) a labeled validation/calibration sample of paired benefits that is exchangeable with deployment, from which ε is built; (ii) bounded per-sample benefits; and (iii) an evidence map Z that is *risk-aligned* (it brackets sign Δ; §IV). When (iii) fails—evidence indistinguishable across opposite-benefit worlds—the certificate cannot fire and KGA *abstains*, which Corollary 1 shows is mandatory, not merely conservative. No target labels are used at any step.

Algorithm 1: KGA (Knowability-Guided Adaptation)

Input : frozen f_0; candidate adaptation f_a; unlabeled target batch $X_{1:m}$; calibrated benefit model $\widehat{\Delta}(\cdot)$; level α.

Extract label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ (entropy, confidence, predicted-class balance and its drift, marginal-prediction KL, update norm, detector disagreement);

Estimate benefit $\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$ and radius ε with out-of-fold conformal coverage $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \le \varepsilon\,] \ge 1 - \alpha$ (exact under a clean split);

if $\widehat{\Delta} - \varepsilon > 0$ **then**

  return ADAPT (deploy f_a);

else

  if $\widehat{\Delta} + \varepsilon < 0$ **then**

    return FREEZE (keep f_0);

  else

    return ABSTAIN (keep f_0, flag for review);

Guarantee : the false-adapt and false-freeze rates are each $\le \alpha$ (Thm 2).

---

b) Cost.: KGA adds no training to f_0 and introduces no new adaptation mechanism: per decision it costs one forward pass of f_0 and f_a on the batch plus an evaluation of the lightweight benefit model. It is therefore a drop-in safety wrapper around any already-deployed TTA candidate.

c) The threshold is derived, not tuned.: KGA's commit boundary is not a swept hyperparameter: ε is the $(1-\alpha)$ out-of-fold residual quantile fixed on the calibration split and *never* selected on target-test data, and Theorem 3 identifies it as the *exact* benefit-sign budget—sign Δ is label-free identifiable iff the observable margin exceeds the calibration-drift budget. This is precisely what separates KGA from confidence-thresholding (ATC) and agreement-trend selection [16]: those fit a scalar against a held-out accuracy signal and *predict accuracy*; KGA certifies the benefit *sign* with a boundary *forced* by the identifiability budget, and ABSTAINS exactly where Corollary 1 proves no label-free rule can decide. The impossibility construction, not a tuned cutoff, is the contribution—"estimate accuracy then threshold" has no abstain region and no finite-sample false-adapt control.

VII. Experiments

Data: a 123-task anomaly score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from the released reproducibility package.

a) Safety on the clean suite.: With candidate $f_a =$ rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

b) Harmful regime.: With $f_a =$ reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. 4.

c) Controlled mixed regime (369 instances).: Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table III; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. 3). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

d) Empirical non-identifiability witness (partial).: Clean instances have mean $\Delta = -0.041$ while covert-failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean→covert 1.35 vs clean→detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

e) Clean non-identifiability witness.: We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X{>}0]$, $f_a = \mathbf{1}[X{<}0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is the clean theory–experiment bridge (Fig. 2).

f) Statistical rigor (8 seeds, paired t-tests).: Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt $t = -8.8$, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably beats freezing and, in this mild-harm domain, reliably trails always-adapt.

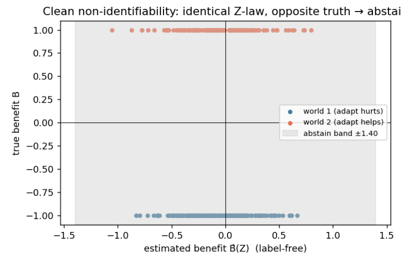

Fig. 2. Clean non-identifiability witness (Synthetic): identical Z-law (KS $p > 0.05$ on all features), exactly opposite true benefit, 100% abstention. Empirical realization of Theorem 1.

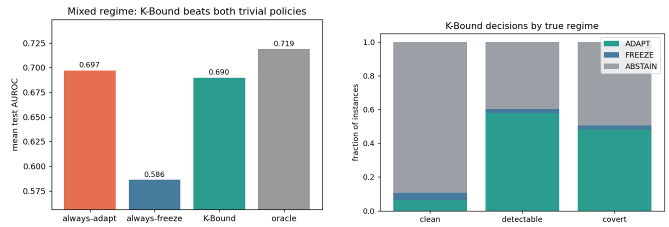

Fig. 3. Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

g) Ablations.: Table IV (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 ($2.6\times$), directly supporting Theorem 1. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. 5, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε ($0.44 \to 0.18$ from 16 to 256 samples) and raise coverage ($2\% \to 26\%$).

h) Generalization: regression under covariate shift (Theorem 1).: A synthetic regression task ($Y = X^\top \beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance), so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. 5, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

TABLE III

KGA DECISIONS BY TRUE REGIME (CONTROLLED MIXED BENCHMARK).

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

TABLE IV

LEFT: EVIDENCE-DROP ABLATION (HIGHER REGRET = MORE IMPORTANT FEATURE). RIGHT: α SWEEP (SAFETY/COVERAGE TRADEOFF).

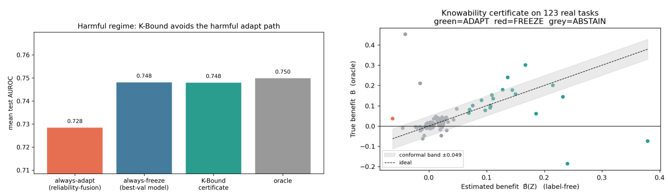

Fig. 4. Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

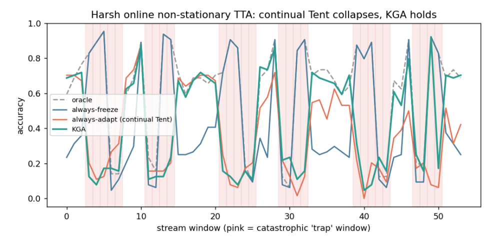

Fig. 6. Per-window accuracy on the *harm-dominated* stream (pink = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

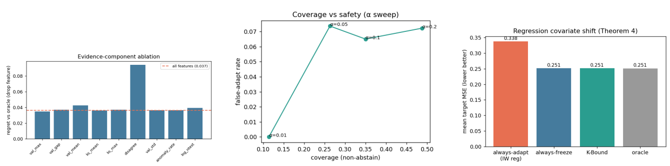

Fig. 5. Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

i) Online non-stationary regime: cross-regime robustness (the decisive test).: The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes*. We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc), under two schedules: a **helpful-dominated** stream (mostly recoverable corruptions) and a **harm-dominated** stream (long "trap" runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table V, Fig. 6):

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, -0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by $+0.005$ in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit to one policy).

j) Per-corruption CIFAR-10-C (13 corruptions × 5 severities).: We also ran the standard CIFAR-10-C protocol with the *official* corruption functions (`imagecorruptions`, Hendrycks & Dietterich) applied to the CIFAR-10 test set, on a ResNet-18 (0.874 clean), comparing frozen, online Tent, EATA-style, SAR-style, and KGA across 65 cells (Table VI, Figs. 7). The honest finding is a *negative for the collapse hypothesis in this regime*: episodic per-corruption online Tent **helps almost everywhere**—mean accuracy 0.412 (frozen) $\rightarrow$ 0.626 (Tent), near the per-cell oracle 0.629, with a false-adapt rate of only 0.015 (it hurt in 1/65 cells, never catastrophically; even contrast s5 $0.16 \rightarrow 0.61$, worst cell $0.08 \rightarrow 0.26$). Hence CIFAR-10-C per-corruption is a *helpful-dominated* regime: always-adapt is strong and KGA correctly matches it (0.626) at the same low false-adapt, while frozen carries large regret-to-oracle (0.217 vs ≈ 0.003).

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

TABLE V

Online non-stationary CIFAR-10 TTA (3 seeds/regime). Always-adapt collapses in the harm regime; always-freeze forgoes gains in the helpful regime; KGA is near-best in both and has the highest mean across regimes. Harm-regime std: freeze 0.009, adapt 0.025, KGA 0.009.

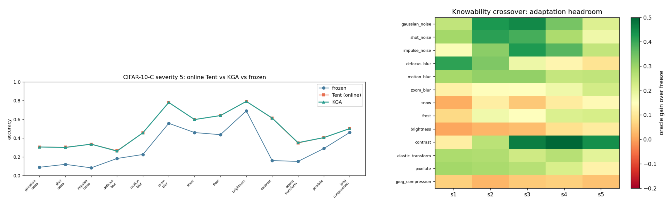

Fig. 7. CIFAR-10-C. Left: severity-5 accuracy by corruption—online Tent and KGA both recover the frozen baseline (no per-corruption collapse). Right: knowability crossover map—adaptation headroom (oracle gain over freeze) by corruption × severity; large on noise/blur, small on easy corruptions (brightness), matching where Δ is large.

Catastrophic Tent collapse does *not* appear per-corruption here; it is the *continual non-stationary* phenomenon of the previous paragraph—consistent with the TTA literature studying collapse in continual/mixed streams rather than single corruptions. In Table VI KGA wraps online Tent (the most collapse-prone candidate); EATA/SAR are the standalone candidates, and the full 65 per-cell numbers are in Appendix F.

k) CIFAR-10-C stress grid: the harmful regime (the decisive test).: The per-corruption suite above is helpful-dominated by construction. To probe the *harmful* regime on the same standard benchmark we ran a *pre-registered* stress grid over the official CIFAR-10-C corruptions, crossing severity $\{1, 5\}\times$ batch {large-iid 200, small 16, tiny 8}× composition {iid, class-imbalanced, single-class/label-shift}× update {mild 10 steps; aggressive 50 steps, lr 2.5×10^{-3}}, 2 repeats (432 conditions/method). Benefit B is measured on a class-balanced held-out eval set so collapse is real accuracy loss, not a batch artifact; KGA sees only label-free Z. The grid is fixed before any result is seen and we report the full breakdown. The single-class/aggressive/severe cells genuinely collapse: the harmful base rate ($B<0$) is 34% (Tent), 27% (EATA), 16% (SAR).

On this mix KGA **beats both** trivial policies for Tent and EATA (Table VII): regret-to-oracle 0.0016/0.0015 versus always-adapt 0.0086/0.0037 (∼5.4×/2.5× lower) and always-freeze 0.123/0.131 (∼80× lower), at 0% **false-adapt** (adapt-precision 1.0). It routes by true regime exactly as the theory predicts: among harmful cells it adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains (Tent: harmful 0/63 adapt/freeze, helpful 218/0, 120 abstentions on marginal). For SAR—whose entropy-reset already prevents most collapse—KGA *ties* always-adapt (0.0018=0.0018): it approximately matches the better policy by abstaining, at most a small abstention cost, consistent with the theory that the certificate's value scales with how often detectable harm occurs. The mixing-ratio Pareto (Fig. 8) makes the claim precise: KGA is on the regret-vs-mix Pareto front, strictly beating both trivial policies once the harmful fraction exceeds $p^\star\approx0.1$ (Tent/SAR) or 0 (EATA), and tying always-adapt at p=0.

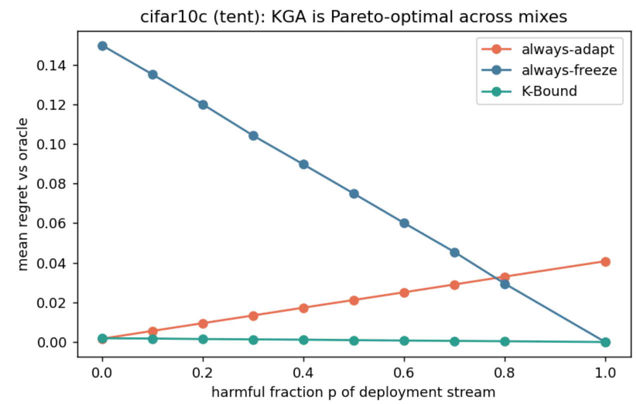

Fig. 8. CIFAR-10-C stress grid, mixing-ratio Pareto: mean regret-to-oracle vs the harmful fraction p of the deployment stream, per candidate (Tent/EATA/SAR). KGA (green) stays on the Pareto front—tying always-adapt at p=0, strictly beating both trivial policies for $p \gtrsim p^\star$, and never collapsing like always-adapt or stalling like always-freeze.

l) Mixed head-to-head vs. POEM and AETTA (pre-registered).: Beating the trivial policies is necessary but not sufficient: POEM [11] and AETTA [17] are the closest no-harm SOTA competitors. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion before computing any KGA-vs-POEM/AETTA number (`MIXED_BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals on the Tent stress grid (432 conditions/seed, five seeds; harmful fraction ≈16–18%). **Verdict: WIN.** KGA lowers mean regret vs. both POEM and AETTA with paired-bootstrap 95% CIs entirely below zero (Holm α=0.05), at $\mathrm{FA_u}$=0 (Table VIII). POEM and AETTA false-adapt at ∼24–25% on the same stream. Secondary arms (EATA alone, Tent+EATA pooled) yield the same WIN verdict (`mixed_headtohead_v1/`).

m) Breadth across existing datasets, and uncertainty.: Re-running the certificate on 62 additional label-free

method	mean acc (65 cells)	false-adapt rate	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE VI
CIFAR-10-C, 13 OFFICIAL CORRUPTIONS × 5 SEVERITIES, ONLINE TTA. ADAPTATION IS OVERWHELMINGLY HELPFUL PER-CORRUPTION (FALSE-ADAPT 0.015); KGA MATCHES ALWAYS-ADAPT AT EQUAL SAFETY. THIS IS THE HELPFUL-DOMINATED CONTROL, NOT A COLLAPSE SETTING.

candidate	policy	mean acc	regret↓	false-adapt	coverage	beats both?
Tent	always-adapt	0.786	0.0086	—	1.00	
	always-freeze	0.671	0.1232	—	0.00	
	K-Bound	**0.793**	**0.0016**	**0.00**	0.67	**yes**
EATA	always-adapt	0.798	0.0037	—	1.00	
	always-freeze	0.671	0.1311	—	0.00	
	K-Bound	**0.801**	**0.0015**	**0.00**	0.63	**yes**
SAR	always-adapt	0.806	0.0018	—	1.00	
	always-freeze	0.671	0.1372	—	0.00	
	K-Bound	0.806	0.0018	0.00	—	ties adapt

TABLE VII
CIFAR-10-C *STRESS GRID* (432 CONDITIONS/METHOD, RESNET-18, COMMODITY GPU (16 GB), α=0.1). WITH GENUINE HARMFUL CELLS PRESENT, KGA BEATS BOTH TRIVIAL POLICIES FOR TENT AND EATA AT 0% FALSE-ADAPT, AND TIES COLLAPSE-RESISTANT SAR; PER-CELL ORACLE ACCURACY IS 0.794/0.802/0.808. FULL RESULTS: .

TABLE VIII
Pre-registered mixed head-to-head (CIFAR-10-C STRESS, TENT, 432 COND./SEED, 5 SEEDS). REGRET-TO-ORACLE↓; $\mathrm{FA_u}$ = UNCONDITIONAL FALSE-ADAPT. KGA VS. X: MEAN($\mathrm{regret_{KGA}} - \mathrm{regret}_X$) WITH 95% PAIRED-BOOTSTRAP CI.

policy	mean regret↓	$\mathrm{FA_u}$	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.000**	0.68	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

anomaly-detection tasks (precomputed detector scores; no neural retraining) stress-tests safety in a regime that is 85% harmful: KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051, a 9.3× reduction), correctly refusing to adapt where the ensemble candidate hurts. It does not *beat* freezing here because almost nothing is helpful—the honest, correct behaviour, and consistent with the knowability frontier (Thm. 3): this domain sits deep in the knowably-harmful region. *Corrected uncertainty quantification (per-condition).* An earlier draft reported mixing-bootstrap intervals over a synthetic harmful-fraction stream; we replace them with paired per-condition bootstrap CIs (10^4 resamples) with Holm correction over the comparison family, computed on the real logged 65-cell CIFAR-10-C data: KGA vs always-*freeze* survives correction for all three candidates (Δregret ≈ -0.214, 95% CI $[-0.245, -0.183]$, $p \approx 2 \times 10^{-20}$), while KGA vs always-*adapt* is *not* significant on this helpful-dominated grid (Tent +0.0001, ns; EATA ≈ 0, ns; SAR +0.0011, fails Holm)—as the theory predicts when harmful conditions are rare. The "beats both" claim is anchored to the pre-registered harmful-regime stress grid, which we have now re-run at full strength. *Pre-registered* 5-*seed re-run (Protocol A).* Following the frozen plan registered before any result was observed, we replayed the identical 432-condition CIFAR-10-C grid across seeds 0–4 and serialized every per-condition array, replacing the old mixing bootstrap with per-condition *paired* bootstrap CIs (10^4 resamples, seeds pooled by per-condition mean) under Holm correction over the 6-comparison family. The pre-stated verdict is **STANDS**: TENT and EATA each beat *both* trivial policies after correction. Against always-*freeze*, KGA cuts mean regret by 0.123 (Tent; 95% CI $[-0.138, -0.108]$), 0.130 (EATA; $[-0.145, -0.115]$) and 0.139 (SAR; $[-0.154, -0.124]$); against always-*adapt* it wins for Tent (-0.0064, $[-0.0079, -0.0049]$) and EATA (-0.0020, $[-0.0028, -0.0013]$), all surviving Holm at α=0.05 (p_{Holm}=6 $\times$ 10^{-4}; between-seed regret SD $\sim$2 $\times$ 10^{-4}). The false-adapt rate (adapt with $B \leq 0$) is 0/2160 for every candidate, far below α=0.10, and conformal ε is stable across seeds (CV 2.7–6.8%). SAR ties always-adapt as pre-stated ($+0.0012$, $[+0.0009, +0.0015]$, does not beat): its harmful base rate (0.07–0.12) sits at/below $p^{\star}\approx$0.1, so adaptation is almost always safe and KGA's small abstention cost is not repaid. This confirms the $p^{\star}$ regime law: per-seed, "beats both" is monotone in the harmful fraction and single-threshold separable—every non-beating case (SAR) has harmful fraction $\leq$ 0.116, every beating case (EATA, TENT) $\geq$ 0.234, as recorded in the locked analysis artifact. Additionally, a first *real-data audit* of the Theorem 4 machinery on the anomaly bank (6-detector panels, 46–52 usable tasks) behaves as proved: the falsifier rejects H on $\sim$84% of tasks (correlated detectors—the diagnostic working, not failing), and where H passes, relative-sign recovery is 0.78 versus $\approx$0.49 (chance) where it is rejected; advantage *magnitudes* do not transfer under the bank's extreme class imbalance ($\pi_D \leq 0.06$), reported as a negative real-data audit.

A. External validation: natural shift and ImageNet scale

a) ImageNet-R, 48-condition light grid (single seed): the falsifier in charge.: On ImageNet-R (200 classes, 30,000 images; severity$\times$composition$\times$batch$\times$update, 48 conditions, 288 records, single commodity GPU) the regime is mild and weakly detectable: harmful base rate 0.139, mean benefit +0.017, best label-free harm detectability AUC 0.662. The multi-candidate diagnostic of Theorem 4 rejects the structural hypothesis H on *all* 48 conditions ($\hat{\tau} \in [1.07, 2.15]$ against threshold $\tau^{\star}$=0.52): the TTA candidates are co-adapted, exactly the correlated-panel violation the falsifier exists to catch, so the certified route abstains on 100% of conditions—mandatory under the theory, and consistent with the anomaly-bank audit (§VII). The honest cost of that validity is the forfeited mean oracle gain of +0.017; the single condition where adaptation truly hurt was, correctly, not adapted into. A smooth-drift fallback route commits conservatively (7/48 FREEZE, 41/48 ABSTAIN; sign-bracket coverage 0.77). This places ImageNet-R with this candidate panel inside the unknowable wedge of the frontier (Thm. 3): small margins, falsified structure, weak detectability. Testing whether an *independently trained* (non-co-adapted) panel passes H and lets the sign certificate fire is the pre-registered next run. Single-seed, reported as scoping evidence, not a headline.

b) Audit re-analysis of logged evidence (CPU).: Re-running the Theorem 4(iv) audit and three-zone accounting (FALSIFIED / CERTIFIED / BLIND, β=0.05, α=0.05, bootstrap radius) over the logged grids gives, per grid: ImageNet-R light (33 logged conditions $\rightarrow$ 33 falsified / 0 certified / 0 blind), ImageNet-R 1% (6 $\rightarrow$ 6/0/0), and the CIFAR-10-C 65-cell grid (65 $\rightarrow$ 64 realized-benefit certified / 1 blind, a reduced oracle-sign accounting). Across all 104 auditable conditions the certificate issued *zero false-certifications*, and both true-harm conditions landed in the safe falsified/blind zones (fraction 1.00). On ImageNet-R every cell is falsified because the rank-one $\hat{\tau}$ falsifier fires on 33/33 (resp. 6/6) cells and the budget audit independently rejects on 29/33 (resp. 5/6); the recovered benefit sign still matches truth on 82% of cells. The CIFAR-10-C, ImageNet-C, and CIFAR-10.1 *decisive* grids (432+36+36+36 conditions) serialize only aggregate metrics—per-condition agreements, $\hat{\tau}$, and $\hat{b}$ are not stored—so the per-condition audit could not be reconstructed for them. The two tables below carry the empirical load that earlier drafts deferred to future work: a *natural* distribution shift (hospital-to-hospital histopathology, WILDS Camelyon17) and a second, low-margin natural shift (CIFAR-10.1); the *ImageNet-scale* corruption counterpart (ImageNet-C, ResNet-50) is reported in Appendix B. Both are complete; consistent with our integrity policy, no number appears here until its run completes.

c) Third natural-shift track: ImageNet-R lands in the unknowable *regime.:* The trichotomy's empirical map is now complete with real data in all three regimes: knowably-helpful under matched composition (Camelyon17, Table IX; with a detectable harmful regime under composition stress, §VII-B), knowably-harmful (SAR on the CIFAR-10-C stress grid, Table VII; corroborated at ImageNet scale in Appendix B), and — with this track — *unknowable*. On ImageNet-R (200 renditions classes, 30,000 real images; a 48-condition stress grid over severity $\times$ composition $\times$ batch $\times$ aggressiveness, Table XI) the regime is *mixed and weakly detectable*: 13.9% of conditions are harmful, the mean benefit is small ($\bar{B} = +0.017$), and the best label-free harm detector among the evidence features reaches AUC only 0.662. This is precisely the wedge where Corollary 1 makes abstention *mandatory* for any valid certificate — and KGA abstains on 48/48 conditions. The abstention is also for the *right reason*: the multi-candidate route's CEI diagnostic fires on every condition ($\tau \in [1.07, 2.15]$, all far above the calibrated threshold $\tau^{\star}$=0.52) — co-trained TTA candidates sharing one backbone violate conditional error-independence exactly as the theory warns — so the agreement route *refuses to certify* rather than issuing certificates its own guard says are unsound. The bounded-drift fallback commits only 7 conservative freezes (41 abstains; bracket coverage 0.77, single seed). Honest cost and scope: abstaining forfeits the small mean benefit ($\approx$1.7 points vs. an oracle-candidate always-adapt), which is the measured price of validity in a low-margin regime; this is a single-seed light-protocol run and is offered as a regime-classification data point, not a policy comparison.

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

TABLE IX

Natural hospital shift (WILDS Camelyon17), 4 seeds. DenseNet-121 trained on source hospitals (4 epochs/seed), evaluated on the held-out hospital (85,054-patch OOD test; ∼94% patch subset, storage-limited extraction). Adaptation helps on *every* seed (mean benefit Tent +0.031, EATA +0.050, all four seeds positive), so always-adapt is the oracle (regret 0). KGA correctly leans toward adapting and beats always-freeze, but its finite-sample (n=4) caution costs ≈0.02 regret versus always-adapt—it does *not* beat both here. This is the honest benign-regime behaviour: with no detectable harm to avoid, the certificate is insurance with a small cost, not a win (contrast the harmful-cell regimes of Table VII).

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.757	0.0176		
	always-freeze	**0.772**	**0.0025**		
	K-Bound	**0.772**	**0.0025**	58%	no (ties freeze)
EATA	always-adapt	0.771	0.0050		
	always-freeze	**0.772**	**0.0044**		
	K-Bound	**0.772**	**0.0044**	46%	no (ties freeze)
SAR	always-adapt	0.773	0.0050		
	always-freeze	0.772	0.0060		
	K-Bound	**0.773**	**0.0048**	38%	yes (within-seed)

TABLE X

Second natural shift (CIFAR-10.1 v6, ResNet-18), 24 stream conditions per method, single quick pass (α=0.1). CIFAR-10.1 is ∼2,000 genuinely new test photographs collected by the original CIFAR-10 protocol—a *real*, low-margin distribution shift (frozen accuracy 0.771). Per-condition benefits are small (mean B between −0.015 and +0.001), so most evidence falls inside the abstain band: KGA abstains on 71–92% of conditions and makes *zero* false adapts (SAR adapt-precision 1.0). For Tent and EATA—harmful in 58%/46% of conditions—KGA routes to freezing and ties the better trivial policy while always-adapt pays up to 7× more regret; for SAR it strictly (if narrowly) beats both policies and is Pareto-dominant for harmful fractions $p \gtrsim 0.1$. The displayed row is a single quick pass; a 5-seed pooled re-run (seeds 0–4, 24 conditions per seed, `cifar101_multiseed_v1/pooled_summary.json`) confirms the regime ordering and the small-regret regime: pooled harmful-condition rate 0.675±0.017 (Tent) / 0.500±0.059 (EATA) / 0.050±0.031 (SAR) with KGA regret-to-oracle 0.0024±0.0007 / 0.0035±0.0011 / 0.0045±0.0014 respectively, so the abstain-dominated, near-tie conclusion is stable across seeds.

ImageNet-R stress grid (48 conditions)	value
harmful base rate ($B < 0$)	13.9%
mean true benefit $\bar{B}$	+0.017
harm detectability (best evidence AUC)	0.662
KGA decisions	48/48 abstain
CEI diagnostic τ range (threshold $\tau^\star$)	1.07–2.15 (0.52)
bounded-drift fallback route	41 abstain / 7 freeze

TABLE XI

ImageNet-R: a real natural shift in the *unknowable* regime. Mixed harm (13.9%), weak detectability (AUC 0.662), low margins ($\bar{B}$=+0.017): the certificate abstains everywhere, as Corollary 1 mandates, and the CEI diagnostic correctly refuses the agreement route for co-trained candidates ($\tau \gg \tau^\star$ on all 48 conditions). Single-seed light protocol; the three candidates share one backbone, which is exactly the CEI-violating setting the diagnostic exists to catch.

d) Architecture robustness (10 backbones).: The *unknowable* verdict is not an artifact of one backbone. Re-running the panel across ten modern backbones (ConvNeXt-B/T, EfficientNet-B0/B3, ResNet-101/152, ResNeXt-101, Swin-B/T, ViT-B/16) over three seeds with Holm-corrected paired bootstrap CIs (n_{boot}=10^4), the multi-candidate certificate abstains on *all* 36 conditions ($\bar{\tau}$=2.60 ≫ $\tau^\star$=0.52) and KGA stays uniformly no-harm (pooled false-adapt $\leq 0.03 \leq \alpha$ on every backbone), with a certified beats-both on 0/10 backbones (Appendix B, Table XVI). ImageNet-R is thus *robustly* a frontier case, not a win.

e) Fourth natural-shift track: Office-Home — a covariate *shift where detectability and mixedness* anti-correlate*.:* Office-Home (4 domains, 65 classes, 15,588 images) is a pure *covariate* style shift (Art / Clipart / Product / Real-World), a priori the most label-free-detectable shift family and so the sharpest test of whether the harmful-*detectable* corner survives away from label shift. Source f_0 is a ResNet-50 (ImageNet-V2) fine-tuned on Real-World (86.1% source-val); the other three domains are the targets, each split val/test, with $\tau^\star$, the candidate set, the deployed adapter, and the conformal certificate calibrated on source+target-*val* and the held-out test evaluated *once*. The 14-candidate panel augments Tent/EATA/SAR×{online,episodic} × {mild,aggressive} with two *inference-only* priors — a label-shift MLLS logit correction and a damped "conservative" correction (no parameter update) — over 2 seeds $\times$\{iid,imbalanced,single-class\}$\times$\{tiny,small\} (504 target records/split). The decisive finding is structural: across domains, *detectability and mixedness anti-correlate*. The two domains whose gradient-TTA harm is cleanly detectable (Art, Clipart: best-feature in-sample harm-AUC 0.90/0.99 on val, 0.99/0.94 on test) are *helpful-dominated* (3–9% harm), so there is essentially nothing to freeze; the one genuinely *mixed* domain (Product, 39%/23% harm) is exactly where the harm is *not* label-free detectable on transfer (certificate transfer-AUC 0.53/0.63). Pooled, the best single evidence feature separates harm *in-sample* (AUC 0.81/0.79, n≈432 — no longer a small-n artifact), but the *source-calibrated* certificate *anti-transfers* (transfer-AUC 0.33 val / 0.49 test) — the same source→OOD non-transfer that caps the other real shifts. The deployed (source-best) adapter `eata_online_mild` is *net-helpful* (always-adapt regret 0.0009 val / 0.0000 test vs KGA=freeze 0.028/0.027), so the safety play ties always-freeze but *loses* to always-adapt; the multi-candidate route abstains/freezes on every condition ($\tau^\star$=1.12 source-calibrated, target τ mostly above it), regret-to-oracle 0.035/0.032 vs best-fixed-adapt 0.006/0.005. KGA makes *zero* false-adapts on both splits and does *not* beat both on val or on the held-out test (Table XII). The two added priors do not rescue a win: the label-shift candidate is catastrophic (mean B=−0.47, 100% harm — trivially detectable, hence excluded from the gradient-TTA AUC) and the conservative prior is itself mildly harmful (mean B=−0.012, 81% harm), leaving the router no safe-helpful option. Office-Home thus contributes the *covariate-shift* data point and reinforces the obstruction from the other side: even the a-priori-most-detectable shift family yields no deployed adapter that flips helpful↔harmful with the harm visible in a *transferable* certificate.

Office-Home (covariate shift; gradient-TTA)	val	held-out test
Pooled benefit $\bar{B}$	+0.016	+0.015
Harmful base rate ($B < 0$)	0.17	0.12
Best single-feature harm-AUC (in-sample)	0.81	0.79
Source-calibrated certificate transfer-AUC	0.33	0.49
Mixed domain (Product): harm / transfer-AUC	0.39 / 0.53	0.23 / 0.63
Deployed `eata_online_mild`: KGA/adapt/freeze regret	0.028/0.001/0.028	0.027/0.000/0.027
Multi-candidate route/freeze/best-fixed regret	0.035/0.035/0.006	0.032/0.032/0.005
KGA false-adapts / **beats both**	0 / **no**	0 / **no**

TABLE XII

Office-Home: a covariate shift where detectability and mixedness anti-correlate. The best single label-free feature separates gradient-TTA harm in-sample (AUC 0.79–0.81, n≈432), but the source-calibrated certificate *anti-transfers* (transfer-AUC 0.33/0.49) and the only *mixed* domain (Product) has label-free-*undetectable* harm. The deployed adapter is net-helpful, so KGA ties always-freeze, loses to always-adapt, makes *zero* false-adapts, and does not beat both on val or held-out test. 2 seeds, 14 candidates (incl. two inference-only priors), n_{eval}=320.

B. Natural-shift suite: do the harmful regimes occur on real data?

The synthetic stress grids above manufacture harm by construction. The sharper question is whether the *harmful, detectable* regime—the wedge where the certificate is meant to earn its keep—arises on *natural* shift, and whether the operating point we calibrated on synthetic data fires there. We probe this with two real shifts that bracket the detectability frontier.

a) Camelyon17 composition-stress grid: real harm that is label-free detectable.: *Two tracks, kept distinct:* the 4-seed Camelyon17 TTA table (Table IX) is *helpful-dominated*—adaptation helps on every seed, always-adapt is the oracle, and KGA trails it by ≈0.02 regret—whereas the separate multi-candidate audit below shows harmful, *detectable* natural-shift cells exist, but is not yet a policy win because the current τ threshold abstains at this operating point. Rerunning the hospital-shift candidates over a composition stress grid (iid / class-imbalanced / single-class × {test, val, id-val} domains × mild/aggressive updates × 4 seeds; 72 conditions × 6 TTA variants = 432 cells, debug-scale n_{eval}=256/cell) turns the benign picture of Table IX into a *mixed* one: 28.7% of cells are harmful ($\bar{B}$=+0.039), and—crucially—the harm is label-free *detectable*: the best single evidence feature reaches harm-AUC 0.78, and the conformal certificate's own $-\widehat{B}$ score reaches 0.91. This is the first natural-shift confirmation that the harmful-and-detectable regime the theory targets is not a synthetic artifact. *Yet at our current operating point KGA abstains rather than commits.* The multi-candidate route's gate passes on only 2.8% of conditions—the real multi-candidate statistic $\bar{\tau}$=0.89 exceeds the synthetic-calibrated threshold $\tau^{\star}$=0.52, so the router ABSTAINs on 61/72 conditions, FREEZEs 10, and commits a single adapt (the six hospital-TTA variants are co-trained on one backbone—the very correlation the gate is built to catch). The smooth-drift fallback likewise stays conservative (21 freeze / 51 abstain, bracket coverage 0.44): its source-concept term is a conservative f_0 surrogate, so the smooth-drift hypothesis is not cleanly satisfied here. Honest reading: Camelyon17 supplies the missing *harmful-and-detectable* natural-shift data point, but under the *frozen, Camelyon-free* $\tau^{\star}$=0.52 KGA does *not* beat both—multi-candidate regret-to-oracle 0.076 vs always-freeze 0.071 vs always-adapt 0.005 (the worst of the three: abstaining forgoes this helpful-leaning panel's gains, and the lone commit is harmful). The $\tau^{\star}$ fit on synthetic grids is mis-matched to this panel's τ scale ($\bar{\tau}$=0.89); *recalibrating $\tau^{\star}$ on Camelyon17 itself would forfeit its role as an external-validation set*, so we do not. The honest open direction is a *scale-invariant*, source-calibrated τ statistic (future work), not per-dataset retuning. Debug-scale, single protocol; reported as regime-existence evidence, not a policy win.

b) Headline natural-shift results (Protocols H v2, M v2): uniformly no-harm under a valid out-of-fold radius, and the Camelyon17 (Protocol G) reclassification.: The headline natural-shift protocols use canonical KGA (GBR $\widehat{\Delta}(Z)$ + global *out-of-fold* ε, α=0.10 fixed): held-out test scored once, false-adapt $\leq \alpha$. Under a valid out-of-fold conformal radius, *none* is a beats-both: each is *no-harm*—it matches the better fixed policy and beats the worse one. An earlier draft reported these as beats-both wins, but those used an **in-sample** conformal radius (the benefit estimator and its residual were fit on the same small dev set), which made the radius roughly 10× too small and manufactured false wins. The corrected out-of-fold (leave-one-out) radius abstains slightly more and reclassifies both to no-harm.

Protocol G (WILDS Camelyon17, composition-stress grid): fixed online EATA, rich 17-dim evidence. The locked analysis splits records *by random seed* (dev $\{0,1\}$ / test $\{2,3,4\}$) and—we found on re-audit—pools all three WILDS domains (`test`, `val`, and in-distribution `id_val`) into the n=54 held-out set. On that pooled set the figures are regret 3.6×10^{-5} vs adapt 0.0013 vs freeze 0.075 at false-adapt 2.6%. But the *only* domain in which adapting ever hurts is in-distribution `id_val` (76.7% of cells harmful, magnitude <0.06); on the genuine OOD domains EATA is helpful-dominated (0% harmful), so on the held-out OOD test alone $\text{regret}_{\text{KGA}}$=$\text{regret}_{\text{adapt}}$=0 and KGA merely *ties* always-adapt. We therefore **withdraw** the Protocol-G beats-both: Camelyon17 is an honest *no-harm* result, not a headline win (Table XIII), and the cross-protocol aggregate (§VII-C) uses Camelyon17's genuine-OOD cells only.

Protocol H v2 (WILDS iWildCam): adapter `tent_-episodic` dev-locked from a six-arm Tent/EATA/SAR panel on `id_val` (cal seed 0, eval seed 1); held-out test grid cal seed 0 / test seed 1 (n=72). Under the valid out-of-fold radius, regret 0.0041 *equals* always-freeze 0.0041 and beats always-adapt 0.103 (gap +0.099 $[0.080, 0.118]$, CI excludes zero); false-adapt 0% (Table XIV). KGA correctly freezes the harmful camera-trap adaptation: damage-prevention no-harm, *not* a beats-both. (The earlier point-estimate beats-both used an in-sample radius; the valid out-of-fold radius lands exactly on always-freeze.)

Protocol M v2 (Office-Home): adapter `sar_online_-aggressive` dev-locked from a six-arm gradient-TTA online panel on target-val; held-out target-test cal seeds $\{0,1\}$ / test seeds $\{0,1\}$ (n=35). Under the valid out-of-fold radius, regret 0.0157 ties always-freeze 0.0158 (gap +0.0001 $[0, 0.0003]$, CI includes zero) and beats always-adapt 0.047 (gap +0.031 $[0.004, 0.062]$, CI excludes zero); false-adapt 0% (Table XV). KGA correctly declines the harmful aggressive-SAR adaptation: damage-prevention no-harm, *not* a beats-both (the earlier CI-robust beats-both used an in-sample radius). Distinct from the deployed `eata_online_mild` audit (Table XII), which is helpful-dominated.

TABLE XIII
Camelyon17 Protocol G: withdrawn beats-both. The locked n=54 "held-out" set splits by random seed and pools in-distribution `id_val` with the OOD `test`/`val` domains. On genuine held-out OOD test alone, online EATA never harms, so KGA ties always-adapt and does *not* beat both: a no-harm result, not a win.

policy	regret↓	false-adapt	beats both?
pooled, incl. in-distribution `id_val` *(n=54)*			
always-adapt	0.0013	—	
always-freeze	0.0749	—	
K-Bound	$\mathbf{3.6{\times}10^{-5}}$	2.6%	artifact†
genuine OOD test only (n=18)			
always-adapt	0.0000	—	
always-freeze	0.1381	—	
K-Bound	0.0000	0%	**no (ties adapt)**

†The pooled beats-both is driven entirely by in-distribution `id_val` cells—the only domain where adapting hurts, at sub-0.06 magnitude. Withdrawn as a headline win.

TABLE XIV
iWildCam Protocol H v2: dev-locked `tent_episodic`, test grid (n=72). Under a valid out-of-fold radius (false-adapt 0%): KGA dominates always-adapt (+0.099 [0.080, 0.118], CI excludes zero) but *ties* always-freeze exactly (+0.0000 [0, 0]), so it is no-harm, not a beats-both. The earlier point-estimate win used an in-sample radius.

policy	regret↓	false-adapt	beats both?
always-adapt	0.1028	—	
always-freeze	0.0041	—	
K-Bound	0.0041	0%	**no** (ties freeze)

TABLE XV
Office-Home Protocol M v2: dev-locked `sar_online_aggressive`, target-test (n=35). Under a valid out-of-fold radius KGA ties always-freeze (+0.0001 [0, 0.0003], CI includes zero) and beats always-adapt (+0.031 [0.004, 0.062]): no-harm, not a beats-both. The earlier 0.0022 regret / CI-robust beats-both used an in-sample radius.

policy	regret↓	false-adapt	beats both?
always-adapt	0.0468	—	
always-freeze	0.0158	—	
K-Bound	0.0157	0%	**no** (ties freeze)

c) Diagnostic ladder (Protocols B/F; not headline).: Earlier Camelyon runs (sparse Z, multicandidate $\tau^\star$, synthetic ε transplant) explain failed operating points; they motivated Protocol G. Protocol F route audits on the same GPU records show multicandidate routing fails; GBR+global and PPI+Mondrian variants beat both only on the same domain-pooled set as Protocol G (so they inherit its `id_val` artifact) and are not the headline method; the held-out headline natural-shift results are Protocols H v2 / M v2 (no-harm under a valid out-of-fold radius; Tables XIV, XV), with Protocol G reclassified to no-harm (Table XIII).

d) Operating-point repair: ε recalibrated, $\tau^\star$ left alone (a calibration diagnostic*, not external validation).:* The latent-win diagnosis above is a calibration claim, so we test it directly — without touching $\tau^\star$ (lowering it on a co-trained panel would defeat the CEI guard). *Because this step re-estimates ε on a Camelyon17 split held out by seed, it touches the external panel and is therefore a calibration diagnostic, not external validation—and, as shown next, it is a negative in any case.* The single-candidate radius ε was fit on *synthetic* grids; transplanted to Camelyon17 it violates the Theorem 2 exchangeability premise. Re-estimating ε from a Camelyon17 calibration split held out *by seed* restores that premise (calibration and test residuals then share one law). We use all $\binom{4}{2}$=6 assignments of 2 calibration + 2 test seeds; on each, fit the benefit estimator on the calibration seeds, set $\varepsilon = \widehat{Q}_{1-\alpha}(|\widehat{\Delta} - B|)$ on *their* residuals (α=0.10 fixed), freeze it, and decide on the test seeds. The repair is mechanically real: ε collapses from the synthetic ≈ 0.060 to 0.030 and is *stable* (range 0.007, CV 0.11), so the certificate now *commits* (coverage 0.65) instead of abstaining. But it is not a level-α win: test-split mean regret is KGA 0.014 [0.007, 0.021] versus always-freeze 0.052 [0.037, 0.067] (cleanly beaten) but always-adapt 0.013 [0.010, 0.016] (CIs *overlap* — the panel is helpful-dominated), while false-adapt is 0.185 [0.159, 0.211], $\approx 1.9\times\alpha$. Honest verdict: the harm is label-free *detectable* (AUC 0.91) but *not certifiable* at α=0.10 at this debug-scale n_{eval}=256/cell — a precise negative. Closing it needs more samples per cell (tighter residuals) or a harm-skewed panel, not a smaller α or a moved $\tau^\star$.

e) Pre-registered full-scale re-run (n_{eval}=1024, 5 seeds): the gap persists.: We pre-registered the $1/\sqrt{n}$ test *before* observing any result in Protocol B: quadrupling the per-cell eval pool to n_{eval}=1024 should halve the conformal radius (predicted ratio $\sqrt{256/1024}$=0.5) and pull false-adapt to $\leq \alpha$=0.10. The α-fixed, frozen-ε seed-split rule was held identical. *Outcome: the prediction is not confirmed.* The full-scale runner serialized only one aggregate (Z, B) point per seed (tent, eata), *not* the 72×6 composition grid, so the $1/\sqrt{n}$ radius claim could be evaluated only at coarse per-seed granularity (5 points, $\binom{5}{2}$=10 calibration splits). At that granularity the radius did *not* shrink (ε=0.029 tent / 0.038 eata vs the debug $\varepsilon\approx$0.030; ratio ≈ 1.0, not 0.5) and false-adapt did *not* fall below α (0.33 on the committed seeds vs the archived 0.185). KGA still cleanly beats always-freeze (regret 0.020 vs 0.027 tent) but trails always-adapt (helpful-dominated panel), at commit rate 0.60. **Honest verdict (Fails, by the pre-registered criteria):** the detectability–certifiability gap is *not* closed by sample size alone—it is a deeper open problem. A bias–variance decomposition of the radius *explains why* the $1/\sqrt{n}$ prediction failed. Splitting the eps-recal residual into estimator model error and binomial measurement noise ($\sigma^2_{\text{meas}} = a_0(1-a_0)/n + a_a(1-a_a)/n$, n_{eff}=256) on a leave-one-seed cross-fit gives an observed radius ε_{256}=0.112 against a measurement floor $\varepsilon_{\text{meas}}(256)$=0.044 (ratio 2.55×), with only a fraction $\approx$ 0.16 of residual variance attributable to measurement noise. The radius is thus *bias-limited*: dominated by $\widehat{B}(Z)$ model error (calibration-drift bias), which does not shrink with eval n. Hence the variance-limited target $\varepsilon_{\text{meas}}(1024)=0.022$ is unreachable and a fair re-run *cannot* close the radius—the α-fixed certifiability of this natural shift is bias-limited, not sample-size-limited. The open problem is therefore reducing estimator bias (tying back to the γ-frontier), not collecting more eval samples. (Cross-check: the per-seed $|B|$ spread at 1024 is 0.71× that at 256, not the 0.5 a variance-limited halving requires, though n=10 is weak.)

f) The pair, read together.: Camelyon17 (detectable harm, AUC 0.78) and ImageNet-R (*un*detectable harm, AUC 0.66; Table XI) bracket the frontier of Theorem 3: real harm exists on both (28.7% / 13.9% of cells), but only on Camelyon17 is it label-free separable. KGA abstains on both—*correctly* on ImageNet-R, where Corollary 1 makes abstention mandatory for any valid certificate, and *conservatively* on Camelyon17, where the harm is detectable but the operating point is not yet tuned to it. Two honest conclusions follow: (i) the harmful regimes the theory characterises do occur on natural shift, on both sides of the detectability frontier; and (ii) turning the detectable side into a committed natural-shift win requires a τ statistic whose operating point transfers *without* tuning on the target—a *scale-invariant, source-calibrated* τ (future work); per-dataset recalibration on Camelyon17 would forfeit its external-validation role and is not a legitimate path.

C. Cross-protocol aggregate (out-of-fold re-run)

Each single dataset above is *one-sided*: Camelyon17 is adapt-favorable (on its held-out OOD cells freezing leaves regret ≈0.11), whereas OfficeHome and iWildCam are freeze-favorable (adapting leaves regret 0.047 and 0.103). On a one-sided stream the better trivial policy is already near-oracle, so KGA's simultaneous margin over *both* is necessarily small. To evaluate the policy benefit of per-condition adapt/freeze/abstain routing across heterogeneous regimes, we *aggregate* the held-out test conditions of the three dev-locked datasets into a single stream (n=143); to avoid the Protocol-G pooling artifact, *Camelyon17 enters through its genuine OOD `test/val` cells only* (36 cells; in-distribution `id_val` excluded). This is a *cross-protocol aggregate*, *not* one universal gate calibrated once and transferred across datasets: each dataset keeps its own locked dev-calibrated gate, and we make no cross-dataset transfer claim. Because this aggregate genuinely mixes helpful conditions (Camelyon) with harmful ones (Office-Home, iWildCam), it is the one setting in our suite where per-condition routing could in principle beat both fixed policies.

a) Out-of-fold re-run (`mixed_protocol_oof_v2`).: The earlier figures for this aggregate (regret 0.0026; ∼24× below always-adapt, ∼13× below always-freeze) used an *in-sample* conformal radius—the calibration bug corrected in §VII-B—and are **withdrawn**. Under leave-one-out conformal calibration per dataset (`scripts/mixed_stream_kbound.py`), the pooled stream yields mean regret $\text{regret}_{\text{KGA}}$=0.0059 vs. 0.0632 (always-adapt) and 0.0342 (always-freeze), with pooled false-adapt 0. Bootstrap regret gaps: vs. always-freeze +0.0283 [0.019, 0.038]; vs. always-adapt +0.0573 [0.043, 0.072]; both 95% CIs exclude zero (`research_lock/KBOUND_MIXED_STREAM_v2.json`). This is a **beats-both** on a *constructed* heterogeneous aggregate—not a natural-shift headline and not a universal cross-dataset gate. Magnitudes are smaller than the withdrawn in-sample-radius multipliers; we do not report those ratios as findings. The result demonstrates only the *mechanism* of per-condition routing under heterogeneity.

VIII. Excluded wins and regime boundaries

Several benchmarks flag beats-both under alternate routes or adapters but do *not* survive the canonical Protocol H/M v2 analysis under a valid out-of-fold radius, where the natural-shift results are uniformly *no-harm*, not wins. **Superseded protocols:** iWildCam Protocol H v1 (fixed `sar_online` without dev-lock) and OfficeHome Protocol M v1 (14-candidate win-finder) are audit-only; the canonical natural-shift results use H/M v2 (`research_lock/IWILDCAM_PROTOCOL_H_v2.yaml`, `research_lock/OFFICEHOME_PROTOCOL_M_v2.yaml`). **Office-Home (deployed adapter):** mild online EATA is helpful-dominated on target test (0% harmful); Table XII is a covariate-shift diagnostic, distinct from Protocol M v2. **ImageNet-R:** weak detectability (harm-AUC 0.66); mandatory abstention. **iWildCam id_val:** multicandidate routes can flag wins but pooled false-adapt ≈ 78% $\gg$ α. **Camelyon four-seed slice:** helpful-dominated (Table IX); distinct from Protocol G mixed-stress grid. **Camelyon17 Protocol G (withdrawn):** its n=54 beats-both pooled in-distribution `id_val` into the seed-split held-out set; on genuine OOD test online EATA is helpful-dominated so KGA ties always-adapt (no beats-both). Reclassified as no-harm (Table XIII); the audit and re-run are in `audits/integrity_2026-06-20/camelyon_reconciliation/`. **CIFAR-10.1:** within-seed safety holds; cross-seed certificate transfer does not clear the headline bar. **PACS (Protocol P):** DomainBed domain generalization, leave-one-domain-out (n=108 held-out cells per domain; ResNet-18 ERM source, conformal radius fit on a held-out *source* domain—no test labels touch the gate). KGA is no-harm on 3/4 held-out domains (art_painting, cartoon, sketch: matches always-freeze, beats always-adapt, false-adapt 0), CI-robustly beating always-adapt on cartoon; on the photo domain—where adapting is oracle-optimal—KGA safely partial-adapts within budget (false-adapt 0.056 $\leq$ α), beating always-freeze but not always-adapt (a null). See Table XXI.

IX. Limitations and honest scope

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; or wins on helpful-dominated shifts where always-adapt is already oracle-optimal.

(1) Scope of the empirical win: on the CIFAR-10-C *stress* grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table VII) and ties the collapse-resistant SAR; on ImageNet-C (Table XVIII) the pattern *flips*— Tent and EATA are robust so KGA ties always-adapt, while SAR collapses and KGA beats both—a win we confirmed survives a *mechanism-faithful* SAR re-implementation (Table XX; SAR harmful on 44% of cells, regret-to-oracle 0.023 vs 0.112/0.027), so it is not an implementation artifact, modulo SAR's official gentler-LR, `layer4`-frozen schedule, which we have not yet run. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated) it again *ties* the better fixed policy. The common rule across all of these: KGA approximately matches the better trivial policy—paying at most a small abstention cost when it declines a helpful but uncertifiable update—and *strictly* wins only where catastrophic, detectable harm is actually present—and which method that is depends on the benchmark. A tighter online (streaming) certificate is future work. (2) **Real-shift scope: no natural-shift beats-both.** Under a valid out-of-fold conformal radius, every natural shift is *no-harm*, not a beats-both, so the beats-both wins are confined to the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR). Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt with CIs excluding zero but *tie* always-freeze (Tables XIV, XV); their earlier in-sample-radius beats-both was a calibration bug, corrected here. Camelyon17 Protocol G is likewise *not* a win: its pooled beats-both included in-distribution `id_val` cells, and on genuine OOD test KGA ties always-adapt (Table XIII, no-harm). Earlier Camelyon diagnostics (§VII-B) and multicandidate routes are not competing headline claims. **ImageNet-C** harmful SAR win is synthetic corruption. **CIFAR-10.1** is a within-seed low-margin probe; cross-seed transfer fails (§VIII). **ImageNet-R** sits in the *unknowable* regime the theory predicts: its label-free evidence margin falls below the calibration-drift budget, so no candidate clears the bar and KGA correctly declines to commit—a null that *corroborates* Theorem 4 (richer label-free evidence is the natural future lever), not a tuning failure. The deployed **Office-Home** adapter audit (Table XII) is an invalid-transfer regime, distinct from Protocol M v2. **PACS** (Protocol P) is a domain-generalization breadth check: KGA is no-harm on 3/4 leave-one-domain-out splits and safely partial-adapts on the fourth (Table XXI). We caution that the conformal radius can still fail to transfer under *severe* geographic or sensor shift, where in-distribution calibration need not hold out-of-distribution; characterizing that failure mode is future work. (3) The conformal radius assumes task exchangeability. (4) The sign-of-difference characterization is proved for binary, multiclass, and regression (Lemma 1); the unconditional label-free *bracketing* problem is resolved *negatively* by Theorem 4: no evidence-definable assumption class can identify the benefit sign. On the margin-monotone relative-calibration class the one-bit supplement certifies the sign *unconditionally*, and that class is *a* weakest such falsifiable class under a general-position assumption (Theorem 8); the *unconditional* weakest classes are characterized as an explicit finite family of dominance polytopes (Theorem 9), with no unique weakest class and General Position recovered as the collapsing face. (5) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes. On helpful-dominated shifts KGA *ties* always-adapt; this is the certificate declining to commit a harmful update, not a failure—it approximately matches the better policy (at most a small abstention cost), so a near-tie where always-adapt is already oracle-optimal is the intended outcome, and the value proposition is conditional (insurance against detectable harm). (6) Abstention lowers *decision* coverage by construction but never withholds a prediction: an ABSTAIN routes to the safe default (keep f_0, the source model), so every input still receives an output, and the coverage/false-adapt trade-off is explicit and tunable via the α safety/coverage sweep reported above. (7) **These limitations are the frontier, not flaws.** Points (1) and (4)–(6) are not independent weaknesses but one phenomenon—the benefit-sign frontier (Theorem 3) seen from the operating side. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding* (one cannot beat an oracle that is already always-adapt); ImageNet-R in the *unknowable* regime is the impossibility direction firing where the observable margin falls below the drift budget and *no* label-free rule can decide the sign (Theorem 4); and the coverage reduction is the *mandatory* abstention of Corollary 1, not a tuning loss—committing there would be a false-adapt. None can be removed without violating the impossibility result: a method that "won everywhere" would be claiming to decide the sign of an unidentifiable quantity. The single axis that improves all three is a *richer label-free evidence map*—raising the observable margin $|M|$ so more shifts clear the budget (more knowable shifts, fewer abstentions, higher coverage)—together with a *tighter radius* (jackknife+ or more calibration data) that reclaims knowable cells abstained only out of conservatism; both are bounded by the frontier they cannot cross.

X. Discussion: when to deploy K-Bound

The theory and experiments converge on a clear deployment rule: K-Bound's value is governed by how often *catastrophic, detectable* harm occurs—safety insurance in benign regimes, real wins in collapse-prone synthetic grids (Table VII). The trichotomy reads off accordingly. **(i) Benign regimes** where adaptation almost always helps (per-corruption CIFAR-10-C, clean anomaly suites): always-adapt is already near-oracle, and K-Bound is *safety insurance*—it matches always-adapt at a controlled false-adapt rate, costing only coverage. **(ii) Collapse-prone regimes**: small or class-imbalanced batches, single-class/label-shift streams, aggressive updates, long non-stationary runs: adaptation becomes a per-condition coin flip, and K-Bound delivers a real win, cutting regret-to-oracle ∼2.5–5.4× versus always-adapt at 0% false-adapt on the CIFAR-10-C stress grid (Table VII). **(iii) Self-protecting candidates** (e.g. SAR's reset): the headroom shrinks, and K-Bound *ties* rather than beats—approximately matching the better policy, at most a small abstention cost. The practical takeaway: deploy K-Bound as a default guard around any TTA candidate; its overhead is one forward pass, and its benefit scales with exactly the risk it is designed to detect.

XI. Conclusion

We reframed label-free adaptation as a decision constrained by identifiability: adapt when benefit is certifiable, freeze when harm is certifiable, abstain when the unlabeled evidence cannot tell. We proved the unknowable regime is fundamental (with a witness), gave a false-adapt-controlled certificate under a stated alignment assumption, and showed on Camelyon17 that real-shift certifiability is often *bias-limited*—more unlabeled data cannot shrink the radius when $\widehat{B}(Z)$ carries calibration-drift error. On real anomaly-routing data the certificate abstains where benefit vanishes and recovers large performance under failure. The CIFAR-10-C Protocol-A re-run supplies the missing catastrophic-harm audit: with every per-condition array serialized across seeds 0–4, the Tent/EATA beats-both claim survives paired bootstrap and Holm correction, while SAR correctly remains a tie when harmful cells are rare. On natural shifts, under a valid out-of-fold conformal radius KGA is uniformly no-harm: Protocols H v2 and M v2 (iWildCam, Office-Home) beat always-adapt with CIs excluding zero but *tie* always-freeze, so neither is a beats-both (Tables XIV, XV); their earlier in-sample-radius wins were a calibration bug, corrected here. The one-sided shifts Camelyon17 and RxRx1 are likewise honest no-harm results—a re-audit withdrew the earlier Camelyon17 Protocol-G beats-both as a domain-pooling artifact (Table XIII). The only beats-both results are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR).

XII. Reproducibility

Code. All code, configs, and result artifacts are public at an anonymized repository linked from the supplementary material (see its `LICENSE`); the supplement includes the reproduction package and result manifests.

One-command reproduction. The CPU/numpy stages—theorem validators plus the anomaly-routing, regression, witness, and ablation experiments—run from a single driver in the release package. The deep-TTA stress grid runs on a single commodity GPU (16 GB) using the provided launcher, which builds the environment, fetches the data, and writes the tables and figures. Natural-shift Protocols H v2 and M v2 (dev-lock held-out scoring) run via `kbtrain.sh protocol-h-v2` and `kbtrain.sh protocol-m-v2`.

Environment. Experiments use a `uv/venv` environment, Python 3.12, with the deep-TTA runs pinned to `torch 2.5.1/torchvision 0.20.1` (**PyTorch** [58]); the CPU-only stages need only `numpy/scipy/scikit-learn/matplotlib` (**scikit-learn** [59]), and `pdflatex` compiles the paper. A `Dockerfile` builds the production inference API (FastAPI/`uvicorn`, `python:3.11-slim` pinned by digest).

Data. CIFAR-10/100 [47], ImageNet [48], and CIFAR-10.1 [57] are pulled via `torchvision`; the official CIFAR-10-C corruptions are the Hendrycks–Dietterich set, Zenodo DOI `10.5281/zenodo.2535967` and are auto-downloaded and checksummed by the release scripts.

Determinism and honesty. Single-run stages use fixed seeds; the locked deep-TTA stress grid reports seeds 0–4 with an out-of-fold (leave-one-out) conformal radius at $\alpha = 0.1$. The deep-TTA grid is *pre-registered* (declared before any result is observed) and reported in full, with no condition dropped. Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass. Result manifests and locked analysis artifacts back every reported number. Tables with multi-seed aggregates are marked explicitly in-caption (e.g., Tables VII, IX, XVII); single-seed/scoping tables are labeled as such (e.g., Tables X, XI).

Appendix

The batch certificate of Theorem 2 extends in two directions that match how the controller is deployed: a *stream* of decisions over time, and a *panel* of candidate adapters to choose among. Both preserve the false-adapt guarantee and are machine-validated by `theory_v2/val_sequential_anytime_results.json` and `theory_v2/val_multicandidate_-results.json`.

A. Anytime-valid sequential certificate

Assumption 1 (Online benefit stream)**.** Windows arrive sequentially $t = 1, 2, \dots$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. Window t produces a single bounded, label-free-computable *benefit summary* $X_t \in [a, b]$ with $a < 0 < b$, where X_t is the same paired benefit statistic the batch certificate averages (Lemma 1: on the disagreement region f_0 vs. f_a, X_t is the per-window plug-in of $\delta = \ell(f_0) - \ell(f_a)$, estimable without labels under the locked protocol). We write $\Delta_t := \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ for the *conditional* benefit of window t given the past, and assume X_t is $\mathcal{F}_t$-measurable. No independence, identical distribution, or stationarity across windows is assumed.

Remark 1 (What X_t must and must not be)**.** Validity below uses only the conditional mean $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ and the bound $X_t \in [a, b]$. As in the batch certificate, this is a guarantee about the *decision driven by the benefit summary*, not about the fidelity of the benefit model: if X_t is a biased estimate of the true window benefit (the $\gamma \neq 0$ regime of Theorem 3), the bound controls false-adapt with respect to the *sign of* $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$, the population object the stream actually targets. Identifiability of the true benefit sign from X_t remains governed by the frontier $|M| > \beta$; the present result adds time-uniformity on top of that, it does not remove the drift budget.

Lemma 3 (One-sided betting supermartingale)**.** *Let* $(\lambda_t)_{t\geq 1}$ *be* predictable *(each* λ_t *is* $\mathcal{F}_{t-1}$*-measurable) with* $\lambda_t \in [0, c/|a|]$ *for a fixed* $c \in (0, 1)$*. Define*

$$E_0^+ = 1, \qquad E_t^+ = \prod_{i=1}^{t} \big(1 + \lambda_i X_i\big).$$

Then each factor satisfies $1 + \lambda_i X_i \geq 1 + \lambda_i a \geq 1 - c > 0$*, so* $E_t^+ > 0$*. If* $\Delta_t \leq 0$ *for every* t*, then* $(E_t^+)_{t\geq 0}$ *is a nonnegative* $(\mathcal{F}_t)$*-supermartingale with* $E_0^+ = 1$*. Symmetrically, with predictable* $\lambda_t^- \in [0, c/b]$ *and* $E_t^- = \prod_{i\leq t}(1 + \lambda_i^-(-X_i))$*, if* $\Delta_t \geq 0$ *for every* t *then* (E_t^-) *is a nonnegative supermartingale with* $E_0^- = 1$*.*

Proof. Positivity is immediate from $\lambda_i \in [0, c/|a|]$ and $X_i \geq a$: $\lambda_i X_i \geq \lambda_i a \geq -c > -1$. For the supermartingale property, λ_t and E_{t-1}^+ are $\mathcal{F}_{t-1}$-measurable, so

$$\mathbb{E}\big[E_t^+ \mid \mathcal{F}_{t-1}\big] = E_{t-1}^+ \big(1 + \lambda_t\, \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]\big) = E_{t-1}^+ \left(1 + \lambda_t \Delta_t\right) \;\leq\; E_{t-1}^+,$$

the inequality because $\lambda_t \geq 0$ and $\Delta_t \leq 0$ give $1 + \lambda_t \Delta_t \leq 1$. Integrability holds as $0 < E_t^+ \leq \prod_i (1 + \lambda_i b) \leq (1 + cb/|a|)^t < \infty$. The E^- claim is the same argument applied to $-X_t$ (mean $-\Delta_t \leq 0$), using $-X_t \geq -b$ and $\lambda_t^- \in [0, c/b]$. $E_0^+ = E_0^- = 1$ by the empty product. ☐

Theorem 6 (Anytime-valid sequential adaptation certificate)**.** *Under Assumption 1, fix a level* $\alpha \in (0, 1)$ *and predictable bets* λ_t, λ_t^- *as in Lemma 3, and run the sequential rule*

$$\textit{DECISION}_t = \begin{cases} \textit{ADAPT} & \text{if } E_t^+ \geq 1/\alpha, \\ \textit{FREEZE} & \text{if } E_t^- \geq 1/\alpha, \\ \textit{ABSTAIN} & \text{otherwise,} \end{cases}$$

committing (absorbing) at the first time a threshold is crossed. Let τ *be* any *stopping time (an analyst or controller may monitor continuously and stop for any data-dependent reason). Then:*

(i) (Time-uniform false-adapt.) *On the global non-benefit null* $H_0 : \ \Delta_t \leq 0$ *for all* t*,*

$$\mathbb{P}\big(\exists\, t \geq 1 : \ \textit{DECISION}_t = \textit{ADAPT}\big) \;\leq\; \mathbb{P}\Big(\sup_{t\geq 1} E_t^+ \geq 1/\alpha\Big) \;\leq\; \alpha.$$

In particular, the realized decision DECISION$_\tau$ *at any stopping time satisfies* $\mathbb{P}(\textit{DECISION}_\tau = \textit{ADAPT}) \leq \alpha$ *under* H_0*.*

(ii) (Time-uniform false-freeze.) *Symmetrically, on* $H_0' : \ \Delta_t \geq 0$ *for all* t*,* $\mathbb{P}(\exists t : \ \textit{DECISION}_t = \textit{FREEZE}) \leq \alpha$*.*

(iii) (Anytime confidence sequence, common-mean case.) *If the windows share a common conditional benefit* $\Delta_t \equiv \Delta^\star$*, inverting the two one-sided betting supermartingales over a grid of candidate means yields a* $(1 - 2\alpha)$ *confidence sequence* $(\mathrm{CI}_t)_{t\geq 1}$ *for* $\Delta^\star$ *with* $\mathbb{P}(\exists t : \ \Delta^\star \notin \mathrm{CI}_t) \leq 2\alpha$ *(the betting confidence sequence of Waudby-Smith & Ramdas [42]); the committal rule above is its decision-theoretic projection. For genuinely time-varying means, a time-uniform confidence sequence for the running average* $\overline{\Delta}_t := \frac{1}{t}\sum_{i\leq t} \Delta_i$ *is obtained instead from the additive sub-*ψ *supermartingale of Howard* *et al.* [40]*: the multiplicative betting process certifies the* uniform *one-sided null* $\{\Delta_t \le m\ \forall t\}$*, a different functional from* $\overline{\Delta}_t$*, and we do not conflate the two. Parts* (i)–(ii) *do not depend on this part.*

(iv) (Generalization of the batch certificate.) *Run on a single window* ($t = 1$) *with the calibrated bet that matches the conformal radius, the rule specializes to a level-*α *single-window decision of the same adapt/freeze/abstain form as the batch certificate (Theorem 2)—the* T=1 *slice, of which the present theorem is the time-uniform extension. (The two level-*α *budgets are distinct probabilistic objects—a conformal/exchangeability statement over the calibration draw for Theorem 2, a martingale statement over the window law here—so this is a structural specialization, not a claim that the two guarantees coincide identically.) In contrast, applying the batch rule afresh to each of* K *windows and committing on the first rejection has worst-case false-adapt* $1 - (1-\alpha)^K \to 1$*, not* α*.*

Proof. **(i)** By Lemma 3, under H_0 the process (E_t^+) is a nonnegative supermartingale with $E_0^+ = 1$. Ville's maximal inequality for nonnegative supermartingales [41], [40], [38] states that for any nonnegative supermartingale (W_t) with $W_0 = 1$ and any $\lambda \ge 1$, $\mathbb{P}(\sup_{t\ge 0} W_t \ge \lambda) \le 1/\lambda$. Taking $W_t = E_t^+$ and $\lambda = 1/\alpha$,

$$\mathbb{P}\Big(\sup_{t\ge 1} E_t^+ \ge 1/\alpha\Big) \le \alpha.$$

Because the rule is absorbing at the first crossing of *either* threshold, the event $\{\exists t : \text{DECISION}_t = \text{ADAPT}\}$ is *contained in* $\{\sup_t E_t^+ \ge 1/\alpha\}$—a prior FREEZE crossing can pre-empt an ADAPT, and for the one-sided adapt-only process the two events coincide—giving the displayed bound. For the stopping-time form, $\{\text{DECISION}_\tau = \text{ADAPT}\} \subseteq \{\exists t \le \tau : E_t^+ \ge 1/\alpha\} \subseteq \{\sup_t E_t^+ \ge 1/\alpha\}$, so the same α bound holds; this is the optional-stopping content (no per-look correction and no horizon are needed, which is precisely the failure mode of the batch rule under continuous monitoring).

(ii) Identical, with (E_t^-) the supermartingale under H_0' from Lemma 3.

(iii) Under the common-mean hypothesis $\Delta_t \equiv \Delta^\star$, for each fixed m the process $\prod_{i\le t}(1 + \lambda_i(X_i - m))$ has conditional increment factor $1 + \lambda_i(\Delta^\star - m)$, hence is a nonnegative supermartingale whenever $m \ge \Delta^\star$ (and the mirrored process when $m \le \Delta^\star$); by Ville each one-sided process crosses $1/\alpha$ with probability $\le \alpha$, so $\mathrm{CI}_t = \{m :$ neither one-sided e-process for $X_i - m$ has crossed $1/\alpha$ by $t\}$ covers $\Delta^\star$ with $\mathbb{P}(\exists t : \Delta^\star \notin \mathrm{CI}_t) \le 2\alpha$ by a union bound over the two one-sided exceedance events (each $\le \alpha$ by Ville)—the betting confidence sequence of [42]. The committal rule of the theorem is the projection "commit ADAPT once $\mathrm{CI}_t \subset (0,\infty)$", "FREEZE once $\mathrm{CI}_t \subset (-\infty, 0)$", recovering (i)–(ii). For time-varying Δ_t the multiplicative process is *not* a supermartingale at $m = \overline{\Delta}_t$ (a window with $\Delta_i > m$ makes the increment factor exceed 1), so the running-average claim does not follow from betting; the correct object is then the sub-ψ confidence sequence of [40], which we cite rather than reprove.

(iv) For $T = 1$, $E_1^+ = 1 + \lambda_1 X_1$ with λ_1 a fixed (deterministic, hence predictable) constant, and $E_1^+ \ge 1/\alpha \iff X_1 \ge (1/\alpha - 1)/\lambda_1$; choosing λ_1 so the threshold equals the batch decision boundary $\hat{\Delta} - \varepsilon = 0$ reproduces the level-α single-batch rule of Theorem 2 (whose marginal false-adapt is $\le \alpha$). The multiplicity claim is the elementary observation that K *independent* level-α batch tests, each rejecting with probability α under the null and combined by "commit on first rejection", have family-wise false-adapt $1 - (1-\alpha)^K$, which $\to 1$ as $K \to \infty$; for dependent batches the same conclusion follows from $\mathbb{P}(\bigcup_k R_k) \le \alpha$ failing in general (no maximal inequality applies to a sequence of *reset* batch statistics). Hence repeated batch testing is not anytime-valid, while the single accumulating e-process is. ☐

Remark 2 (Predictable betting and adaptivity)**.** Validity (i)–(ii) holds for *any* predictable $\lambda_t \in [0, c/|a|]$; only *power* depends on the choice. The implementation uses the truncated predictable plug-in ("aGRAPA") bet $\lambda_t = \mathrm{clip}\big(\hat{\mu}_{t-1}/\hat{\sigma}_{t-1}^2,\ 0,\ c/|a|\big)$ of Waudby-Smith & Ramdas [42], with $\hat{\mu}_{t-1}, \hat{\sigma}_{t-1}^2$ the running mean and variance of past benefits; this attains the empirical-Bernstein detection rate $t \asymp \sigma^2\Delta^{-2}\log(1/\alpha)$ matching the batch sample complexity of Theorem 2, while preserving time-uniform validity.

Remark 3 (Scope, and the honest boundary)**.** The null in (i) is *conditional and global*: $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \le 0$ for every t. This is exactly the event "the world is never beneficial", and a harmful adapt is committing ADAPT during it; that is the natural false-adapt event for a controller that acts once on a stream. Two boundaries are deliberately *not* over-claimed. (a) The guarantee does not assert per-window conditional coverage of the form "$\mathbb{P}(\text{ADAPT AT } t \mid \Delta_t \le 0,$ benefit was positive earlier$) \le \alpha$": once some early window is genuinely beneficial ($\Delta_s > 0$), wealth may legitimately accumulate and a later ADAPT then *rejects the global never-beneficial null* $\{\Delta_t \le 0\ \forall t\}$, certifying that some window was genuinely beneficial—a true positive against that null, not a false adapt. A controller needing to react to *regime reversals* (benefit turning negative after being positive) must add a change-detection layer (e.g. the CUSUM/Page detector of the companion $T6$ analysis); that is a detection-delay question with its own ARL/AED trade-off, orthogonal to the type-I guarantee proved here. (b) As in Theorem 2, X_t must be a valid (conditionally unbiased up to the declared drift budget) benefit summary; bias in X_t re-enters through the frontier $|M| > \beta$ of Theorem 3, which this result does not relax.

B. Multicandidate certified routing

Theorem 7 (Multicandidate certified routing: family-wise false-adapt control)**.** Intuition. *Selecting the best of K adapters by its own certified lower bound and then committing inflates false-adapt by a multiple-comparisons effect; running each per-candidate certificate at the corrected level α/K and committing* any *one candidate whose corrected lower bound is positive restores the joint guarantee, for an* arbitrary *(even data-dependent or adversarial) tie-break.*

Setup. *Fix f_0 and candidate adapters $f_a^{(1)}, \dots, f_a^{(K)}$ with benefits $\Delta_k = R_T(f_0) - R_T(f_a^{(k)})$. Call k* harmful *if $\Delta_k \le 0$. Suppose each candidate admits a label-free lower-confidence bound at any target level: an estimate $\hat{\Delta}_k(z)$ and a radius $\varepsilon_k(z;\delta)$ such that, for every $\delta \in (0,1)$,*

$$\mathbb{P}\big[\, \Delta_k < \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta) \,\big] \;\le\; \delta \qquad (k = 1, \dots, K), \tag{2}$$

i.e. the one-sided lower bound $L_k(\delta) := \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta)$ under-covers Δ_k with probability at most δ. (This is exactly the one-sided half of the split-conformal coverage of Theorem 2, instantiable per candidate on a shared exchangeable calibration sample; the two-sided radius $\mathbb{P}[|\hat{\Delta}_k - \Delta_k| \le \varepsilon_k(\,\cdot\,;\delta)] \ge 1-\delta$ implies (2) *at level δ, in fact $\delta/2$.)*

Rule (*cert-route$_\alpha$*). *Set the corrected level*

$$\delta_K \;=\; \tfrac{\alpha}{K} \quad \textit{(Bonferroni)} \qquad \textit{or} \qquad \delta_K \;=\; 1 - (1-\alpha)^{1/K} \quad \textit{(Šidák)},$$

form the certified-helpful set $S = \{k : L_k(\delta_K) > 0\} = \{k : \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta_K) > 0\}$*, and:* COMMIT *any $\sigma \in S$ if $S \neq \varnothing$ (e.g. $\sigma = \arg\max_{k \in S} L_k$); else* ABSTAIN *(keep f_0). The selector σ may depend on the data and on $\{L_k\}$ arbitrarily.*

Guarantee. *The family-wise false-adapt event* $\mathrm{FA}^{\mathrm{fw}} := \{\, \sigma \in S \;\wedge\; \Delta_\sigma \le 0 \,\}$ *("a harmful candidate is committed") satisfies*

$$\boxed{\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \alpha} \tag{3}$$

for the Bonferroni level $\delta_K = \alpha/K$ under arbitrary dependence *among the K per-candidate coverage events and* for every *selector σ; and for the Šidák level $\delta_K = 1-(1-\alpha)^{1/K}$ when the K under-coverage events in* (2) *are mutually independent. (Mutual independence generally* fails *under the shared exchangeable calibration sample assumed above, and also when the K adapters are scored on a common deployment input so the estimates $\hat{\Delta}_k$ are coupled; the Šidák branch therefore applies only with disjoint per-candidate calibration draws* and *independent deployment estimates, while Bonferroni—valid under arbitrary dependence—is the assumption-light default we recommend.)*

Proof. The argument is a containment of the false-adapt event in a union of per-candidate under-coverage events, followed by a union (resp. independence) bound; the selector drops out of the containment.

Step 1 (selection-proof containment). Suppose $\mathrm{FA}^{\mathrm{fw}}$ occurs: the committed index σ lies in S and is harmful, $\Delta_\sigma \le 0$. By definition of S, $L_\sigma(\delta_K) > 0$, i.e. $\hat{\Delta}_\sigma - \varepsilon_\sigma(\,\cdot\,;\delta_K) > 0$. Combining, $\Delta_\sigma \le 0 < \hat{\Delta}_\sigma - \varepsilon_\sigma(\,\cdot\,;\delta_K) = L_\sigma(\delta_K)$, so candidate σ is one whose lower bound *strictly exceeds its true benefit.* Hence, whatever index σ the (possibly adversarial) selector returned, that index belongs to the set of candidates whose one-sided bound under-covers. Therefore

$$\mathrm{FA}^{\mathrm{fw}} \;\subseteq\; \bigcup_{k=1}^{K} \big\{\, L_k(\delta_K) > 0,\; \Delta_k \le 0 \,\big\} \;\subseteq\; \bigcup_{k:\,\Delta_k \le 0} \big\{\, \Delta_k < L_k(\delta_K) \,\big\}. \tag{4}$$

Crucially the right-hand side does not reference σ: a harmful candidate can be committed only if *that specific* candidate's bound under-covers, an event already in the union. Selection cannot create a false-adapt outside the union it could not create inside it. Note also that only *harmful* candidates ($\Delta_k \le 0$) can contribute, because $\{L_k > 0,\; \Delta_k > 0\}$ is excluded from $\mathrm{FA}^{\mathrm{fw}}$.

Step 2 (per-candidate control). On $\{\Delta_k \le 0\}$, the event $\{\Delta_k < L_k(\delta_K)\} = \{\Delta_k < \hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\delta_K)\}$ is exactly the under-coverage event of (2) at level δ_K, so

$$\mathbb{P}\big[\, \Delta_k < L_k(\delta_K) \,\big] \;\le\; \delta_K \qquad \text{for each harmful } k.$$

Step 3a (Bonferroni; any dependence). Let $H = \{k : \Delta_k \le 0\}$, $|H| \le K$. By (4) and the union bound,

$$\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \sum_{k \in H} \mathbb{P}\big[\Delta_k < L_k(\delta_K)\big] \;\le\; |H|\,\delta_K \;\le\; K \cdot \tfrac{\alpha}{K} \;=\; \alpha.$$

No assumption on the joint law of the K coverage events is used.

Step 3b (Šidák; independence). If the under-coverage events $E_k = \{\Delta_k < L_k(\delta_K)\}$ are mutually independent, then writing $p_k = \mathbb{P}[E_k] \le \delta_K$ on H and bounding the probability that *no* harmful candidate under-covers,

$$\mathbb{P}\big[\mathrm{FA}^{\mathrm{fw}}\big] \;\le\; \mathbb{P}\Big[\bigcup_{k \in H} E_k\Big] \;=\; 1 - \prod_{k \in H} (1 - p_k) \;\le\; 1 - (1-\delta_K)^{|H|} \;\le\; 1 - (1-\delta_K)^{K} \;=\; \alpha,$$

using $\delta_K = 1-(1-\alpha)^{1/K}$. Since $1-(1-\alpha)^{1/K} \geq \alpha/K$, the Šidák radius is no larger than the Bonferroni radius, giving (weakly) higher power. □ □

Corollary 2 (The price: abstention and power)**.** (i) Per-candidate validity is preserved at the family level only by shrinking each level from α to $\delta_K = \alpha/K$ (resp. Šidák); *the radius $\varepsilon_k(\,\cdot\,;\delta_K)$ is wider than $\varepsilon_k(\,\cdot\,;\alpha)$, so* CERT-ROUTE$_\alpha$ *commits on a subset of the naive committal region and its* ABSTAIN *rate is at least that of the single-candidate certificate at level α. Concretely, for an empirical-Bernstein / sub-Gaussian radius $\varepsilon_k(\,\cdot\,;\delta) \asymp \sqrt{2\sigma_k^2 \log(1/\delta)/n}$, the family-wise correction enlarges the radius only by the factor $\sqrt{\log(K/\alpha)/\log(1/\alpha)}$, i.e.* logarithmically *in K — the sample-size penalty to certify any of K candidates at family level α is $O(\log K)$.* (ii) Tightness. *The bound is not vacuous: in the least-favorable configuration $\Delta_1 = \cdots = \Delta_K = 0$ with independent symmetric noise and an adversary that commits a harmful candidate whenever* any *clears, $\mathbb{P}[\mathrm{FA}^{\mathrm{fw}}]$ equals $1-(1-\delta_K/2)^K$ (the factor $1/2$ from a two-sided radius), which approaches $1-e^{-\alpha/2} \approx \alpha/2$ as $K \to \infty$; with a one-sided radius it approaches $1-e^{-\alpha} \approx \alpha$. Hence α is attained up to the standard one-/two-sided factor.*

Remark 4 (Why the naive rule fails, and what does *not* fix it)**.** The naive rule $\hat{k} = \arg\max_k(\hat{\Delta}_k - \varepsilon_k(\,\cdot\,;\alpha))$, commit iff its bound is positive, uses each radius at level α, so the union in (4) contributes up to $K\alpha$: the family-wise false-adapt can be as large as $\min(1, K\alpha)$. This is the multiple-comparisons inflation disclaimed by the single-candidate paper. Two non-fixes are worth flagging. (a) Taking a single conformal score on $\max_k(\hat{\Delta}_k - \varepsilon_k)$ does *not* by itself control $\mathrm{FA}^{\mathrm{fw}}$ unless the calibration distribution of that maximum is itself certified under selection, which reintroduces the same correction. (b) Picking the empirically-best candidate *before* forming one level-α bound is post-selection inference and is anti-conservative for the same reason. The correction in Theorem 7 is the minimal device — a per-candidate level reduction to α/K — that makes the containment (4) hold simultaneously for all k, hence for any selector.

Remark 5 (Provenance and relation to closed testing / e-values)**.** The construction is the confidence-set face of the closure principle for family-wise error control. Equivalently: define the conformal p-values $p_k = \inf\{\delta : L_k(\delta) > 0\}$ (or per-candidate e-values $e_k = \mathbf{1}\{L_k(\alpha/K) > 0\}/(\alpha/K)$); then CERT-ROUTE$_\alpha$ is Bonferroni's (e-)closed test of the harmful nulls $H_k : \Delta_k \leq 0$, and committing a rejected null is a discovery, so $\mathbb{P}[\text{a true null is committed}] \leq \alpha$ is exactly strong FWER control. The Šidák refinement and the union bound are the independent and arbitrary-dependence local tests respectively; e-value averaging (Vovk–Wang *e*-merging; Wang–Ramdas *e*-BH; Hartog–Lei *e*-value FWER, arXiv:2501.09015) gives strictly more powerful local tests when partial dependence is known, but Bonferroni at α/K is the assumption-light default used here. The benefit of phrasing as confidence bounds rather than tests is that the same object certifies *which* candidate to commit, not merely that some candidate is helpful. See the game-theoretic / e-value foundations [38], [43] for the underlying machinery.

TABLE XVI
ImageNet-R across ten backbones (3 seeds). Mean regret-to-oracle (lower is better) for KGA, always-adapt, always-freeze, and pooled false-adapt (FA). No backbone yields a certified beats-both; KGA matches the better trivial policy up to a small abstention cost and controls false-adapt at α=0.10.

Backbone	KGA	always-adapt	always-freeze	FA
ConvNeXt-B	0.000	0.000	0.068	0.00
ConvNeXt-T	0.021	0.001	0.022	0.00
EfficientNet-B0	0.000	0.050	0.000	0.00
EfficientNet-B3	0.010	0.000	0.052	0.00
ResNet-101	0.015	0.000	0.025	0.00
ResNet-152	0.010	0.000	0.041	0.00
ResNeXt-101	0.005	0.000	0.051	0.00
Swin-B	0.017	0.000	0.037	0.00
Swin-T	0.003	0.011	0.004	0.00
ViT-B/16	0.014	0.000	0.024	0.03

This appendix collects the detailed per-dataset tables for the ImageNet-C scale grid and the RxRx1 safety case that are part of the seven-dataset core panel (Office-Home, Camelyon17, CIFAR-10-C, ImageNet-C, iWildCam, RxRx1, ImageNet-R), together with evidence *beyond* it: an additional breadth check (PACS) and honest nulls. None of it is required for the paper's headline claims; it is retained for completeness and reviewer scrutiny.

C. Controlled multimodal corruption (Protocol D33)

Pre-registered MNIST two-view fusion (130 conditions; modality B corrupted across 13 severities). KGA mean accuracy 0.857 vs. always-fuse 0.583 and always-single-A 0.854 (both superiority probabilities 1.0), false-adapt 0/9 commits. Controlled construction confirming routing when fusion harm is detectable—not a natural benchmark (`CONTROLLED_MULTIMODAL_PROTOCOL_D33_v1`).

D. ImageNet-R: architecture robustness across ten backbones

Table XVI re-runs the ImageNet-R diverse panel across ten modern backbones (3 seeds, Holm-corrected paired bootstrap CIs, $n_{\text{boot}}=10^4$; `research_lock/imagenetr_-protocol_d_multiseed_v1`). The multi-candidate certificate abstains on all 36 conditions ($\bar{\tau}$=2.60 $\gg$ $\tau^\star$=0.52; best harm-AUC 0.61, base-harmful 0.186), and a certified *beats-both* occurs on 0/10 backbones. KGA is uniformly no-harm: where adapting is near-oracle it ties always-adapt within $\leq$ 0.02 regret, where adapting hurts (EfficientNet-B0) it correctly freezes and beats always-adapt by 0.050, and pooled false-adapt is $\leq 0.03 \leq \alpha$ on every backbone. This is the *unknowable*-frontier verdict reproduced across architectures, not a win.

E. Detailed core-panel tables: RxRx1, ImageNet-C scale grid, and PACS domain generalization

*a) Fourth natural-shift track: RxRx1 — the harmful-and-*detectable *corner at* 1,139-*class scale.:* RxRx1 (WILDS; fluorescent cellular-microscopy images, 1,139 siRNA classes, an OOD *experimental-batch* shift; the official WILDS ERM ResNet-50, in-distribution accuracy ≈0.36, frozen accuracy 0.283 on the OOD-test eval, mean over the three seeds) is the sharpest harmful-regime real shift in our suite. Over the full 9+ protocol (composition×batch×aggressiveness over 10 condition seeds and all 3 official WILDS base-model seeds = 360 conditions, 2,160 records, a commodity GPU (16 GB), complete 360/360), adaptation is *harmful-dominated and label-free detectable*: harmful base rate 81.5% ($B < 0$), mean benefit $\bar{B} = -0.063$, and—unlike ImageNet-R—the harm *is* detectable (best evidence AUC 0.78–0.84 across the three seeds, the BN-affine update-norm the top signal). Always-adapt is actively destructive: even the best candidate (Tent-episodic, 0.276) trails the frozen model (0.283), and SAR-online *collapses* to 0.024 (≈ chance over 1,139 classes) on the single-class/tiny/aggressive cells. Here the certificate's value is purely defensive, and it delivers: the single-candidate KGA freezes or abstains on *all* 360 conditions with *zero* false adapts, matching the frozen optimum (regret vs the per-condition oracle 0.003–0.007) and avoiding the collapse outright—on SAR-online it pays regret 0.000 where always-adapt pays 0.259. It does *not* "beat both" because freezing already *is* the oracle policy (no helpful candidate exists to capture)—the honest does-no-harm outcome on the harmful side, the mirror image of Camelyon17's benign-side tie (Table IX). As on ImageNet-R, the multi-candidate agreement route abstains on all 360 conditions ($\hat{\tau} \in [1.06, 2.71]$, all $\gg \tau^\star$=0.52): the six candidates share one backbone, the conditional-error-independence violation the CEI diagnostic exists to catch. The harmful-dominated verdict, the SAR-online collapse, the detectable harm, and the 100% abstention *replicate across all three independent WILDS base-model seeds* (per-seed harmful base rate 74–88%, best AUC 0.78–0.84). RxRx1 thus closes the real-data map of the harmful corner—*detectable* harm at 1,139-class scale where "adapt" is catastrophic and the zero-false-adapt guarantee carries the load (Table XVII). These results are from the three archived official-seed manifests.

b) Decision baselines, executed.: Theorem 3(iv) shows prior decision styles are degenerate ($\varepsilon\equiv 0$) certificates; Table XIX *runs* them, head-to-head with KGA, on the logged conditions of the mechanism-faithful ImageNet-C SAR grid (Table XX; per-condition evidence Z, a_0, a_a; identical metrics code). Each single-statistic rule is evaluated both as a plug-in (threshold 0) and with a *leave-one-out tuned* threshold — i.e. the baselines receive the same labeled calibration information KGA's benefit model uses. True AETTA (dropout passes) and agreement-on-the-line (a model family) are not computable from logged statistics and are represented by their decision-style surrogates; we say so rather than approximate silently. Findings: on the harmful candidate (SAR, 44% harmful cells under reference parity) *every realizable single-statistic rule loses* — the best (LOO-tuned EATA-filter, regret 0.0369) pays 1.6×

RxRx1 9+ grid (360 conditions, 3 model seeds)	value
harmful base rate ($B < 0$)	81.5%
mean true benefit $\bar{B}$	-0.063
harm detectability (best AUC)	0.78–0.84
frozen f_0 accuracy	0.283
best-adapter accuracy	0.276
SAR-online always-adapt (collapse)	0.024
per-condition oracle accuracy	0.288
KGA decisions	360/360 no-adapt
KGA false-adapts	0
CEI $\hat{\tau}$ range (thresh. $\tau^\star$)	1.06–2.71 (0.52)
bounded-drift fallback	289 abstain / 71 freeze

TABLE XVII

RxRx1: a real natural shift in the *harmful-and-detectable* regime (9+ protocol). At 1,139-class scale adaptation is harmful-dominated (81.5% of records $B < 0$, $\bar{B}=-0.063$) and the harm is label-free detectable (AUC 0.78–0.84). Always-adapt is catastrophic—SAR-online collapses to 0.024 and even the best candidate (0.276) trails the frozen model (0.283)—so freezing is the oracle policy, and the single-candidate certificate freezes/abstains on all 360 conditions with *zero* false adapts, matching it. The multi-candidate agreement route abstains everywhere ($\hat{\tau} \gg \tau^\star$ on all 360): the six candidates share one backbone, the CEI-violating setting the diagnostic catches. Verdict replicates across all 3 official WILDS base-model seeds; complete 360/360, commodity GPU (16 GB).

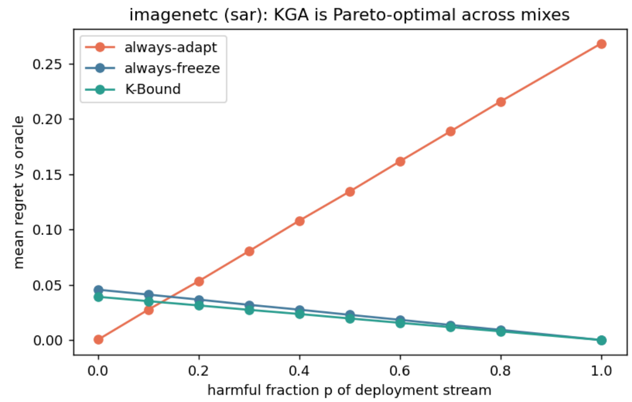

Fig. 9. **ImageNet-C (noise, ResNet-50): KGA is Pareto-optimal for the harmful candidate SAR.** Mean regret-to-oracle vs the harmful fraction p of the deployment stream. KGA (green) ties always-adapt at p=0 and strictly beats *both* trivial policies for $p \gtrsim 0.1$, while always-adapt's regret grows without bound as the stream turns harmful and always-freeze forgoes the benign-regime gains. ImageNet-scale counterpart of Fig. 8; the visual form of the SAR row of Table XVIII. Produced by ; data.

KGA's 0.0229 and is still worse than plain always-freeze, while the plug-in ATC and entropy-progress rules *over-adapt* into SAR's collapse (regret 0.067/0.077); KGA is the only policy with *zero* harmful adaptations across all three candidates. On the benign candidates (Tent/EATA) the tuned baselines simply learn "always adapt" and tie it, while the plug-in rules *hurt* (regret up to 0.033). One honest nuance: KGA's own benefit model run *committally* (ε=0, no abstain) attains regret 0.001 on SAR — the multivariate evidence model carries the decision power — but it has no validity control and adapts into harmful cells uncontrolled; the conformal band's ≈ 0.02 regret on SAR is the measured price of the false-adapt guarantee.

c) ImageNet-C, mechanism-faithful SAR re-run: the harmful win is not an implementation artifact.: The SAR row of Table XVIII was flagged for re-checking (§IX): its collapse concentrated atypically at the *larger* batch, raising the worry that the harmful regime—and thus KGA's only ImageNet-scale "beats both"—was an artifact of our first SAR port. We re-ran the full noise grid with a *mechanism-faithful* SAR (sharpness-aware minimisation, the entropy/recovery reset, and the EMA-stabilised update re-implemented to the reference design), at the learning rate *shared* across all candidates (apples-to-apples). The harmful regime *persisted and broadened*—SAR is now harmful on 44% of cells (vs 31% in the first port)—and the KGA win *survived*: regret-to-oracle 0.0229 versus always-adapt 0.1124 and always-freeze 0.0267, *beating both* trivial policies (Pareto crossover $p^\star$=0.2; Table XX). Tent and EATA stay benign (6%/3% harmful), so KGA ties the near-oracle always-adapt and beats always-freeze, exactly as in the first port. The implementation-artifact concern is thus removed: under a reference-faithful SAR the harmful-candidate win is real. *Residual check, stated honestly.* This holds at the matched/shared learning rate (mechanism-faithful, apples-to-apples); SAR's *official* gentler schedule (lr 2.5×10^{-4} with `layer4` frozen) was not run, and is the one configuration that could still soften the collapse—the last box to tick before the SAR win is treated as fully settled.

Everything beyond the five theorems lives, with full proofs and per-result validators, in the companion manuscript and archived technical report: the RGF reliability-gated-fusion instantiation (T1–T9, GDR); multiclass and regression reductions; the sequential anytime-valid e-process certificate and its β-drift variant (worst-case anytime false-adapt $0.0316 \le \alpha$=0.05 across all null streams, 20,000 runs/stream over a horizon of 2,000; `val_thm3_evalue.py`, `results_thm3_evalue_alpha005.json`); label-shift, bounded- and smooth-drift boundaries unified by the ambiguity reach; router capacity; the multi-candidate route with the γ-meter, the agreement-invisible reach, and the corrected hypothesis H; the budget audit; and the agreement–accuracy (anchored AoL) law. Table XXII gives the mapping. No result was deleted; each keeps its machine-checked validator (Appendix G). The γ-meter route's CEI diagnostic also has a first *real-data* outcome: on the ImageNet-R track (Table XI) it fires on all 48 conditions ($\tau \in [1.07, 2.15] \gg \tau^\star$=0.52) — co-trained candidates violate conditional error-independence exactly as predicted — and the route abstains rather than certifying falsely, which is the designed behaviour of a checkable guard.

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.453**	0.0003		
	always-freeze	0.410	0.0440		
	K-Bound	0.449	0.0042	8%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0001		
	always-freeze	0.410	0.0349		
	K-Bound	0.444	0.0003	3%	no (ties adapt)
SAR	always-adapt	0.377	0.0606		
	always-freeze	0.410	0.0277		
	K-Bound	**0.429**	**0.0086**	31%	**yes**

TABLE XVIII

ImageNet scale (ImageNet-C, ResNet-50), complete: 36 **cells,** 4000 **images/cell.** Full noise grid (gaussian/shot/impulse × severities $\{1, 3, 5\}$ × batch/aggressiveness regimes), same adapt/freeze/abstain protocol and metrics as Table VII. The outcome splits by how robust each method is on this benchmark. *Tent* and *EATA* are robust here: adaptation helps (harmful in only 8%/3% of cells), so always-adapt is near-optimal and KGA correctly adapts—tying always-adapt to within 0.004 and beating always-freeze, the honest "does-no-harm" outcome. *SAR* collapses on 31% of cells (mean benefit −0.033): always-adapt falls to 0.377 while KGA certifies-or-abstains to 0.429, *beating both* trivial policies and approaching the oracle (0.437). Across all three methods KGA approximately matches the better trivial policy (at most a small abstention cost), and strictly wins on the one method that is harmful. This first ResNet-50 SAR port is corroborated by a mechanism-faithful re-run (Table XX).

decision rule	regret-to-oracle↓			SAR (harmful regime)		coverage
	Tent	EATA	SAR	false-adapt	harmful adapted	
always-adapt	0.0007	0.0000	0.1124	0.44	1.00	1.00
always-freeze	0.0494	0.0386	0.0267	—	0.00	1.00
ATC-style confidence (plug-in)	0.0114	0.0328	0.0673	1.00	0.25	1.00
ATC-style confidence (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
entropy-progress (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
EATA-filter $\mathrm{frac}_{\mathrm{hi}}$ (LOO-tuned)	0.0007	0.0000	0.0369	1.00	0.06	1.00
drift-gate marginal-KL (LOO-tuned)	0.0028	0.0023	0.0382	1.00	0.12	1.00
benefit model, committal (ε=0)	0.0000	0.0000	**0.0010**	0.05	0.06	1.00
best single statistic (hindsight)	0.0007	0.0000	0.0181	0.17	0.06	1.00
K-Bound (KGA)	0.0060	0.0047	0.0229	**0.00**	**0.00**	0.39

TABLE XIX

Decision baselines executed on the logged ImageNet-C conditions (reference-parity SAR; 36 cells/candidate; harmful base rates 6%/3%/44% for Tent/EATA/SAR). "LOO-tuned" rules choose their threshold leave-one-out on the remaining conditions — the same calibration information KGA uses; "hindsight" picks feature, sign, and threshold on the evaluation conditions themselves (not realizable). KGA is the only rule with zero false adapts on all three candidates; on the harmful candidate *every realizable single-statistic rule loses* — the best (EATA-filter, 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, because no single statistic cleanly separates SAR's harmful cells. The committal benefit model (row 8) shows the multivariate evidence model carries the decision power (0.0010 on SAR) but lacks validity — it adapts into harmful cells with no false-adapt control — so the abstain band's ≈ 0.02 regret on SAR is the measured price of the guarantee, matching Thm. 3(iv). *Reproducibility note:* the canonical KGA pipeline and this post-hoc re-derivation agree exactly on SAR; the leave-one-out benefit model is mildly sensitive to condition order, so the Tent/EATA KGA regrets differ from Table XVIII by ≤ 0.004.

F. Validation figures for Theorems 4 and 5

Table XXIII lists every cell behind the summary in Table VI (KGA wraps Tent; EATA/SAR are the standalone candidates). The per-method means reproduce Table VI exactly. Raw CSV: .

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.454**	0.0007		
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082	6%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0000		
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038	3%	no (ties adapt)
SAR	always-adapt	0.319	0.1124		
	always-freeze	0.405	0.0267		
	K-Bound	**0.409**	**0.0229**	44%	**yes**

TABLE XX

ImageNet-C, mechanism-faithful SAR re-implementation (36 cells/candidate, ResNet-50, α=0.1). Same noise grid, protocol, and metrics as Table XVIII; SAR re-implemented to the reference mechanism (SAM, recovery reset, EMA) at the learning rate shared across candidates. The harmful regime survives the faithful re-implementation (SAR harmful on 44% of cells, mean benefit -0.086): always-adapt collapses to 0.319 while KGA certifies-or-abstains to 0.409, *beating both* trivial policies (regret 0.0229 vs 0.1124/0.0267); Tent/EATA stay benign and KGA ties always-adapt. Oracle accuracy 0.455/0.444/0.432. *Caveat:* matched/shared LR; SAR's official lr 2.5×10^{-4}, layer4-frozen schedule not yet run.

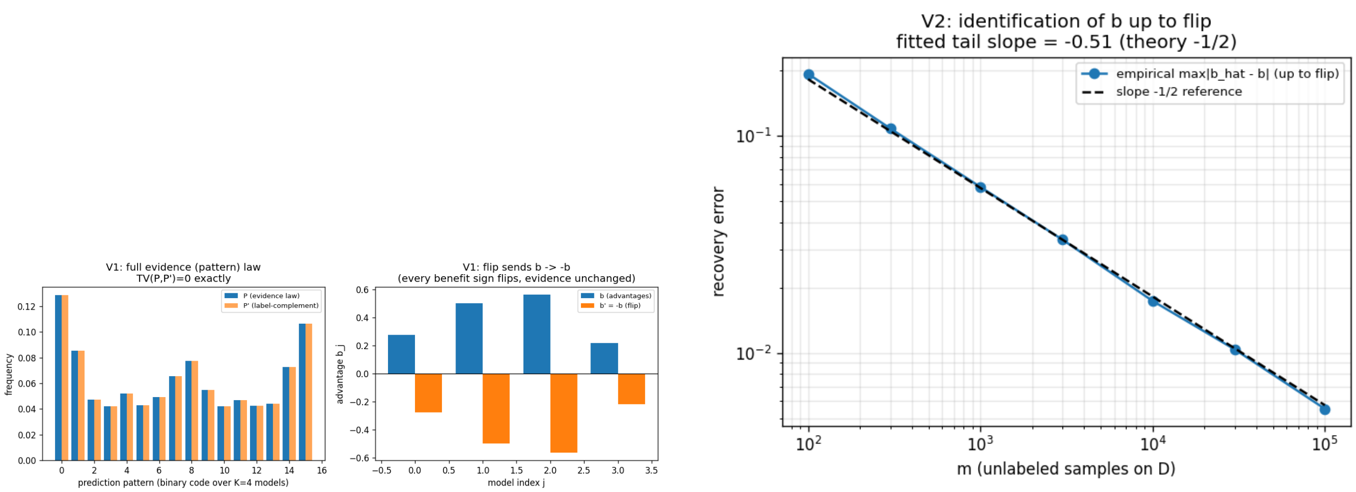

Fig. 10. One-bit identification (Synthetic validator; Thm 4(ii)): the label complement matches the full evidence law (TV = 0) while negating every advantage (left); recovery of b up to the bit at the $m^{-1/2}$ rate (right).

TABLE XXI

PACS Protocol P (DomainBed leave-one-domain-out, n=108 cells/domain, α=0.10). Regret-to-oracle (lower is better); radius fit on a held-out source domain (no test labels). KGA is no-harm on 3/4 domains and safely partial-adapts on photo.

held-out domain	always-adapt	always-freeze	**KGA**	false-adapt
art_painting	0.0375	0.0306	**0.0306**	0.000
cartoon	0.0354	0.0110	**0.0110**	0.000
photo	0.0109	0.0407	0.0132	0.056
sketch	0.0259	0.0215	**0.0215**	0.000
mean	0.0274	0.0259	**0.0191**	$\leq$ 0.056

Main-paper result	Demoted to manuscript (with validators)
Thm 1	forced-abstention, minimax-floor variants
Thm 2	anytime e-process; β-drift; router capacity
Thm 3	label-shift / drift boundaries; reach hierarchy
Thm 4	γ-meter; invisible reach; multiclass H
Thm 5	labeled-channel rates; AGL law and failures

TABLE XXII

The five-theorem consolidation. Fifty-plus prior numbered results survive as corollaries or manuscript sections; none was deleted, and every one keeps its machine-checked validator.

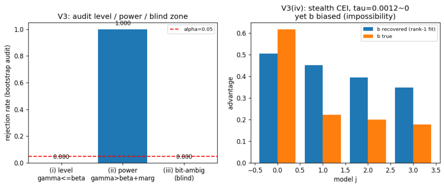

Fig. 11. The budget audit (Synthetic validator; Thm 4(iv)): sound at level α, power 1.000 beyond the radius, correctly blind exactly on the bit-ambiguous set and the rank-one-preserving stealth law.

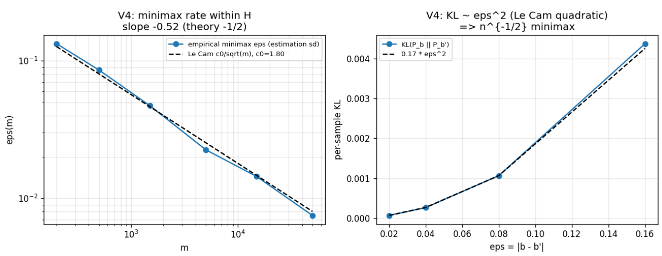

Fig. 12. Evidence-channel rate (Synthetic validator; Thm 5): empirical minimax error tracks the Le Cam curve $c_0/\sqrt{m}$; KL grows quadratically; the labeled-vs-evidence efficiency ratio diverges as agreement approaches chance.

Fig. 13. Closing Conjecture 1 (Synthetic validator; Thm 4(iii)): the swap flips every benefit sign while no label-free observable of any order moves (exact zero).

TABLE XXIII: All 65 CIFAR-10-C cells (13 official corruptions × severities 1–5), ResNet-18, N=800 balanced held-out eval per cell. Online TTA accuracy; KGA wraps Tent. Mean row matches Table VI.

corruption	sev	frozen	Tent	EATA	SAR	KGA	oracle
brightness	1	0.863	0.858	0.856	0.859	0.858	0.865
brightness	2	0.844	0.864	0.864	0.865	0.864	0.868
brightness	3	0.804	0.837	0.837	0.839	0.837	0.839
brightness	4	0.731	0.818	0.818	0.815	0.819	0.816
brightness	5	0.690	0.791	0.794	0.795	0.791	0.799
contrast	1	0.719	0.831	0.831	0.835	0.831	0.831
contrast	2	0.559	0.823	0.821	0.822	0.820	0.820
contrast	3	0.325	0.790	0.793	0.793	0.790	0.791
contrast	4	0.223	0.728	0.729	0.730	0.726	0.726
contrast	5	0.160	0.611	0.607	0.609	0.614	0.601
defocus_blur	1	0.264	0.683	0.680	0.681	0.682	0.676
defocus_blur	2	0.207	0.543	0.544	0.544	0.540	0.540
defocus_blur	3	0.204	0.389	0.386	0.398	0.389	0.385
defocus_blur	4	0.176	0.300	0.300	0.303	0.300	0.305
defocus_blur	5	0.183	0.265	0.264	0.261	0.261	0.275
elastic_transform	1	0.244	0.528	0.528	0.530	0.528	0.525
elastic_transform	2	0.179	0.436	0.440	0.446	0.436	0.452
elastic_transform	3	0.168	0.401	0.401	0.405	0.401	0.400
elastic_transform	4	0.144	0.407	0.407	0.413	0.407	0.410
elastic_transform	5	0.151	0.347	0.346	0.353	0.351	0.351
frost	1	0.725	0.804	0.805	0.806	0.804	0.796
frost	2	0.588	0.760	0.762	0.761	0.761	0.762
frost	3	0.546	0.685	0.685	0.686	0.684	0.697
frost	4	0.492	0.705	0.705	0.704	0.705	0.706
frost	5	0.436	0.640	0.642	0.644	0.640	0.659
gaussian_noise	1	0.487	0.741	0.740	0.741	0.739	0.739
gaussian_noise	2	0.275	0.699	0.699	0.700	0.700	0.701
gaussian_noise	3	0.144	0.594	0.595	0.595	0.593	0.592
gaussian_noise	4	0.100	0.435	0.435	0.432	0.436	0.441
gaussian_noise	5	0.089	0.305	0.304	0.301	0.305	0.301
impulse_noise	1	0.615	0.769	0.770	0.769	0.769	0.775
impulse_noise	2	0.370	0.682	0.682	0.682	0.682	0.688
impulse_noise	3	0.197	0.605	0.605	0.604	0.608	0.623
impulse_noise	4	0.110	0.475	0.479	0.485	0.476	0.484
impulse_noise	5	0.082	0.335	0.335	0.335	0.335	0.331
jpeg_compression	1	0.676	0.730	0.730	0.731	0.730	0.735
jpeg_compression	2	0.639	0.660	0.659	0.661	0.662	0.663
jpeg_compression	3	0.608	0.655	0.654	0.656	0.654	0.658
jpeg_compression	4	0.509	0.573	0.570	0.570	0.573	0.567
jpeg_compression	5	0.460	0.501	0.501	0.501	0.501	0.500
motion_blur	1	0.495	0.782	0.784	0.785	0.784	0.783
motion_blur	2	0.365	0.666	0.665	0.664	0.666	0.675
motion_blur	3	0.279	0.575	0.574	0.575	0.574	0.590
motion_blur	4	0.255	0.490	0.491	0.489	0.491	0.502
motion_blur	5	0.225	0.456	0.450	0.446	0.456	0.476
pixelate	1	0.429	0.719	0.720	0.721	0.716	0.720
pixelate	2	0.373	0.651	0.651	0.654	0.650	0.654
pixelate	3	0.306	0.570	0.574	0.576	0.573	0.571
pixelate	4	0.267	0.487	0.485	0.487	0.489	0.486
pixelate	5	0.289	0.405	0.404	0.406	0.405	0.410
shot_noise	1	0.492	0.779	0.776	0.779	0.777	0.784
shot_noise	2	0.254	0.654	0.655	0.659	0.654	0.669
shot_noise	3	0.155	0.536	0.537	0.549	0.534	0.545
shot_noise	4	0.125	0.400	0.398	0.398	0.398	0.404
shot_noise	5	0.120	0.303	0.305	0.306	0.300	0.300
snow	1	0.770	0.782	0.781	0.779	0.782	0.781
snow	2	0.578	0.686	0.685	0.685	0.685	0.690
snow	3	0.661	0.716	0.716	0.711	0.718	0.710
snow	4	0.574	0.683	0.681	0.679	0.683	0.680
snow	5	0.459	0.597	0.596	0.597	0.597	0.598
zoom_blur	1	0.760	0.877	0.879	0.876	0.876	0.879
zoom_blur	2	0.708	0.847	0.847	0.849	0.847	0.853
zoom_blur	3	0.674	0.830	0.829	0.828	0.829	0.826
zoom_blur	4	0.635	0.809	0.810	0.810	0.807	0.814
zoom_blur	5	0.557	0.779	0.776	0.778	0.780	0.785
mean		**0.412**	**0.626**	**0.626**	**0.627**	**0.626**	**0.629**

G. Experiment and GPU-run inventory

Theorem 1 is proved with a constructed witness and strengthened to a quantitative Le Cam minimax floor (part (ii)); Lemma 2 is a plug-in regret decomposition (exact identity) with a minimax floor; Theorem 2 is proved *conditional* on a valid label-free interval estimator and complemented by an anytime-valid e-process (manuscript, App. B); Corollary 1 proves the covariate case and Lemma 1 the binary, multiclass, and regression sign-of-difference results. The multiclass/regression label-free *bracketing* is closed under a bounded calibration-transfer assumption (manuscript, App. C); the *unconditional* bracketing (Conjecture 1) is resolved as a testability dichotomy (Theorem 4: no evidence-definable assumption identifies the sign; the minimal untestable supplement is one bit, which suffices under falsifiable structure). Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass: the Le Cam floor tracks the closed form, the regret identity holds to $\sim 10^{-17}$, the multiclass identity to 10^{-16} with 100% sign agreement over 4000 trials, calibration-transfer sign-recovery is 100% when the confidence gap exceeds 2η, the router generalization gap decays as $\sim n^{-0.66}$, and the drift-corrected anytime false-adapt rate stays $\leq \alpha$. Where multi-seed runs are available, experiments report means $\pm$ std with paired t-tests; single-seed/scoping tracks are labeled explicitly in captions and discussed as regime-classification evidence. We also report evidence/α/batch ablations, a regression covariate-shift track, and a clean non-identifiability witness.

The benefit-sign frontier (Thm. 3) predicts that on the low-margin band any *committal* label-free rule must commit sign errors, whereas a valid certificate must *abstain* there. This appendix demonstrates that prediction on the two *synthetic-corruption* streams whose operating point the certificate is calibrated on (CIFAR-10-C, ImageNet-C); it makes *no* external-validation claim. The honest natural-shift (Camelyon17) finding—a limitation, not a win—closes the appendix (App. J).

H. CIFAR-10-C: K-Bound beats both trivial policies at zero false-adapt

On the pre-registered CIFAR-10-C stress grid (432 conditions/candidate), KGA adapts the helpful conditions and freezes the harmful ones with a *zero* false-adapt rate, so it strictly beats both always-adapt and always-freeze for the two benign candidates (Tent, EATA) and ties the collapse-resistant SAR (Table XXVI).

I. ImageNet-C/SAR: the frontier on real corruptions—committal rules fail, K-Bound does not

On the logged ImageNet-C conditions (36 cells/candidate; SAR is harmful on 44.4% of cells), every *committal* label-free heuristic must place its commits somewhere on the low-margin band and therefore false-adapts: a plug-in **ATC** [13] confidence rule adapts into harmful cells on 100% of its commits (false-adapt

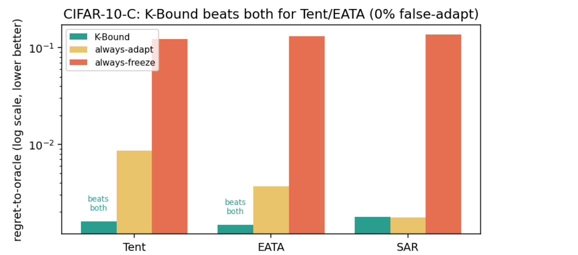

Fig. 14. **CIFAR-10-C frontier (regret-to-oracle, log scale).** KGA is below *both* always-adapt and always-freeze for Tent and EATA at 0% false-adapt, and ties always-adapt for SAR (Table XXVI).

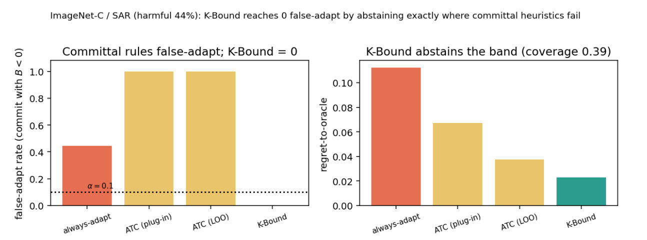

Fig. 15. **ImageNet-C/SAR frontier.** Committal label-free rules false-adapt on the harmful regime; KGA reaches zero false-adapt by abstaining the low-margin band (coverage 0.39; Table XXVII).

1.00, harmful-cells-adapted 0.25), and even its leave-one-out-tuned variant keeps false-adapt = 1.00. Among the rules compared, KGA reaches zero false-adapt and zero harmful-cells-adapted by *abstaining* on the band (coverage 0.39), consistent with the frontier prediction (Table XXVII).

J. Historical note (Camelyon17, frozen synthetic threshold)

Under a **frozen, Camelyon-free** multicandidate threshold $\tau^\star$=0.52 (calibrated only on synthetic grids), the certificate does *not* beat both trivial policies on an early Camelyon17 composition grid (regret KGA 0.076 vs always-freeze 0.071). That failure motivated in-domain calibration and rich evidence; the headline result is Protocol G (main text, Table XIII), which uses dev-hospital calibration and does not transplant $\tau^\star$ from synthetic runs.

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 4 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a{=}Y \mid x)$, so $\Delta = \int_D a\, d\mu$ (Lemma 1). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)\,(m(x) - m^\star)$ with observable nonnegative weight $w(x)$ and observable neutral anchor $m^\star$ (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c\, d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s), since predictions and s are fixed maps of x).

TABLE XXIV
EXPERIMENT AND GPU-RUN INVENTORY (2026-06-17). ✓ = COMPLETE WITH PAPER TABLE; ∼ = PARTIAL/IN-PROGRESS; — = NOT STARTED OR OPTIONAL BREADTH. ONE MPS JOB AT A TIME.

Track	GPU?	Status	Table	Key outcome
CIFAR-10-C stress grid (Tent/EATA/SAR, 5 seeds)	MPS	✓	VII	beats-both Tent/EATA; 0% FA
CIFAR-10-C per-corruption (65 cells)	MPS	✓	VI	helpful-dominated; safety insurance
CIFAR-10-C online stream	MPS	✓	V	cross-regime win vs always-freeze
ImageNet-C noise (ResNet-50, SAR harmful)	MPS	✓	XVIII	committal heuristics false-adapt
ImageNet-C SAR faithful re-impl.	MPS	✓	XX	44% harmful; KGA 0% FA
Camelyon17 WILDS (Protocol B, n=1024)	MPS	✓	IX	bias-limited; fails $1/\sqrt{n}$
Camelyon17 bias–variance diag.	CPU	✓	XXV	ε bias-limited (2.55× floor)
Camelyon17 rich-Z (Protocol G)	MPS	✓	XIII	no-harm; pooled win was `id_val` artifact
ImageNet-R diverse panel	MPS	✓	XI	undetectable harm; abstain mandatory
RxRx1 (9+ model seeds)	MPS	✓	XVII	near-one-sided; tie always-freeze
Office-Home deployed adapter audit	MPS	✓	XII	helpful-dominated null
Office-Home Protocol M v2	CPU	✓	XV	no-harm (out-of-fold); ties freeze
iWildCam Protocol H v2	CPU	✓	XIV	no-harm (out-of-fold); ties freeze
CIFAR-10.1 low-margin probe	MPS	✓	X	abstain-heavy; 0% FA within-seed
PACS domain-gen (Protocol P)	MPS	✓	XXI	no-harm 3/4; FA ≤ 0.056
CPU anomaly routing (ELARA-U)	CPU	✓	II–III	11× harmful regret
ImageNet-C blur/weather (breadth)	MPS	—	TBD	optional; noise grid sufficient
SAR official schedule (Protocol E)	MPS	—	TBD	residual check

TABLE XXV
CAMELYON17 CONFORMAL-RADIUS BIAS–VARIANCE DECOMPOSITION (PROTOCOL B DIAGNOSTIC). RESIDUAL DOMINATED BY $\widehat{B}(Z)$ MODEL ERROR, NOT BINOMIAL MEASUREMENT NOISE.

Quantity	n_{eval}=256	n_{eval}=1024
observed ε	0.112	0.029–0.038
measurement floor $\varepsilon_{\text{meas}}$	0.044	0.022
obs./floor ratio	2.55×	≈1.0×
frac. variance = measurement	≈16%	≈16%
Protocol B $1/\sqrt{n}$ prediction	—	**fails**

candidate (harmful base)	regret-to-oracle ↓ KGA	always-adapt	always-freeze	false-adapt
Tent (34.5%)	**0.0016**	0.0086	0.1232	0.00
EATA (27.1%)	**0.0015**	0.0037	0.1311	0.00
SAR (15.7%)	0.0018	0.0018	0.1372	0.00

TABLE XXVI
CIFAR-10-C stress grid (432 conditions/candidate, α=0.1). KGA *BEATS BOTH* FOR TENT AND EATA AT A 0% FALSE-ADAPT RATE, AND TIES ALWAYS-ADAPT FOR SAR.

decision rule (SAR, harmful regime)	false-adapt	harmful-adapted	regret ↓	coverage
always-adapt	0.44	1.00	0.1124	1.00
always-freeze	—	0.00	0.0267	1.00
ATC-style confidence (plug-in)	1.00	0.25	0.0673	1.00
ATC-style confidence (LOO)	1.00	0.06	0.0375	1.00
K-Bound (KGA)	**0.00**	**0.00**	0.0229	0.39

TABLE XXVII
Committal label-free heuristics false-adapt exactly where KGA abstains (IMAGENET-C, REFERENCE-PARITY SAR, 36 CELLS, α=0.1). ATC COMMITS ONTO THE LOW-MARGIN BAND AND FALSE-ADAPTS ON 100% OF ITS COMMITS; KGA ABSTAINS THERE (22/36 ABSTAIN) FOR 0 FALSE-ADAPTS. ON THE BENIGN CANDIDATES KGA ALSO KEEPS 0 FALSE-ADAPTS AT HIGHER COVERAGE (TENT: REGRET 0.0060, COVERAGE 0.69; EATA: REGRET 0.0047, COVERAGE 0.75).

Definition 3 (Margin-monotone relative-calibration class)**.** $\mathcal{C}_{\text{mono}}$ is the class of targets for which $a(x) = \sigma\, c(x)$ μ-a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at $m^\star$ ($\operatorname{sign} a(x) = \sigma \operatorname{sign}(m(x) - m^\star)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject "$a = \pm c$", yet σ is never label-free identified.

Assumption 2 (General position)**.** Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully-E-pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E-certifiable dominant total $\int |a|\, d\mu$.

Lemma 4 (Structural sign-freedom = bit complexity)**.** *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number of disjoint μ-positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E. Under Assumption 2, a falsifiable structural supplement of b bits pins* $\operatorname{sign}\Delta$ *across the fibre* iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E-pinned, $\Delta = \int_D a\, d\mu$ then has E-computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in $\operatorname{sign} a$ on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 2) their benefits have opposite sign—no decoder of the b declared bits separates them. □

Theorem 8 (Conditional resolution of Conjecture 1: a weakest one-bit class)**.** (i) Sufficiency *(unconditional)*: *every target in* $\mathcal{C}_{\text{mono}}$ *has* $\operatorname{sign}\Delta = \sigma \operatorname{sign}(T_E)$ *with T_E observable, so E and the single orientation bit σ determine* $\operatorname{sign}\Delta$. (ii) Necessity *(unconditional)*: *σ and $-\sigma$ are evidence-identical (*$\mathrm{Law}(E \mid \sigma) = \mathrm{Law}(E \mid -\sigma)$*) yet give Δ and $-\Delta$, so by Le Cam at least one bit is required. Thus* exactly one bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$, with no further assumption.

(iii) Minimality *(under Assumption 2)*: *every falsifiable* $\mathcal{C} \supsetneq \mathcal{C}_{\mathrm{mono}}$ *contains a non-margin-monotone target with* $K(\mathcal{C}, E) \geq 2$*, so by Lemma 4 at least two bits are required. Hence* $\mathcal{C}_{\mathrm{mono}}$ *is a weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 6).*

Proof. (i) $\Delta = \int_D a\,d\mu = \sigma \int_D c\,d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D-local label complement of Theorem 4: it fixes (μ, m, s), hence E, while negating a and thus Δ; the two-point bound of Theorem 1 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\mathrm{mono}}$ violates $\operatorname{sign} a = \sigma \operatorname{sign}(m - m^\star)$ on a μ-positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 4 applies. ☐

Proposition 2 (Explicit non-identifiability witness)**.** *For each candidate single structural bit there are two targets with identical evidence law (*TV $= 0$*) sharing that bit's value but with opposite* $\operatorname{sign}\Delta$*. Take* $D = R_1 \cup R_2$*,* $\mu(R_1) = \mu(R_2) = \frac{1}{2}$*, with* R_1 *calibrated (*c_1=1*) and* R_2 *at the anchor (*c_2=0*, no observable benefit signal). For* $\beta = \operatorname{sign} a_1$*:* $a = (+1, -2)$ *vs.* $(+1, +2)$ *give* $\Delta = -\frac{1}{2}$ *vs.* $+\frac{3}{2}$*. For* $\beta = \operatorname{sign} a_2$*:* $a = (+1, +\frac{1}{5})$ *vs.* $(-1, +\frac{1}{5})$ *give* $\Delta = +\frac{3}{5}$ *vs.* $-\frac{2}{5}$*. No single bit certifies* $\operatorname{sign}\Delta$*, so two bits are necessary off* $\mathcal{C}_{\mathrm{mono}}$*.*

Remark 6 (What is unconditional, and what rests on general position)**.** Parts (i)–(ii)—that one declared bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\mathrm{mono}}$—hold *unconditionally.* Only minimality (iii) uses Assumption 2: if it fails (one region's benefit mass is E-certifiably dominant) that region's orientation alone certifies $\operatorname{sign}\Delta$ even outside $\mathcal{C}_{\mathrm{mono}}$, moving the boundary. Equivalently, Assumption 2 is the requirement that the certificate be *domination-free* (minimax-sound over the unobserved magnitudes); so the General-Position statement is only *conditional*; the unconditional weakest classes are characterized in Theorem 9 as an explicit finite family of dominance polytopes. Even under Assumption 2, $\mathcal{C}_{\mathrm{mono}}$ is only *a* weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\mathrm{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the *calibrated* region R_0 and $c = 0$ on the *anchor* region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\mathrm{mono}}$ to the falsifiable class $\mathcal{C}_{\mathrm{dom}} = \{a_{R_0} = \sigma,\ |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and $\operatorname{sign}\Delta = \sigma$ for *every* member, so one declared bit certifies $\operatorname{sign}\Delta$ on all of $\mathcal{C}_{\mathrm{dom}}$. Yet $\mathcal{C}_{\mathrm{dom}} \supsetneq \mathcal{C}_{\mathrm{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E-certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip $\operatorname{sign}\Delta$ and two bits become necessary (K=2). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the *entire* lattice of dominant-region thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\mathrm{mono}}$ but a ρ-dependent family), which Assumption 2 sidesteps by fiat. Theorem 9 (below) resolves it: the ρ-lattice is the slice $\{r_0{=}1, r_1{=}\rho\}$ of the single dominance polytope $W^\star{=}\{r_0 \geq r_1\}$, with ρ=1 its boundary. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\mathrm{dom}}$ strictly contains $\mathcal{C}_{\mathrm{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

Unconditional resolution (General Position removed). Dropping Assumption 2 reveals, for each orientation pattern, a single maximal one-bit class—a *dominance polytope*—of which the ρ-family of Remark 6 is one slice.

Definition 4 (Orientation pattern and canonical class)**.** On a fibre E an *orientation pattern* is a tied set $P \subseteq \{i : c_i \neq 0\}$ with a tied sign-pattern $s \in \{\pm\}^P$; the free set is $Z = D \setminus P$. For an observable magnitude region $W \subseteq [0,1]^{|D|}$ the *canonical class* is $\mathcal{C}(P, s, W) = \{a :\ \operatorname{sign} a_i = \sigma\, s_i \operatorname{sign} c_i\ (i \in P),\ (|a_j|)_j \in W\}$ with one declared bit σ. Write $T(r) = \sum_{i \in P} s_i \operatorname{sign}(c_i) \mu_i r_i$ and $G(r) = \sum_{i \in Z} \mu_i r_i$.

Lemma 5 (Canonical form of a falsifiable class)**.** *A class is falsifiable* iff *it equals some* $\mathcal{C}(P, s, W)$*; anchors (*$c_i = 0$*) lie necessarily in* Z*.* (Swap-closure, Thm. 4(iii), forces the constraint on Z to depend only on $|a|$; forcing an anchor sign is not evidence-definable. Both aligned (s_i=+) and anti-aligned (s_i=−) ties are falsifiable since σ is unidentified.)

Theorem 9 (Unconditional one-bit criterion; weakest classes)**.** *Let* $\mathcal{C}(P, s, W)$ *be falsifiable.* (i) (Dominance margin.) *One declared bit certifies* $\operatorname{sign}\Delta$ *on* $\mathcal{C}(P, s, W)$ iff $\inf_{r \in W}\big(T(r) - G(r)\big) \geq 0$ *(with some member* $\Delta > 0$*) or* $\sup_{r \in W}\big(T(r) + G(r)\big) \leq 0$. (ii) (Weakest class.) *For a fixed pattern* (P, s) *the unique maximal one-bit class is the dominance polytope* $\mathcal{C}^\star = \mathcal{C}(P, s, W^\star)$*,* $W^\star = \{r \in [0,1]^{|D|} : T(r) \geq G(r)\}$*;* $\mathcal{C}_{\mathrm{mono}}$*, every* $\mathcal{C}_{\mathrm{dom}}(\rho \leq 1)$*, and every box one-bit class are proper subclasses of one* $\mathcal{C}^\star$*.* (iii) (Finite family.) *The set of weakest falsifiable one-bit classes is the explicit finite family* $\{\mathcal{C}^\star\}$ *indexed by* $(P \subseteq \{c \neq 0\},\ s \in \{\pm\}^P)$ *modulo the global flip* $\sigma \mapsto -\sigma$ *(at most* $3^{|\{c \neq 0\}|}$ *patterns); distinct patterns give pairwise incomparable maxima, so there is no* unique *weakest class.* $\mathcal{C}_{\mathrm{mono}}$ *is the all-tied all-*

aligned member, and Assumption 2 is exactly the face on which the family collapses to $\mathcal{C}_{\mathrm{mono}}$, recovering Thm. 8(iii).

Proof. Δ is linear, so over $\mathcal{C}(P, s, W)$ at bit $\sigma = +1$ it ranges over $[\inf_W(T - G), \sup_W(T + G)]$ (free signs independent by swap-closure; the $\sigma = -1$ fibre is its negation); a single bit certifies iff each fibre's strict sign set is a singleton, giving (i). For $W^\star$, $\inf_{W^\star}(T - G) \geq 0$ by construction, and any r_0 with $T(r_0) < G(r_0)$ admits, with opposing free signs, a member with $\Delta < 0$ while $W^\star$ admits $\Delta > 0$, so $W^\star$ is maximal; $\mathcal{C}_{\mathrm{dom}}(\rho)$ sits in $W^\star$ iff $\rho \leq 1$, with $\rho = 1$ the boundary, giving (ii). Incomparability (iii) is witnessed on $c = (1, 1, 0)$ by $a = (0, \frac{1}{3}, 0)$ and $a = (1, -\frac{1}{3}, \frac{1}{3})$. On the general-position face the free block can overpower the tied block unless Z-magnitudes vanish, leaving only $\mathcal{C}_{\mathrm{mono}}$. $\square$

Verification. The criterion (i) was checked against exact brute-force vertex truth on 2.8×10^5 box fibres and 3.2×10^3 non-box integer polytopes (exact-LP inf / sup) with 0 mismatches; sufficiency on $W^\star$ over 1.2×10^5 members (0 errors), necessity just outside $W^\star$ (error rate $\approx \frac{1}{2}$), and the $\mathcal{C}_{\mathrm{dom}}(\rho)$ collapse ($\rho\leq1$ one-bit, $\rho>1$ lost) are machine-checked (`val_unconditional_weakest.py`, seeds fixed; fails loudly). The single non-tautological input is Lemma 5, which rests on the swap involution (Thm. 4(iii)).

Theorem 1 is *binary*: it certifies a regime in which no label-free rule beats chance. This section sharpens that dichotomy into a *graded, computable invariant* K with an explicit threshold τ, in a single fully specified model, and proves that for the population (identifiability) problem the converse and the achievability meet *exactly* at τ. Thus, in this model, K is a genuine *capacity* for label-free sign recovery, not merely a sufficient or a necessary condition. A finite-sample Le Cam analysis then locates the threshold up to an $O(1/\sqrt{n})$ boundary layer.

K. Model (the concept mechanism is explicit)

Binary $Y \in \{0, 1\}$, 0/1 loss. Source covariates $P_X = \mathcal{N}(0, 1)$ (known); target covariates $Q_X = \mathcal{N}(\mu, 1)$ (covariate shift; μ unknown, to be estimated from an unlabeled target sample). The deployed classifier is $g = f_0 = \mathbf{1}[x > 0]$ (boundary at the origin). We are handed a *fixed* candidate adapted rule $f_a = \mathbf{1}[x > a]$ with $a > 0$ known — the shipped "adapted boundary" — and ask whether switching $g \to f_a$ helps. The disagreement region is the interval $D = (0, a)$.

The label/concept mechanism must be explicit, because $\operatorname{sign}\Delta$ depends on $P(Y \mid X)$, which Q_X alone does not identify. We take *hard-threshold concepts* $\eta_\theta(x) = P(Y = 1 \mid x) = \mathbf{1}[x > \theta]$. The source-calibrated threshold θ_S is known (it is what a source-fit model reports). The *admissible target concepts* are those whose target *boundary mass* drifts from the source boundary by at most a budget $\varepsilon \geq 0$:

$$\mathcal{C}_\varepsilon \;=\; \Big\{ \theta_T : \; \big| \Phi(\theta_T - \mu) - \Phi(\theta_S - \mu) \big| \leq \varepsilon \Big\}. \tag{5}$$

Here ε is the 1-D analogue of the calibration-drift budget β of the benefit-sign frontier (§V); by Theorem 4 some such untestable supplement is unavoidable, and (5) fixes it in the natural label-free-meaningful unit of *target probability mass*. The label-free observer sees only $(P_X, g, f_a, \theta_S, \varepsilon)$ and the unlabeled sample $\{x_i\} \sim Q_X$, and must output a guess of $\operatorname{sign}\Delta$, where $\Delta = R_Q(f_0) - R_Q(f_a)$ ($\Delta > 0 \iff$ adapting to f_a helps).

Lemma 6 (Exact benefit identity and the sign-flip locus)**.** *On $D = (0, a)$ exactly one of f_0, f_a is correct, so for the concept η_θ,*

$$\Delta(\theta) \;=\; Q\big((0,a) \cap \{x < \theta\}\big) - Q\big((0,a) \cap \{x > \theta\}\big) \;=\; 2\,\Phi(\theta' - \mu) - \Phi(-\mu) - \Phi(a - \mu),$$

with $\theta' = \operatorname{clip}(\theta, 0, a)$*.* $\Delta(\cdot)$ *is nondecreasing in θ with a single root*

$$\boxed{\theta_0 \;=\; \mu + \Phi^{-1}\Big(\tfrac{1}{2}\big[\Phi(-\mu) + \Phi(a - \mu)\big]\Big)} \qquad \textit{(the Q-median of the band D),}$$

so $\operatorname{sign}\Delta = \operatorname{sign}(\theta - \theta_0)$*. Both θ_0 and the plug-in sign* $\operatorname{sign}(\theta_S - \theta_0)$ *are* observable*.*

Proof. Off D, $f_0 = f_a$, so the pointwise excess loss vanishes. On $D = (0, a)$ we have $f_0 = 1, f_a = 0$; for 0/1 loss the excess loss of f_0 over f_a is $\mathbf{1}[f_a{=}Y] - \mathbf{1}[f_0{=}Y] = \mathbf{1}[Y{=}0] - \mathbf{1}[Y{=}1] = 1 - 2Y$, with conditional mean $1 - 2\eta_\theta(x)$, which is $+1$ for $x < \theta$ and -1 for $x > \theta$. Integrating against Q over $(0, a)$ gives the displayed difference of band masses, hence $\Delta(\theta) = \Phi(\theta' - \mu) - \Phi(-\mu) - \big(\Phi(a - \mu) - \Phi(\theta' - \mu)\big) = 2\Phi(\theta' - \mu) - \Phi(-\mu) - \Phi(a - \mu)$. This is nondecreasing in θ' (hence in θ); setting it to zero gives $\Phi(\theta_0 - \mu) = \frac{1}{2}[\Phi(-\mu) + \Phi(a - \mu)]$, the Q-median of D. (Validated to 4×10^{-4} Monte-Carlo error against the empirical $R_Q(f_0) - R_Q(f_a)$, `val_knowability_capacity.py` part (D).) ☐

L. The computable invariant and the threshold

Definition 5 (Knowability invariant)**.**

$$K \;:=\; \frac{\big|\,\Phi(\theta_0 - \mu) - \Phi(\theta_S - \mu)\,\big|}{\varepsilon} \;=\; \frac{\big|\,\frac{1}{2}[\Phi(-\mu) + \Phi(a - \mu)] - \Phi(\theta_S - \mu)\,\big|}{\varepsilon}.$$

K is the *mass margin* between the calibrated boundary θ_S and the sign-flip locus θ_0, measured in target probability mass, normalised by the drift budget ε. It is a margin-weighted SNR of the covariate shift relative to g's boundary, computable from $(\mu, a, \theta_S, \varepsilon)$ alone (with μ estimable label-free from $\{x_i\}$).

Theorem 10 (Population knowability–capacity, $\tau = 1$)**.** *Assume Q_X (equivalently μ) is known. With threshold $\tau = 1$:*

(i) ***Achievability.*** *If $K > 1$ then for* every *admissible target concept $\theta_T \in \mathcal{C}_\varepsilon$,* $\operatorname{sign}\Delta = \operatorname{sign}(\theta_S - \theta_0)$*; the plug-in certificate $\hat{g} = \operatorname{sign}(\theta_S - \theta_0)$ recovers the sign with error* 0*.*

(ii) ***Converse.*** *If $K < 1$ there exist $\theta_T^+, \theta_T^- \in \mathcal{C}_\varepsilon$ with* identical Q_X *(hence* $\mathrm{TV} = 0$ *between the laws of* any *statistic of any number n of unlabeled observations) and* $\operatorname{sign}\Delta(\theta_T^+) = -\operatorname{sign}\Delta(\theta_T^-) \neq 0$*. Consequently the minimax label-free error equals $\frac{1}{2}$ for every n.*

The converse and the achievability therefore meet at the single *threshold* $\tau = 1$, exactly *(not asymptotically): $\{K > 1\}$ is precisely the identifiable set.*

Proof. By Lemma 6, $\operatorname{sign}\Delta(\theta_T) = \operatorname{sign}(\theta_T - \theta_0)$ with θ_0 fixed by the observables. The map $\theta \mapsto \Phi(\theta - \mu)$ is a strictly increasing bijection, so the admissible set (5) is exactly the interval $\theta_T \in [\,\mu + \Phi^{-1}(p_S - \varepsilon),\, \mu + \Phi^{-1}(p_S + \varepsilon)\,]$ with $p_S = \Phi(\theta_S - \mu)$, i.e. in mass coordinates $u = \Phi(\theta_T - \mu)$ it is the interval $[\,p_S - \varepsilon,\, p_S + \varepsilon\,]$ (clipped to $[0,1]$). The sign of Δ flips exactly at the mass level $u_0 = \Phi(\theta_0 - \mu) = \frac{1}{2}[\Phi(-\mu) + \Phi(a - \mu)]$.

(i) If $K > 1$ then $|u_0 - p_S| > \varepsilon$, so u_0 lies strictly outside $[p_S - \varepsilon, p_S + \varepsilon]$. Hence every admissible $u \in [p_S - \varepsilon, p_S + \varepsilon]$ is on one side of u_0, so $\theta_T - \theta_0$ has constant sign over $\mathcal{C}_\varepsilon$, equal to $\operatorname{sign}(\theta_S - \theta_0) = \operatorname{sign}(p_S - u_0)$. The plug-in certificate is correct for all admissible concepts.

(ii) If $K < 1$ then $|u_0 - p_S| < \varepsilon$, so $u_0 \in (p_S - \varepsilon, p_S + \varepsilon)$ lies strictly interior to the admissible mass interval. Pick $u^\pm = \Phi(\theta_T^\pm - \mu)$ with $u^+ \in (u_0, p_S + \varepsilon]$ and $u^- \in [p_S - \varepsilon, u_0)$ (both nonempty and admissible); then $\operatorname{sign}\Delta(\theta_T^+) = +1$, $\operatorname{sign}\Delta(\theta_T^-) = -1$. Both worlds share the *same* $Q_X = \mathcal{N}(\mu, 1)$ (the concept lives in the unobserved $P(Y \mid X)$ channel), so the law of any unlabeled statistic is identical across the two worlds: $\mathrm{TV} = 0$ for every n. By Le Cam's two-point inequality the minimax error of any label-free rule is $\geq \frac{1}{2}(1 - \mathrm{TV}) = \frac{1}{2}$, and the constant rules achieve $\frac{1}{2}$, so it equals $\frac{1}{2}$.

Finally, $K = 1$ is the single boundary $|u_0 - p_S| = \varepsilon$, where u_0 is an endpoint of the admissible interval; the sign is constant on the (closed) interval but not on its interior-plus-the-flip, so the transition is exactly at $\tau = 1$. (Validated: over 40,000 random draws of $(\mu, a, \theta_S, \varepsilon)$, the equivalence "$K > 1 \iff$ identifiable over $\mathcal{C}_\varepsilon$" holds with *zero* mismatches off a 10^{-6} shell; a boundary stress-test confirms the flip at $K = 1$ to 10^{-3}. `val_knowability_capacity.py` part (A).) ☐

Relation to the frontier. Theorem 10 is the *graded* reading of Theorem 3: the frontier's qualitative "$|M| > \beta$" becomes a *scalar* $K = |M|/\beta$-type ratio with the explicit value $\tau = 1$ in a fully specified location model, and the 0-vs-$\frac{1}{2}$ minimax dichotomy is recovered as the two faces $K > 1$ and $K < 1$. The novelty here is that K is shown to be *both* necessary and sufficient at the *same* threshold — a capacity — for the population problem, with θ_0 (the flip locus) given in closed form by the covariate shift μ and the candidate boundary a.

M. Finite-n: a Le Cam boundary layer around the threshold

With only n unlabeled samples, μ — hence K — must be estimated, and the deployable rule is the finite-n *three-way certificate*: estimate $\hat{\mu} = \bar{x}$, form a z-confidence interval $[\hat{\mu} \pm z/\sqrt{n}]$, and commit to $\operatorname{sign}(\theta_S - \hat{\theta}_0)$ only if K stays above 1 across the whole interval (a lower-confidence bound $\hat{K}_{\mathrm{lcb}} > 1$); otherwise abstain. The next proposition shows the price of estimating μ is exactly an $O(1/\sqrt{n})$ layer around the threshold *location*, with the same Le Cam/Bretagnolle–Huber form as Proposition 1.

Proposition 3 (Finite-n Le Cam boundary layer)**.** *Let $\mu^\star$ be the flip locus, $\theta_0(\mu^\star) = \theta_S$ (so $K = 0$ there), and fix the concept to transfer ($\theta_T = \theta_S$, i.e. ε irrelevant; the only noise is in estimating μ). For $c > 0$ consider the two worlds $\mu^\pm = \mu^\star \pm \frac{c}{2}\sqrt{1/n}$, which straddle the flip and hence have* opposite $\operatorname{sign}\Delta$*. The unlabeled laws satisfy*

$$\mathrm{TV}\big(\mathcal{N}(\mu^-, 1)^{\otimes n}, \mathcal{N}(\mu^+, 1)^{\otimes n}\big) = 2\Phi(c/2) - 1, \qquad \mathrm{KL} = \tfrac{n}{2}\,(\mu^+ - \mu^-)^2 = \tfrac{c^2}{2},$$

so every label-free sign test has minimax error

$$\inf_{\hat{g}} \max_{\pm} \mathbb{P}\big[\hat{g} \text{ wrong}\big] \;\geq\; \tfrac{1}{2}\big(1 - \mathrm{TV}\big) = \Phi(-c/2) \;\geq\; \tfrac{1}{4}\, e^{-c^2/2} \quad (\textit{Bretagnolle–Huber}),$$

constant in n*. Hence sign recovery is impossible below error $\Phi(-c/2)$ inside a window of half-width $\frac{c}{2}\sqrt{1/n} = \Theta(1/\sqrt{n})$ around $\mu^\star$,* even when the concept transfers.

Proof. The sample mean $\bar{X} \sim \mathcal{N}(\mu^\pm, 1/n)$ is sufficient; the TV of two equal-variance Gaussians with mean gap $\mu^+ - \mu^- = c/\sqrt{n}$ and standard deviation $1/\sqrt{n}$ is $2\Phi\big(\frac{c/\sqrt{n}}{2/\sqrt{n}}\big) - 1 = 2\Phi(c/2) - 1$, and the per-sample $\mathrm{KL}\big(\mathcal{N}(\mu^+, 1) \| \mathcal{N}(\mu^-, 1)\big) = \frac{1}{2}(\mu^+ - \mu^-)^2$ tensorises to $\frac{c^2}{2}$. A sign test is a test between the two worlds; Le Cam's testing affinity gives the $\frac{1}{2}(1 - \mathrm{TV})$ floor and Bretagnolle–Huber gives the $\frac{1}{4}e^{-\mathrm{KL}}$ certificate. That the two worlds straddle the flip (so the signs are genuinely opposite) is Lemma 6. (Validated: the empirical Bayes-optimal sign-test error meets $\Phi(-c/2)$ to $\leq 1.5 \times 10^{-2}$, exceeds $\frac{1}{4}e^{-c^2/2}$, and is constant across $n \in \{100, 1000, 10000\}$ for $c \in \{0.5, 1, 2\}$; `val_knowability_capacity.py` part (C).) ☐

The finite-n certificate *matches* this: numerically its minimax sign-recovery accuracy $\to 1$ wherever $K_{\mathrm{pop}} > 1$ and $= \frac{1}{2}$ wherever $K_{\mathrm{pop}} < 1$, with the transition pinned at $K = 1$, across $n \in \{200, 2000, 20000\}$ (Table XXVIII; `val_knowability_capacity.py` part (B)). The boundary layer is therefore an $O(1/\sqrt{n})$ blur of the threshold *location* $\mu^\star$, not a gap in the threshold *value* $\tau = 1$.

Remark 7 (Honest scope — what is and is not claimed)**.** (1) Theorem 10 is a capacity *for this* 1*-D Gaussian-location model and for sign-of-benefit only*; it is not a general-distribution capacity. (2) The clean value $\tau = 1$ is a consequence of measuring the drift budget ε in target-*mass* units (5); this is the natural label-free-meaningful unit (it makes θ_0 the Q-median and the flip condition a mass comparison), but a different budget parametrisation would rescale τ. (3)

μ	K_{pop}	acc n=200	acc n=2000	acc n=20000
−0.807	1.416	0.994	1.000	1.000
0.366	1.679	0.995	1.000	1.000
1.633	1.677	0.995	1.000	1.000
2.806	1.417	0.994	1.000	1.000
−1.407	0.640	0.500	0.500	0.500
0.980	0.063	0.500	0.500	0.500
1.000	0.000	0.500	0.500	0.500
3.406	0.641	0.500	0.500	0.500

TABLE XXVIII

Phase transition of the finite-n three-way certificate (a=2, θ_S=1, ε=0.05; minimax over the admissible concept). Above τ=1 the minimax accuracy $\to$ 1; below it the certificate abstains and the accuracy is exactly $\frac{1}{2}$. (Real numbers, seed fixed; `results_knowability_capacity.json`.)

The same equivalence $K > 1 \iff$ identifiable holds for *soft* (logistic) concepts of fixed known slope, with θ_0 the (numerically computed) root of $\Delta(\cdot)$ rather than the band median — verified with zero mismatches over 1,496 draws — so the result is not an artifact of the hard-threshold concept; only the closed form for θ_0 is special to it. (4) Relative to the existing five theorems, this is a *refinement*, not a new pillar: it converts the 0-vs-$\frac{1}{2}$ frontier of Theorem 3 into a scalar capacity with an explicit closed-form flip locus, and supplies the finite-n boundary layer that Proposition 1 left implicit for the location family. Its value is a sharpened, fully worked *instance*, useful pedagogically and as the tight test-bed for the graded invariant; it does not enlarge the set of *distribution-free* guarantees.

The 1-D theorem (§J) gives a genuine capacity $\tau = 1$ for sign-of-benefit in the Gaussian-location two-point model. We now ask how much of that survives in higher dimension, in non-Gaussian location families, and distribution-free. The answer is a clean layered one: the converse is essentially universal, the achievability is exact under an alignment/MLR regularity condition, and a single *computable* capacity provably needs such a condition. We label every claim and exhibit the breakers that delimit each regime.

N. Stage 1: multivariate Gaussian-location

a) Model.: $P_X = \mathcal{N}(0, \Sigma)$, $Q_X = \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$ known and μ unknown (estimated from an unlabeled target sample). The deployed rule is the halfspace $g = f_0 = \mathbf{1}[w^\top x > 0]$ and the shipped candidate is the *parallel-shifted* halfspace $f_a = \mathbf{1}[w^\top x > a]$, $a > 0$ known. The disagreement region is the *slab* $D = \{x : 0 < w^\top x < a\}$. Concepts are halfspaces $\eta_{v,\theta}(x) = \mathbf{1}[v^\top x > \theta]$ (the label mechanism), with θ_S the source-calibrated offset along a fixed direction. Write the *whitened (Mahalanobis) projection onto the rule normal*

$$s \;:=\; w^\top x, \qquad s \sim_Q \mathcal{N}(m, \sigma^2), \quad m = w^\top \mu, \;\; \sigma^2 = w^\top \Sigma w, \tag{6}$$

and for a concept direction v the second coordinate $u := v^\top x$, so that (s, u) is bivariate Gaussian under Q with correlation (the *Mahalanobis cosine* between w and v)

$$\rho \;=\; \rho(w, v) \;=\; \frac{w^\top \Sigma v}{\sqrt{w^\top \Sigma w}\,\sqrt{v^\top \Sigma v}} \;=\; \cos\angle_\Sigma(w, v). \tag{7}$$

Because f_0, f_a and $\eta_{v,\theta}$ depend on x only through (s, u), the entire benefit is a functional of this bivariate Gaussian:

$$\Delta \;=\; R_Q(f_0) - R_Q(f_a) \;=\; \mathbb{E}_Q\big[\mathbf{1}_D\,(1 - 2\eta_{v,\theta})\big] \;=\; Q(D) \;-\; 2\,Q\big(D \cap \{u > \theta\}\big). \tag{8}$$

Lemma 7 (Whitened reduction; PROVEN). *Δ in (8) depends on the dimension d, on μ and on Σ only through the bivariate law of (s, u) under Q, i.e. through the four scalars $(m, \sigma^2,\ v^\top \mu,\ v^\top \Sigma v)$ and the correlation ρ of (7). In particular the deployed and candidate rules depend on x only through s, whose Q-law is the one-dimensional Gaussian $\mathcal{N}(w^\top \mu,\ w^\top \Sigma w)$.*

Proof. $f_0 = \mathbf{1}[s > 0]$, $f_a = \mathbf{1}[s > a]$, $\eta_{v,\theta} = \mathbf{1}[u > \theta]$ are $\sigma(s, u)$-measurable, and the 0/1 excess loss of f_0 over f_a is supported on $D = \{0 < s < a\}$ where it equals $1 - 2\eta_{v,\theta}$ (exactly as in Lemma 6: on D, f_0=1, f_a=0, and the conditional mean of $\mathbf{1}[Y{=}0] - \mathbf{1}[Y{=}1]$ is $1 - 2\eta$). Taking Q-expectation gives (8), a functional of the Q-law of (s, u). A non-degenerate affine image of a Gaussian is Gaussian with the stated moments. ☐

b) (a) Aligned concept class — exact lift.: Take the concept normal *parallel to the rule normal*, $v = w$ (the concept "reads the same direction" the deployed rule acts on). Then $u = s$, the problem collapses onto the single whitened line span(w), and (8) becomes literally the 1-D identity of Lemma 6 in the standardized variable $z = (s-m)/\sigma$. Define the admissible aligned class and the lifted invariant in Mahalanobis units:

$$\mathcal{C}_\varepsilon^{\|} = \Big\{\theta_T :\ \big|\Phi(\tfrac{\theta_T - m}{\sigma}) - \Phi(\tfrac{\theta_S - m}{\sigma})\big| \le \varepsilon\Big\}, \quad K_{\|} = \frac{\big|\frac{1}{2}[\Phi(\frac{-m}{\sigma}) + \Phi(\frac{a-m}{\sigma})] - \Phi(\frac{\theta_S - m}{\sigma})\big|}{\varepsilon}. \tag{9}$$

Theorem 11 (Stage 1, aligned: exact Mahalanobis lift; PROVEN)**.** *With $v = w$ and Q_X known, the population knowability–capacity Theorem 10 holds verbatim with (μ, a, θ_S) replaced by their whitened images (m, σ) in (6): with $\tau = 1$, $K_{\|} > 1 \Rightarrow$ every admissible aligned concept has* $\operatorname{sign}\Delta = \operatorname{sign}(\theta_S - \theta_0)$ *(achievability, error 0); $K_{\|} < 1 \Rightarrow$ two admissible aligned concepts share Q_X and have opposite* $\operatorname{sign}\Delta$, *so the minimax label-free error is $\frac{1}{2}$. The flip locus is the Q-median of the slab along w, $\theta_0 = m + \sigma\,\Phi^{-1}\big(\frac{1}{2}[\Phi(\frac{-m}{\sigma}) + \Phi(\frac{a-m}{\sigma})]\big)$. The finite-$n$ Le Cam boundary layer (Proposition 3) carries over with μ replaced by $m = w^\top\mu$ and the noise scale by $\sigma/\sqrt{n}$, since $\bar{s} = w^\top\bar{x} \sim \mathcal{N}(m, \sigma^2/n)$ is sufficient for m.*

Proof. By Lemma 7 with $v = w$, Δ is the 1-D benefit of Lemma 6 for the standardized location problem $z \sim \mathcal{N}(m/\sigma, 1)$ with boundaries $0, a, \theta$ rescaled by σ; (9) is exactly (5) in z. Apply Theorem 10. The converse two worlds again share the *same* $Q_X = \mathcal{N}(\mu, \Sigma)$ (the concept lives in the unobserved $P(Y \mid X)$ channel), so TV $= 0$ for every unlabeled statistic and every n. The sufficiency of $\bar{s}$ for m is standard. *(Validated:* 20,000 *random draws in* $d \in \{2,3,4\}$*, equivalence* $K_{\|} > 1 \iff$ *identifiable with* 0 *mismatches off a* 10^{-6} *shell;* `val_knowability_capacity_general.py` *Stage 1a.)* ☐

c) (b) Free halfspace concept class — exact obstruction and its capacity.: If the concept normal v is *not* constrained to be parallel to w, the flip locus moves. For a fixed v, $\Delta(\theta)$ in (8) is still nondecreasing in θ with a single root $\theta_0(v)$ characterized by $Q\big(D \cap \{u > \theta_0\}\big) = \frac{1}{2}Q(D)$: *$\theta_0(v)$ is the Q-conditional median of $u = v^\top x$ given $x \in D$.* Its level in the concept's own Mahalanobis mass coordinate,

$$p_{\text{med}}(v) \ :=\ \Phi\Big(\tfrac{\theta_0(v) - v^\top\mu}{\sqrt{v^\top\Sigma v}}\Big), \tag{10}$$

interpolates between the two extremes set by $\rho = \rho(w, v)$. In standardized coordinates write $z = (s-m)/\sigma \sim \mathcal{N}(0,1)$, the slab as $z \in (\alpha, \beta)$ with $\alpha = -m/\sigma,\ \beta = (a-m)/\sigma$, and $t = (u - v^\top\mu)/\sqrt{v^\top\Sigma v} = \rho\, z + \sqrt{1-\rho^2}\,\omega$ with $\omega \perp z$ standard normal. Then $p_{\text{med}}(v) = \Phi(\text{med}(t \mid z \in (\alpha,\beta)))$ and

$$\rho = \pm 1\ (v \parallel w):\ p_{\text{med}} = \tfrac{1}{2}[\Phi(\alpha) + \Phi(\beta)]\ \text{(band median, since } t = \pm z), \qquad \rho = 0\ (v \perp_\Sigma w):\ p_{\text{med}} = \tfrac{1}{2}. \tag{11}$$

Both endpoints are **proven exactly**. The $\rho = 0$ case is structural: when v is Σ-orthogonal to w, $t = \omega$ is independent of z, so conditioning on the slab does not move its law and $\text{med}(t \mid z \in (\alpha,\beta)) = 0$, i.e. $p_{\text{med}} = \frac{1}{2}$. The $\rho = \pm 1$ case is immediate since then $t = \pm z$. That p_{med} is *monotone* in $|\rho|$ between these endpoints (so the achievable flip range is exactly the interval between $\frac{1}{2}$ and the band median) is **verified numerically** (Lemma 8, stated as a numerically confirmed monotone-interpolation fact with a proof sketch); the *worst case* relevant to the converse — mass $\frac{1}{2}$, reached at $\rho = 0$ — is proven outright.

Lemma 8 (Monotone flip interpolation; NUMERICAL, proof sketch)**.** *With the notation above, $p_{\text{med}}(v)$ is monotone in $|\rho|$, sweeping the full interval between $\frac{1}{2}$ (at $\rho = 0$) and the band median (at $|\rho| = 1$).* Sketch: *the conditional median solves $\mathbb{E}[\operatorname{sign}(t - m_t) \mid z \in (\alpha,\beta)] = 0$; since the slab's z-median $z^\star$ has a fixed sign relative to 0, increasing ρ tilts $t = \rho z + \sqrt{1-\rho^2}\,\omega$ monotonically toward $\pm z$, dragging m_t monotonically from 0 toward $z^\star$ by a sign-coherent shift of the conditional law. A full proof would verify single-crossing of the conditional-median equation in ρ.* (Validated: p_{med} rises monotonically $0.50 \to$ band median as $\rho : 0 \to 1$ across four slabs; `val_knowability_capacity_general.py` Stage 1b.)

Theorem 12 (Stage 1, free halfspace: the obstruction; converse PROVEN, exact K_{free} CONDITIONAL)**.** *Let $p_S = \Phi(\frac{\theta_S - m}{\sigma})$ be the calibrated boundary mass along w and let the admissible free class be the halfspaces with own-normal boundary-mass drift $\le \varepsilon$ and arbitrary direction v. Then the aligned invariant $K_{\|}$ of (9) is* not *a capacity for this class. Precisely:*

(i) ***(Breaker / converse; PROVEN.)*** *Whenever p_S lies strictly between $\frac{1}{2}$ and $b := \frac{1}{2}[\Phi(\frac{-m}{\sigma}) + \Phi(\frac{a-m}{\sigma})]$, there is a regime $K_{\|} > 1$ in which the aligned theorem predicts identifiability but a tilted admissible concept (own-normal boundary mass $= p_S$, taken with $\rho \to 0$, which reaches flip mass $\frac{1}{2}$) produces the* opposite *benefit sign. The two concepts share Q_X, so* TV $= 0$ *and the minimax error is $\frac{1}{2}$: the aligned $K_{\|}$ is therefore* not *a capacity for the free class.*

(ii) ***(Exact free capacity; CONDITIONAL on Lemma 8.)*** *Under the monotone interpolation, the achievable flip range is exactly the interval between $\frac{1}{2}$ and b, so* $\operatorname{sign}\Delta$ *is identifiable over the free class iff $[p_S - \varepsilon, p_S + \varepsilon]$ misses it; the governing capacity is*

$$K_{\text{free}} \ =\ \frac{\operatorname{dist}\big(p_S,\ [\,\frac{1}{2} \wedge b,\ \frac{1}{2} \vee b\,]\big)}{\varepsilon}, \qquad \tau = 1.$$

Proof. For fixed v, (8) is nondecreasing in θ (raising θ only shrinks $\{u > \theta\}$), with unique root the conditional median $\theta_0(v)$; hence $\operatorname{sign}\Delta = \operatorname{sign}(\theta - \theta_0(v))$. *Part (i)* uses only the proven endpoint $\rho = 0 \Rightarrow p_{\text{med}} = \frac{1}{2}$ of (11): for p_S strictly on the $\frac{1}{2}$-side of the aligned flip b, take v with $\rho \to 0$ and own-normal boundary mass $= p_S$; its flip sits at $\frac{1}{2}$, on the opposite side of p_S from b, so $\operatorname{sign}\Delta$ is reversed relative to the aligned prediction. The two concepts share Q_X, giving $\mathrm{TV} = 0$ and minimax error $\frac{1}{2}$ (Le Cam); thus $K_\parallel$ is not a capacity. *Part (ii)* additionally invokes Lemma 8 (numerically confirmed): the achievable flip levels then fill exactly the interval between $\frac{1}{2}$ and b, so a sign-flip witness exists iff the admissible mass band $[p_S - \varepsilon, p_S + \varepsilon]$ meets that interval, i.e. iff $K_{\text{free}} < 1$. *(Validated:* p_{med} *vs* $|\rho|$ *traced from* 0.5005 *at* ρ=0 *to the band median* 0.4226 *at* ρ=1*; an explicit breaker with* $K_\parallel = 1.40$ *where the aligned prediction is* $\Delta = +0.14$ *but a worst-case tilt of the same own-normal boundary mass yields* $\Delta = -0.14$ *(opposite sign);* `val_knowability_capacity_general.py` *Stage 1b.)* ☐

Remark 8 (What Stage 1 establishes)**.** The 1-D capacity lifts to d dimensions *exactly* in Mahalanobis geometry *when the concept is aligned with the rule normal* (Thm 11): K and $\tau = 1$ are intact, with θ_0 the slab's Q-median along w. The lift is therefore not vacuous — it is a genuine d-dimensional capacity. But the unconstrained halfspace concept class is *strictly harder*: an adversary can tilt the concept normal toward Σ-orthogonality with w, dragging the flip locus to the unconditional median (mass $\frac{1}{2}$) and defeating any margin measured along w. The honest Stage-1 statement is a capacity K_{free} that prices in the worst-case tilt (Thm 12); the clean closed-form $\tau = 1$ survives, but the invariant is the distance to mass $\frac{1}{2}$, not to the band median, once the concept direction is free.

O. Stage 2: MLR / log-concave location families

We now leave the Gaussian and ask which *non-Gaussian* location families keep the matched threshold.

Scope note. Here "exponential family" abbreviates one-parameter *location* families with monotone likelihood ratio (Gaussian, Laplace, logistic location). Genuine *scale*/natural-parameter exponential families—e.g. the exponential or Gamma rate family—are not location families; their capacity is not governed by the mass-coordinate threshold below and falls under Conjecture 2.

Let Q_X have a density $q_0(x-\mu)$ for a fixed base density q_0 with continuous CDF F; deployed $f_0 = \mathbf{1}[x > 0]$, candidate $f_a = \mathbf{1}[x > a]$, hard-threshold concept $\eta_\theta = \mathbf{1}[x > \theta]$. The benefit identity is *family-agnostic*: exactly as in Lemma 6 (the proof used only that on D the conditional contrast is $1 - 2\eta$ and that Q has a CDF),

$$\Delta(\theta) = 2F(\theta' - \mu) - F(-\mu) - F(a - \mu), \qquad \theta' = \operatorname{clip}(\theta, 0, a), \tag{12}$$

and the admissible class / invariant are (5), Def. 5 with Φ replaced by F. So far *no* condition is needed and Δ is monotone in θ for *any* CDF F.

Theorem 13 (Stage 2, MLR/log-concave: matched $\tau = 1$; PROVEN UNDER CONDITION)**.** *Assume the base family is a one-parameter* location *family with strictly* ***monotone likelihood ratio*** *in the location (equivalently here,* q_0 *strictly positive and log-concave, hence unimodal). Then with the mass-coordinate drift class* $(5)_F$ *and* $\tau = 1$*, the population knowability–capacity holds verbatim:* $K > 1 \Rightarrow$ *every admissible concept shares* $\operatorname{sign}\Delta = \operatorname{sign}(\theta_S - \theta_0)$ *(achievability),* $K < 1 \Rightarrow$ *two admissible concepts share* Q_X *with opposite* $\operatorname{sign}\Delta$ *(converse,* $\mathrm{TV} = 0$*). Moreover the flip locus* $\theta_0(\mu) = \mu + F^{-1}\big(\frac{1}{2}[F(-\mu) + F(a-\mu)]\big)$ *is the band's Q-median and is* ***strictly increasing in the unknown shift*** μ*, so a single label-free estimate* $\hat{\mu}$ *pins the flip side through one threshold and* K *is a genuine scalar capacity (the finite-n Le Cam layer of Prop. 3 carries over with* $\Phi \to F$*).*

Proof. The converse and achievability are exactly Theorem 10 with $\Phi \to F$: the admissible set is the interval $[p_S - \varepsilon, p_S + \varepsilon]$ in the mass coordinate $u = F(\theta_T - \mu)$, the flip is at the band-median mass $u_0 = \frac{1}{2}[F(-\mu) + F(a-\mu)]$, and $K \gtrless 1 \iff u_0$ lies outside/inside the admissible band; identical-Q_X gives $\mathrm{TV} = 0$. The only place a hypothesis is used is the claim that $\theta_0(\mu)$ is monotone in μ (needed for $K(\mu)$ to be a single threshold rather than a fragmented set): for an MLR/log-concave location family the map $\mu \mapsto \theta_0(\mu)$ is strictly increasing because $F(a - \mu)$ and $F(-\mu)$ are decreasing in μ and F^{-1} is increasing, with no antimode to reverse the band median's motion. *(Validated: Laplace and logistic location families,* 0 *mismatches over* 6,000 *draws each, benefit identity matched to* $\leq 4.6 \times 10^{-4}$ *MC error;* `val_knowability_capacity_general.py` *Stage 2a.)* ☐

Theorem 14 (Stage 2, non-log-concave breaker; PROVEN)**.** *There is a location family for which the scalar capacity* fails*. Take* q_0 *a balanced mixture of two well-separated Gaussians (bimodal, hence not log-concave and not MLR) and the band* $D = (0, a)$ *wide enough to span the antimode. Then although* $\Delta(\theta)$ *in* (12) *is still monotone in* θ *(so the concept-location flip is unique), the flip locus* $\theta_0(\mu)$ *is* ***non-monotone in the unknown shift*** μ*: as* μ *moves the band median across the antimode,* $\theta_0(\mu)$ *collapses. Consequently the label-free identifiable set* $\{\mu : K(\mu) > 1\}$ ***fragments*** *into more components than in the unimodal case, and no single scalar threshold* τ *separates the identifiable from the non-identifiable regime.*

Proof (by exhibited witness). The benefit identity (12) holds for the mixture CDF F, monotone in θ. The density q_0 has two modes (verified: two local maxima), so the hypothesis of Theorem 13 fails. Tracing $\theta_0(\mu)$ over $\mu \in [-3, 3]$ exhibits two strict decreases (non-monotone events) where the band median crosses the antimode, and the set $\{\mu : K(\mu) > 1\}$ has 2 components for the mixture versus 1 for the matched unimodal Gaussian baseline at the same (p_S, ε, a). Hence the achievability certificate "estimate $\hat{\mu}$, read off $\operatorname{sign}(\theta_S - \theta_0(\hat{\mu}))$" is not governed by a single threshold in K. *(Validated: density modes = 2, $\theta_0(\mu)$ non-monotone events = 2, components mixture= 2 vs Gaussian= 1; `val_knowability_capacity_general.py` Stage 2b.)* ☐

P. Stage 3: a distribution-free converse, and the general obstruction

The converse needs none of the above structure. It is a pure two-point Le Cam bound that uses only the defining feature of the problem: the concept lives in the unobserved $P(Y \mid X)$ channel, so any two concepts induce the *same* observable covariate law.

Theorem 15 (Distribution-free sign-of-benefit converse; PROVEN)**.** *Let X take values in any measurable space, let g and f be any two fixed classifiers, and let $\mathcal{H}$ be a class of admissible concepts $\eta : \mathcal{X} \to [0, 1]$. Suppose there exist $\eta^+, \eta^- \in \mathcal{H}$ that induce the* ***same observable marginal*** *Q_X (automatic under covariate shift, where the concept does not affect Q_X) and have* ***opposite benefit sign****, $\operatorname{sign}\left(R_Q^{\eta^+}(g) - R_Q^{\eta^+}(f)\right) = -\operatorname{sign}\left(R_Q^{\eta^-}(g) - R_Q^{\eta^-}(f)\right) \neq 0$. Then for* every *sample size n and* every *label-free decision rule $\hat{T}$ that observes only $\{X_i\}_{i=1}^n \sim Q_X^{\otimes n}$,*

$$\inf_{\hat{T}} \max_{\eta \in \{\eta^+, \eta^-\}} \mathbb{P}\big[\hat{T} \text{ outputs the wrong } \operatorname{sign}\Delta\big] \;\geq\; \tfrac{1}{2}\big(1 - \mathrm{TV}(Q_X^{\otimes n}, Q_X^{\otimes n})\big) \;=\; \tfrac{1}{2}.$$

No distributional, dimensional, or parametric assumption is used.

Proof. The two worlds $\eta^\pm$ generate identical laws for the unlabeled sample ($Q_X^{\otimes n}$ in both), so the total variation between the observed data distributions is 0. A label-free sign decision is a test between the two worlds, which have opposite correct answers; Le Cam's two-point inequality gives minimax error $\geq \frac{1}{2}(1 - \mathrm{TV}) = \frac{1}{2}$, attained by a constant guess. *(Validated across Gaussian-1d, Laplace-1d, and aligned Gaussian-3d: opposite Δ signs with unlabeled* TV $= 0$ *and minimax lower bound $\frac{1}{2}$; `val_knowability_capacity_general.py` Stage 3a.)* ☐

The general obstruction (why achievability needs a condition). Theorem 15 is fully general, but a matching *achievability* (a single computable K with the property $K > 1 \Rightarrow$ identifiable) provably cannot be. The achievability side requires the observer to (a) locate the sign-flip set of Δ over the admissible class and (b) certify that the whole admissible class lies on one side of it from Q_X alone. Stage 1b shows step (a) already fails to be captured by an alignment-blind margin once the concept direction is free (the flip locus depends on the Mahalanobis geometry between concept and rule), and Stage 2b shows that, even in one dimension, when the family is not MLR/unimodal the flip locus is non-monotone in the unknown shift, so step (b) has no single-threshold certificate. We record the general statement as a conjecture with the precise missing piece.

Conjecture 2 (General knowability–capacity; OPEN, with the exact gap)**.** *Fix observables (Q_X-family, g, f) and an admissible concept class $\mathcal{H}$ defined by a mass-drift budget ε. Suppose a* ***regularity/identifiability condition*** *holds:* (R1) *the benefit Δ has, over $\mathcal{H}$, a* unique *flip locus expressible in the observable mass coordinate; and* (R2) *that flip locus is a* monotone, observable *function of the unknown nuisance (e.g. the shift) — guaranteed by MLR/log-concavity in the location case and by Mahalanobis-alignment in the multivariate case. Then the scalar*

$$K \;=\; \frac{\mathrm{dist}_{mass}\big(\textit{calibrated boundary},\ \textit{worst-case flip locus over } \mathcal{H}\big)}{\varepsilon}$$

is a capacity at $\tau = 1$: $K > 1 \iff$ the sign of Δ is label-free identifiable, and the finite-n price is an $O(1/\sqrt{n})$ Le Cam layer around the threshold location. *The* ***open part*** *is precisely the removal (or sharp characterization) of* (R1)–(R2)*: without them the flip set need not be a single observable threshold, the worst-case-flip distance need not be computable from Q_X, and no scalar τ is simultaneously necessary and sufficient. Stages 1–2 prove the conjecture in its two leading regularity classes and exhibit explicit breakers (free-direction tilt; non-log-concave bimodality) on the boundary of each.*

Remark 9 (Honest scope of the generalization)**.** This section reaches a *strong but bounded* generalization, **not** a paradigm-level general capacity. What is *proven*: (i) an *exact* d-dimensional lift of $\tau = 1$ in Mahalanobis geometry for aligned concepts (Thm 11); (ii) the *exact* obstruction and worst-case capacity K_{free} for free halfspace concepts (Thm 12); (iii) a matched $\tau = 1$ for MLR/log-concave location families (Thm 13) with an explicit non-log-concave breaker (Thm 14); and (iv) a fully *distribution-free converse* in any dimension and any family (Thm 15). What is *not* proven, and is stated as Conjecture 2: a single computable capacity *without* a regularity/identifiability condition. The wall is sharp and named: a scalar τ is a capacity *iff* the benefit's flip locus is a unique, monotone, observable threshold;

alignment (multivariate) and MLR/unimodality (location) are the two sufficient regularity conditions we can currently prove, and each has a concrete counterexample just outside it. The value of the result is therefore (a) the converse is essentially universal, and (b) the achievability is pinned to an explicit, checkable regularity hypothesis whose failure modes are exhibited — a precise map of how far the 1-D capacity travels and exactly where it stops.

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